

IMPACT BETWEEN A PLUNGING BREAKER AND A STRUCTURE

by

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ABSTRACT

The study presented in this dissertation proposal experimentally investigates the interaction of a deep-water plunging breaker with a vertical-faced structure of finite draft. The structure, a cube with one face exposed to the incoming breaker, is held at a constant submergence (z) below the mean water level while its streamwise position (x/λ_0 , where λ_0 is the characteristic wavelength of the breaker) is systematically varied. The objective is to examine how the cube's position relative to the plunging crest influences both the free-surface evolution and the spatial and temporal distribution of impact pressures. Based on synchronized visual observations and pressure measurements obtained from 24 sensors on the cube face, the interactions across the eight streamwise positions are categorized into three distinct regimes. Zone I, referred to as the “*no-aeration*” impact zone, corresponds to early, underdeveloped breakers that produce relatively mild, non-aerated impacts characterized by a single sharp pressure peak (P_m) and negligible post-impact oscillations. In this regime, the velocity of the upward-moving contact line follows a power-law increase with time. Zone II, associated with “*low-aeration*” impacts, involves the entrapment of a small air pocket between the plunging jet and the cube face at first contact. This results in higher and more variable pressures, accompanied by strong post-impact oscillations—features consistent with air-cushion compression and subsequent rebound. These violent low-aeration impacts, although stochastic in their peak magnitude, can be reasonably represented by the idealized pressure-impulse framework. Zone III encompasses “*broken-wave*” impacts, in which the plunging crest collapses before reaching the cube, and a highly aerated, turbulent air–water mixture impinges on the structure.

In this case, the impact region shifts noticeably downward, and significant differences arise in local wave period (T_i), vertical velocity near the wall (V_c), and post-impact pressure dynamics compared with the earlier regimes. Together, these observations clarify how the streamwise placement of a finite-draft structure governs the hydrodynamic regime, impact severity, and near-wall flow evolution during plunging-breaker interactions.

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Chapter 1

Introduction

1.1 Background

When the steepness of a wave becomes sufficiently large—typically when its amplitude reaches approximately one-seventh of its wavelength—the crest loses stability, steepens further, and eventually breaks. The interaction of such a breaking wave with a structure represents one of the most violent and short-lived events in fluid mechanics, characterized by rapid energy transfer and highly unsteady flow behavior. Such phenomena are central to marine and coastal engineering, where structures are repeatedly exposed to powerful and impulsive forces generated by waves. Coastal defenses, bridge piers, and offshore installations, in particular, must withstand sudden pressure surges that can impose significant mechanical and hydrodynamic loads. Comparable situations arise in naval applications when steep or breaking waves strike a vessel's deck or hull, producing what is commonly known as slamming or green-water events. These transient interactions often result in sharp, localized pressure peaks that pose serious design and safety challenges for both fixed and floating structures operating in harsh marine environments.

During the instant of impact, a substantial portion of the wave's kinetic energy is transferred to the structure, resulting in rapid pressure rise and strong fluid accelerations. Depending on the local flow geometry, air may become trapped or entrained at the water–structure interface. The entrapped air alters the effective compressibility of the fluid, giving rise to oscillations and modifying the overall shape of the pressure pulse. Understanding the mechanisms governing these interactions is essential for

predicting the magnitude and duration of impulsive forces and for developing reliable design criteria for coastal and offshore structures.

The fluid motion during impact is typically modeled under the assumptions of incompressible, inviscid, and irrotational flow. Under these assumptions, a velocity potential $\phi(x, y, z, t)$ exists such that the velocity field is given by

$$\mathbf{u} = \nabla\phi, \quad (1.1.1)$$

and continuity requires that ϕ satisfy Laplace's equation,

$$\nabla^2\phi = 0. \quad (1.1.2)$$

Although this governing equation is linear, the associated boundary conditions are nonlinear and time-dependent, making the overall problem highly complex.

The kinematic condition on the free surface ensures that a fluid particle on the surface remains there for all time:

$$\frac{\partial\eta}{\partial t} + \frac{\partial\phi}{\partial x}\frac{\partial\eta}{\partial x} + \frac{\partial\phi}{\partial y}\frac{\partial\eta}{\partial y} = \frac{\partial\phi}{\partial z}, \quad (1.1.3)$$

where $\eta(x, y, t)$ represents the free-surface elevation. The dynamic boundary condition is obtained from Bernoulli's equation, which for unsteady potential flow can be written as

$$\frac{\partial\phi}{\partial t} + \frac{1}{2}|\nabla\phi|^2 + gz + \frac{p}{\rho} = C(t), \quad (1.1.4)$$

where p is the pressure, ρ the fluid density, g the acceleration due to gravity, and $C(t)$ a function of time that can be set based on the reference pressure. On the free surface, where the air pressure can be approximated as constant, the dynamic

condition simplifies to $p = p_a$, and the temporal evolution of ϕ directly governs the local free-surface motion.

At the interface between the fluid and a solid boundary, the normal velocity of the fluid equals the velocity of the structure surface:

$$\frac{\partial \phi}{\partial n} = \mathbf{V}_s \cdot \hat{\mathbf{n}}, \quad (1.1.5)$$

where $\mathbf{V}_s$ is the velocity of the structural surface and $\hat{\mathbf{n}}$ is its unit normal. These boundary conditions together form the mathematical basis for potential-flow impact models, which have been employed to study a wide variety of wave–structure interactions.

Despite the apparent simplicity of the governing equations, the nonlinear coupling between the free-surface motion and the impact dynamics produces rich and complex flow behavior. The transient nature of the impact, combined with the strong inertial forces, leads to very large Reynolds numbers and negligible viscous effects in the bulk flow. Air entrapment, free-surface overturning, and localized jet formation introduce additional challenges that are difficult to capture using conventional theoretical or numerical models. Although numerous theoretical and experimental studies have been carried out to understand this process, the interplay between wave kinematics, aeration, and transient pressure dynamics remains a central challenge in the field.

1.2 Review of relevant literature

Wave breaking involves energy dissipation across scales that span from meters (structure or wavelength scale) down to microns (bubble and droplet diameters) (Bagnold, 1939). In shallow water, breaking is mainly driven by depth-induced steepening

(Peregrine et al., 1983; Li and Fuhrman, 2022b), whereas in deep water, nonlinear processes such as spectral focusing (Derakhti and Kirby, 2016) and modulational instability (Melville, 1982; Li and Fuhrman, 2022a) produce strongly deformed crests. The breaking process is typically accompanied by intense air entrainment and spray generation (Deane and Stokes, 2002; Deike et al., 2016), which give rise to localized pressure fluctuations and high-speed impact jets. In deep water, the crest often overturns to form a plunging jet that impacts ahead of the crest, creating a cavity that entrains air whose subsequent collapse generates droplets, bubbles, and acoustic signatures (Ting and Kirby, 1995). Plunging breakers have been studied extensively through experiments (Rapp and Melville, 1990; Lim et al., 2015) and numerical simulations (Dommermuth et al., 1988; Hu et al., 2024). Modern Navier–Stokes and LES computations capture these processes in detail, reproducing vortex–bubble interactions (Lubin and Glockner, 2015) and related acoustic activity (Deane, 1997).

The impingement of a plunging jet on a solid surface can generate abrupt, large-amplitude, short-duration loads resulting from the rapid deceleration of the jet and the associated impact pressures (Smith et al., 1998; Ibrahim, 2020). Such impulsive loads have been implicated in severe damage to LNG carriers (often triggering hull whipping and structural fatigue), flexible or perforated barriers, wave-energy devices, monopile-type offshore installations, and various coastal structures (Oumeraci, 1994; Faltinsen and Timokha, 2009; Dias and Ghidaglia, 2018; Tiron et al., 2015; Giorgi and Ringwood, 2018). Although the duration of these loads is short, their magnitudes are governed by intense momentum exchange during jet impact, and experiments have shown that near-breaking and breaking-wave components can contribute substantially to the total hydrodynamic load. Oumeraci’s (1994) synthesis of breakwater performance in the UK, Japan, and Italy demonstrated that breaking-wave impacts are often the dominant source of structural damage and underscored the importance

of accurately quantifying these transient forces.

Recognition of extreme pressures from steep-fronted breakers striking vertical faces dates back to Stevenson (1886), De Rouville (1938), and Bagnold (1939). Reliable measurements became possible with the introduction of piezoelectric instrumentation (De Rouville, 1938; Bagnold, 1939; Denny, 1951), enabling subsequent experimental programs and field campaigns such as those by Richert (1968), Blackmore and Hewson (1984), Kirkgöz (1995), Hattori et al. (1994), Bullock et al. (1999), and the EU MAST projects MCS and PROVERBS (Oumeraci et al., 2001). Bagnold’s (1939) carefully controlled experiments showed that the largest peaks occurred when very little air was trapped and highlighted the dominant role of the incoming fluid inertia. Because linear theory contributes little near impact, numerical advances later used overturning-wave simulations (Longuet-Higgins and Cokelet, 1976) and boundary-integral formulations for steep waves at walls (Dold and Peregrine, 1986; Dold, 1992).

The presence of a wall modifies the breaking evolution. In deep water, multi-frequency focusing can lead to plunging, but reflections from a vertical face alter local focusing so that the near-wall free-surface history differs from that in open water at the same upstream position (Chan and Melville, 1988; Peregrine, 2003a; Wang et al., 2018). The relative wall position dictates the impact regime: when the wall lies far downstream of the natural breaking point, the primary jet impacts the free surface before the crest arrives, entraining large air volumes that modify compressibility and the pressure field, often producing double pressure peaks as the secondary jet reaches the wall ahead of the turbulent core (Chan and Melville, 1988); when the wall is positioned too far upstream, the breaker is underdeveloped and impacts are milder; and at an intermediate, critical position—the flip-through limit—the run-up sheet meets the oncoming crest at a single point, launching a high-speed vertical jet and

producing intense near-surface pressures with minimal air capture (Peregrine, 2003a; Moalemi et al., 2024). Such strongly focused motion yields highly localized pressures and jet accelerations of order $O(10^3\text{--}10^4 g)$ (Lugni et al., 2006).

Air entrainment and compressibility play a decisive role in shaping impact dynamics. A thin air layer at the interface can cushion the impact and lower the peak pressure, whereas a larger trapped pocket may redistribute the load over a broader area so that the total impulse is not necessarily reduced; in some cases, rebound of the compressed air can even amplify the overall momentum transfer (Peregrine et al., 2005). The ensuing oscillations of the entrapped air—arising from alternating compression and expansion—govern the characteristic post-impact pressure fluctuations, with frequencies increasing as aeration decreases; smaller bubbles oscillate faster than larger ones, and near-surface bubbles resonate at higher frequencies than those deeper in the water column (Topliss et al., 1993). The geometry of the air pocket, controlled by the instantaneous crest curvature at impact, influences the magnitude and spatial distribution of these pressures (Peregrine, 2003b). Because the presence of air alters the effective compressibility of the mixture, Froude scaling alone often overpredicts prototype pressures. To account for this, both spring–mass (Bagnold–Mitsuyasu-type) models and homogenized mixture approaches have been developed, linking the pressure response to void fraction, approach Mach number, and the relative magnitudes of U^2 and p_0 (Peregrine and Thais, 1996; Bullock et al., 2001). Experimental evidence further indicates that at low peak pressures (below about 318 kPa), a Froude-like trend persists, whereas at higher intensities, the Bagnold–Mitsuyasu scaling provides a better description (Bredmose et al., 2015). Laboratory and field observations also show that the most extreme loads often arise when a small air pocket is trapped between a nearly vertical or slightly curled front and the wall, or when a plunging breaker engulfs a sizable cavity (Hattori et al., 1994; Cooker and Peregrine, 1990;

Bredmose et al., 2009; Tsaousis and Chatjigeorgiou, 2020; Hull and Müller, 2002; Hu et al., 2024). Such aerated impacts commonly exhibit post-peak oscillations and even sub-atmospheric phases (Hattori et al., 1994; Bullock et al., 2007; Hu and Li, 2023), and at full scale, discrepancies between measurements and model predictions have led to empirical adjustments that explicitly account for the degree of incident aeration (Blackmore and Hewson, 1984).

Despite these advances, significant event-to-event variability remains a hallmark of wave-impact phenomena. While the pressure impulse (the time-integrated pressure) is often repeatable (Cooker and Peregrine, 1995), the instantaneous peaks and their spatial distributions exhibit notable variability between realizations due to their sensitivity to subtle variations in the pre-impact breaker geometry (Bullock et al., 2007; Raby et al., 2022; Moalemi et al., 2024). Such irregularities arise from minor differences in crest position, near-wall kinematics, or air-pocket formation—often influenced by residual disturbances from the preceding impact, wave-maker motion, or partial breaking as the wave travels toward the structure. Broader numerical studies of slamming and impact processes (Hirt and Nichols, 1981; Zhang et al., 1996; Nielsen and Mayer, 2004) and visualizations of approach flow (van Meerkerk et al., 2020) corroborate this stochasticity, with impact-location variability reported to be on the order of 0.5% in repeated experiments.

1.3 Research gap and Objectives

Although breaking-wave impacts have been investigated for many decades (e.g., Bagnold, 1939; Shen et al., 2025), their nonlinear and highly transient nature remains only partially understood. Most laboratory studies employ a wall extending to the bed, which enforces a quasi-two-dimensional, fully confined approach flow. In contrast,

many real-world offshore and coastal structures expose only a partially submerged vertical face to incoming waves—for instance, the upper portions of jacket legs and transition pieces, the freeboards of caissons and column-stabilized units, deck edges on pile-supported bridges, wave-energy converters, and auxiliary components attached to monopiles. As offshore infrastructure continues to extend into deeper waters, an important question arises: do deep-water plunging impacts occur on such finite-draft faces, and if so, how do their mechanics differ from the well-documented shoaling impacts on full-depth walls?

Despite the prevalence of shallow-draft components relative to wavelength and depth, detailed investigations addressing this configuration remain limited. To address this gap, the present experimental study examines the interaction between a plunging breaking wave and a two-dimensional structure with a finite draft of 0.91 m below the still water level. The primary focus is on the interaction of the plunging breaker at different stages of its evolution, achieved through systematic variations in the streamwise position of the cube’s front face relative to the nominal breaking location. This parameter is critical because the streamwise offset determines the impact regime at the face (no-aeration, low-aeration, or high-aeration), governs air entrainment, and consequently influences the peak pressure, impulse, and post-impact oscillations. The resulting dataset provides benchmark measurements of pressure, near-wall kinematics, and spectral content relevant to structures whose upper portions are exposed to waves. The objective of the present study is not to propose a new impact mechanism but to provide geometry-specific experimental evidence that clarifies when results from bed-mounted walls can be extended to finite-draft faces.

This proposal is organized into four main chapters. Chapter 2 describes the experimental setup. Chapter 3 discusses the kinematics of free-surface evolution as the wave approaches and interacts with the cube, as well as the post-impact evolution.

Chapter 4 presents the dynamic response, with a detailed analysis of the associated pressure characteristics. Chapter 5 summarizes the key findings, and finally, Chapter 6 outlines the plan for subsequent analyses and future work.

Chapter 2

Experimental Details

2.1 Experimental setup

The experiments were conducted in a 14.8-m-long and 1.22-m-wide wave tank, shown in Figure 2.1. The water depth, i.e., vertical distance from the still water level to the bottom plane of the tank, is 0.91 m throughout the experiments. The tank consists of steel frames that support transparent glass and acrylic side walls, allowing optical access from the sides and above the tank. A wedge wave maker is installed at one end of the tank. The wedge consists of a vertical face and an inclined face that is 30° from the vertical plane. Both faces span the entire width of the tank. During wave generation, the wedge moves vertically, driven by an electric servo motor. The instantaneous position of the wedge is controlled by a feedback system to follow the prescribed profile in real time. When the wedge is at equilibrium position, its vertex is located 44.3 cm below the still water level.

A cube with side length $\ell = 30.48$ cm is installed with its center positioned at an equal distance from the two side walls of the tank. The front face of the cube is kept vertical and perpendicular to the axis of wave propagation (streamwise direction). The bottom face of the cube is placed at $z_b = -133.4$ mm, relative to the still water level throughout the present experiments. The distance from the front face of the cube to the vertical face of the wave maker, x_c , called the streamwise position of the cube herein, is varied in a range from 640.5 cm to 670.8 cm for different conditions.

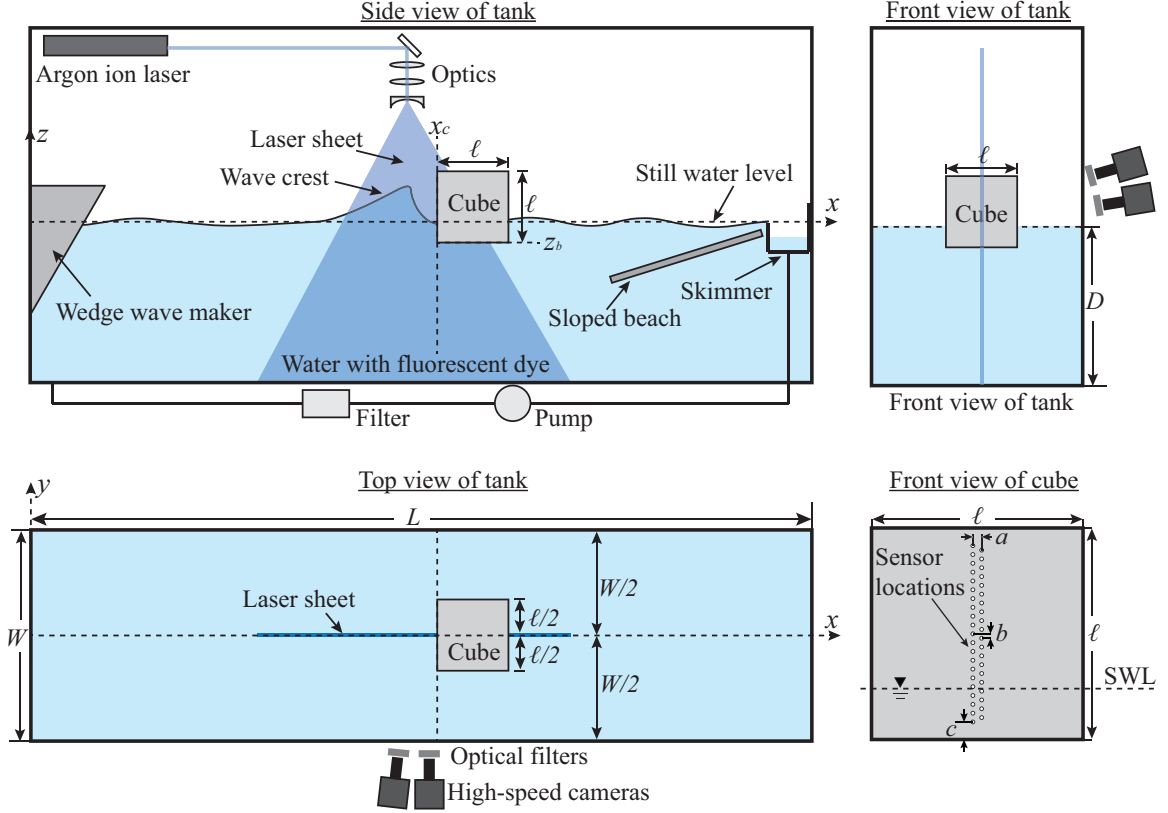


Figure 2.1: Schematic of the experimental setup. The cube front face is located at a streamwise position x_c relative to the back of the wave maker ($x = 0$). The value of x_c is varied during the experiments. The bottom face of the cube is located at vertical position $z_b = -133.4$ mm relative to the still water level ($z = 0$) throughout the experiments. The center of the cube is located at the cross-stream centerplane of the tank ($y = 0$). The length of all cube edges is $\ell = 304.8$ mm. Other notable dimensions in the figure: tank length, $L = 14.80$ m; tank width, $W = 1.22$ m; water depth, $D = 0.91$ m; horizontal spacing between adjacent pressure sensors, $a = 12.70$ mm; vertical spacing between adjacent pressure sensors, $b = 6.35$ mm; distance from the lowest sensor to the cube bottom edge, $c = 25.40$ mm. Note that the schematic is not drawn to scale in order to highlight the key features of the experiments.

2.2 Measurements of surface profiles and pressures

The time-resolved water surface profiles are measured with a cinematic laser induced fluorescence (LIF) technique. An argon ion laser operated at 5 W in multi-line mode is used to generate a beam that is spread into a laser sheet by a set of mirrors, spherical and cylindrical lenses. The laser sheet is positioned in a vertical plane parallel to the side walls and is focused at the mean water surface with the light sheet thickness approximately 1 mm. For most of the conditions, the plane of the laser sheet passes the center of the cube, while for some conditions, the distance from the cube's center to the plane of the laser sheet is varied. Fluorescent sodium salt is added to the tank water at a mass concentration of 1.5 g/m^3 to provide illumination of the tank water in the plane of the laser sheet. Two high-speed cameras (Phantom V640, V641, Vision Research Inc.), with 60 mm and 85 mm lenses and low-pass optical filters, record the evolution of the illuminated water surface through the transparent side wall of the tank. The high-speed cameras were operated at frame rates ranging from 1500 Hz to 4500 Hz.

Fast-response piezoelectric pressure sensors (113B28, PCB Piezotronics Inc.) are installed to the front panel of the cube. The front face of the sensor (the sensing area) has a diameter of 5.54 mm. The front face of each sensor is mounted with a 0.0508 mm recess from the front face of the cube and the recess is filled with insulation grease to prevent thermo shock to the sensors when their sensing area transitions between dry and wet states. These pressure sensors have a measurement range of 345 kPa, a resolution 0.007 kPa and rise time less than $1 \mu\text{s}$. The locations for the pressure sensors are shown in Figure 2.1. The two primary sets of sensor positions are located on each side of the cube's centerline by an offset of 6.35 mm. The sensor positions along the two columns are staggered to allow a higher spatial resolution

for the distribution of pressure. The smallest vertical distance between adjacent pressure sensors is 6.35 mm. Several pressure sensors are also installed between the cube's centerline and the edge to explore the variation of pressure in the cross-stream direction. The cube is sealed to prevent water from entering its interior. The pressure data is captured via an A/D data acquisition system (National Instrument PXI-6133, 14-bit) operated at a sample rate of 900 kHz. The measurements of the pressure and the water surface evolution are synchronized with the wave maker motion via a pulsed-delay generator.

2.3 Breaking wave generation

The experimental waves were produced using a dispersive focusing technique, following the approach of Rapp and Melville (1990); Duncan et al. (1999); Wang et al. (2018), and others. In this procedure, a group of wave components with varying frequencies (with an average frequency of $f_0 = 1.15$ Hz) is generated and propagated along the flume. According to linear wave theory, individual frequency components travel with different phase velocities. As a result, by selecting appropriate frequency spacing, the components can be arranged to converge at a specified point, leading to an amplification of the composite wave amplitude. This process ultimately gives rise to a plunging breaker in deep water.

The motion of the wave paddle for the present configuration is expressed as

$$z_c = w(t) A \lambda_0 \frac{1}{N} \sum_{i=1}^N \left(\frac{k_0}{k_i} \right)^{1.75} \cos \left[x_b \left(\frac{\omega_i}{\bar{c}_g} - k_i \right) - \omega_i t + \frac{\pi}{2} \right], \quad (2.3.1)$$

where N is the total number of frequency components, A is a dimensionless scaling factor controlling overall amplitude, and x_b represents the distance between the

maximum linear amplitude location and the rear surface of the wave maker wedge. The parameters (k_i, ω_i) denote the wavenumber and angular frequency of the i^{th} component, with $k_i = \omega_i^2/g$ according to linear theory. The reference wavenumber $k_0 = \bar{\omega}^2/g$, where $\bar{\omega}$ is the mean angular frequency of all N components. The mean group velocity of the wave components is c_g , and $\pi/2$ is the phase shift. The function $w(t)$ acts as a temporal window controlling the wave maker motion, and q defines how the energy is distributed among the spectral components.

The i^{th} component frequency is given by

$$\omega_i = \bar{\omega} - \frac{\Delta\omega}{2} + (i-1)\frac{\Delta\omega}{N-1},$$

where $\bar{\omega}$ and $\Delta\omega$ are constants ensuring that the frequencies are uniformly spaced in the spectral domain. The window function $w(t)$ ensures that the paddle remains nearly stationary when the sum of component motions tends to zero, and is defined as

$$w(t) = 0.25 [\tanh(\zeta f_0(t - \tau_1)) + 1] [1 - \tanh(\zeta f_0(t - \tau_2))], \quad (2.3.2)$$

where ζ governs the rise rate of the modulation envelope, taken as 5.0 in these experiments. The function remains approximately unity for most of the interval between $t = \tau_1$ and $t = \tau_2$ and vanishes outside this range. The parameters τ_1 and τ_2 are selected such that the highest and lowest frequency components reach the focal point x_b simultaneously:

$$\tau_1 = x_b \left(\frac{1}{\bar{c}_g} - \frac{1}{c_N} \right), \quad (2.3.3)$$

where $\bar{c}_g$ and c_N are the group velocities of the mean and the N^{th} frequency components, respectively.

All tests were performed using the following parameters for the paddle motion:

$$N = 32, \quad A = 0.0688, \quad q = 1.75, \quad \Delta\omega = 0.7\bar{\omega}, \quad \bar{\omega} = 7.226 \text{ s}^{-1}, \quad x_b = 7.15\lambda_0,$$

where $\lambda_0 = 2\pi g/\bar{\omega}^2 = 1.181 \text{ m}$. These settings yield $f_0 = \bar{\omega}/(2\pi) = 1.15 \text{ Hz}$, a mean group velocity of $\bar{c}_g = 0.7138 \text{ m/s}$, and a characteristic phase velocity of $\bar{c}_p = 2\bar{c}_g = 1.437 \text{ m/s}$. In the present experiments, the same wave maker motion is used for all conditions.

2.4 Experimental procedures and data processing

The experiments were performed following the procedures described below. First, at each streamwise position, the cube is secured to ensure no relative motion between the cube and the wave tank during the impact. Subsequently, a flat plate covered by checkerboard pattern is installed on the cube at the plane of the laser sheet, and the water level in the tank is lowered so that the checkerboard pattern is visible within the entire field of view of the cameras. Then, images of the checkerboard pattern were taken by each camera for image calibration purposes. After the calibration images were recorded, the calibration plate is removed and the water level in the tank is raised to the working condition at 0.91 m with the wave maker wedge raised to its equilibrium position. Before each run, the tank was left to rest for 20 min to ensure that all water surface disturbance is damped. Before the wave maker motion starts, each high-speed camera will record an LIF image of the still water surface to be used as the reference of streamwise direction for rotating the water surface profiles back to the earth reference frame. After the images of the still water surface were taken, the wave is generated to produce the impact event and measurements of the water surface profiles and the pressure on the cube are conducted. After the measurement

is completed, the surface filtration and circulation system (consisting of a skimmer, a pump and a filter) is turned on for 10 min to remove possible unsoluble surfactants that enters the tank system during the impact.

For each LIF image, the water surface profile in the plane of light sheet is extracted based on the gradient of image intensity (see method used in Liu and Duncan, 2006; Wang et al., 2018). After the profiles are extracted, a third order polynomial was used to correct the distortions created by the inclined viewing angle between the camera axis and the normal of the laser sheet, as well as the imperfection of the lens system. After the correction, the profiles were rotated to the laboratory coordinate system with the water surface as the x axis, according to the slope of still water surface in the image captured before each experimental run.

2.5 Experimental conditions

To facilitate the description of the experiments, a global Cartesian coordinate system is defined, as shown in Figure 2.1. In this coordinate system, the $x - y$ plane is at the still water surface, with the x - and y - axes parallel and perpendicular to the direction of wave propagation, respectively. The vertical face of the wedge wave maker is located at $x = 0$, and the side walls of the tank are positioned at $y = \pm W/2$, where W is the width of the tank. The z -axis is normal to the still water surface and points upward, with the still water surface at $z = 0$.

The present experiments were conducted with varying streamwise positions of the cube's front face (x_c), as summarized in table 3.1.

Chapter 3

Kinematics of Plunging-Breaker impact

This chapter examines the kinematic evolution of the free surface as the plunging breaker approaches and interacts with the finite-draft structure. The analysis combines high-speed LIF measurements to track the crest motion, contact-point rise, and surface deformation leading up to and following impact. These observations provide the physical basis for distinguishing different impact regimes and interpreting the flow mechanisms that precede pressure generation

3.1 Categories of Impact

The cube face was positioned at eight streamwise locations within the tank to interact with a nominally identical plunging breaker. Depending on the local wave evolution and the resulting impact characteristics, three distinct regimes were identified: no-aeration, low-aeration, and broken-wave impact. The classification was determined through frame-by-frame inspection of the LIF recordings combined with analysis of the corresponding pressure time series. At each location, three to four repeated trials were conducted to assess the repeatability of both the wave evolution and the impact response. The streamwise positions and their corresponding impact categories are summarized in Table 3.1. Each impact type is examined separately in the following sections, where the wave evolution immediately before and after impact is analyzed alongside key pressure characteristics.

Table 3.1: Streamwise positions of the cube front and their corresponding impact classifications. The distance from the vertical face of the wavemaker ($x = 0$) to the front face of the cube is denoted by x_c , and x_c/λ_0 , where $\lambda_0 = 118$ cm, is the nominal wavelength of the wave packet. Impact types are assigned based on visual inspection of the LIF recordings and the associated pressure measurements.

x_c (cm)	x_c/λ_0	Impact category
639.9	5.423	No-aeration
640.8	5.431	
644.9	5.466	Low-aeration
645.9	5.474	
646.9	5.483	
649.0	5.500	Broken-wave
654.2	5.544	
670.2	5.680	

3.1.1 No-aeration impact

For streamwise locations categorized under no-aeration impact, the front face of the cube is positioned where the plunging breaker remains underdeveloped until it reaches the cube. The wave crest does not attain sufficient steepness to form an overturning jet—a defining feature of a fully developed plunging breaker. Consequently, the wave front primarily sloshes against the vertical face of the cube and rises upward along it, producing an impact of comparatively lower intensity than that observed in the low-aeration and broken-wave regimes discussed later.

Among the two streamwise positions classified under the no-aeration category, an experimental run from the condition $x_c = 5.431\lambda_0$ is selected for discussion, as it results in a higher-intensity impact with the cube. This is evident from the highest peak nondimensional pressures (P_m) measured across multiple trials. The parameter P_m represents the peak nondimensional pressure recorded at the elevation (z/λ_0) of each pressure sensor over the entire duration of measurement. Since there are 24

pressure sensors mounted on the front face of the cube, we obtain 24 values of P_m , each corresponding to a different sensor elevation. The highest peak nondimensional pressure is defined as the largest among these 24 values. For the four trials conducted at $x_c = 5.423\lambda_0$, the highest recorded P_m is $0.69\rho\bar{c}_p^2$ at $z/\lambda_0 = 0.107$, whereas at $x_c = 5.431\lambda_0$, the highest P_m reaches $1.07\rho\bar{c}_p^2$ at $z/\lambda_0 = 0.102$ over three trials.

Figure 3.1 presents a sequence of images capturing the evolution of the wave as it approaches and interacts with the front face of the cube for the condition $x_c = 5.431\lambda_0$. The point of wave impact, highlighted in Figure 3.1*b*, corresponds to the moment when the crest impacts the cube front, coinciding with the highest recorded P_m at that instant. Following the impact, as shown in Figure 3.1*c*, an upward-directed jet emerges and propagates at high velocity.

Additionally, Figure 3.2 illustrates the surface profiles for $x_c = 5.431\lambda_0$ within the laboratory reference frame, depicting the wave evolution from the initial rise of the water surface at the contact point to the instant of impact. As the wave crest advances toward the cube, a distinct dip in elevation is observed, followed by a steepening and a subsequent increase in crest height near the impact location. This steepening is accompanied by a rapid rise in surface elevation at the frontal face of the cube, progressing at a velocity comparable to that of the advancing crest. A similar phenomenon was observed for the condition $x_c = 5.423\lambda_0$, with the only distinction being a lower-intensity wave impact on the cube front.

A frame-by-frame analysis of the LIF movies confirms the absence of air entrainment during the wave impact. No evidence of trapped air bubbles is observed as the crest interacts with the front face of the cube, both at the moment of impact and during the subsequent upward jet formation. This observation is further supported by the pressure measurements at elevations $z/\lambda_0 = 0.091$ to 0.107 , shown in Figure 3.3, which exhibit no oscillations in the post-impact pressure records. The absence of such

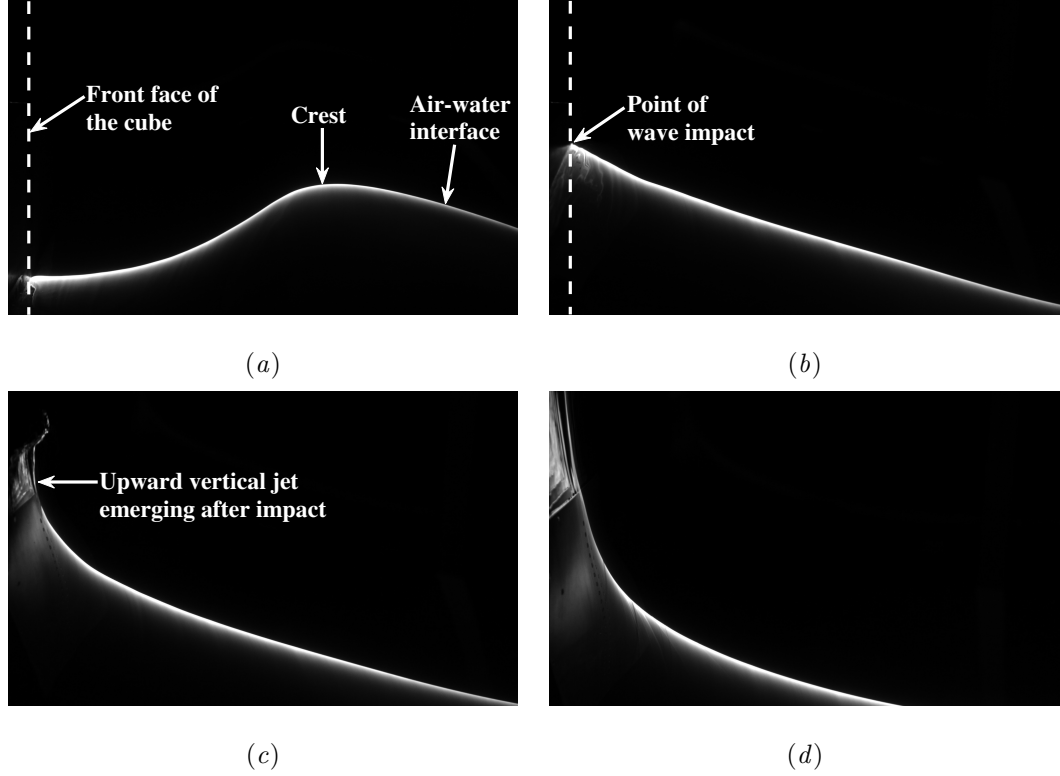


Figure 3.1: A series of frames from a high-speed LIF recording illustrates the evolution of the free surface for $x_c = 5.431\lambda_0$. The field of view spans approximately $54 \text{ cm} \times 35 \text{ cm}$ and the wave travels from right to left in every snapshot. The images were recorded at a frame rate of 1500 fps. Key features of the evolving water surface, including the position of the front face of the structure relative to the incoming wave, are highlighted. The frames from panels (a) to (d) correspond to $tf_0 = -0.164, 0, 0.026$, and 0.130 , respectively. The instant when the wave crest reaches the structure's front face, shown in panel (b), is defined as the time of impact and set as $t = 0$.

oscillations indicates a lack of cyclic compression and expansion, which would otherwise be indicative of entrapped air bubbles. The inset in Figure 3.3 highlights this aspect, showing a distinct pressure peak at each sensor location in the impact zone. Additionally, the progressive delay in the occurrence of peak pressure at increasing elevations signifies the upward propagation of the wave front upon impact with the cube.

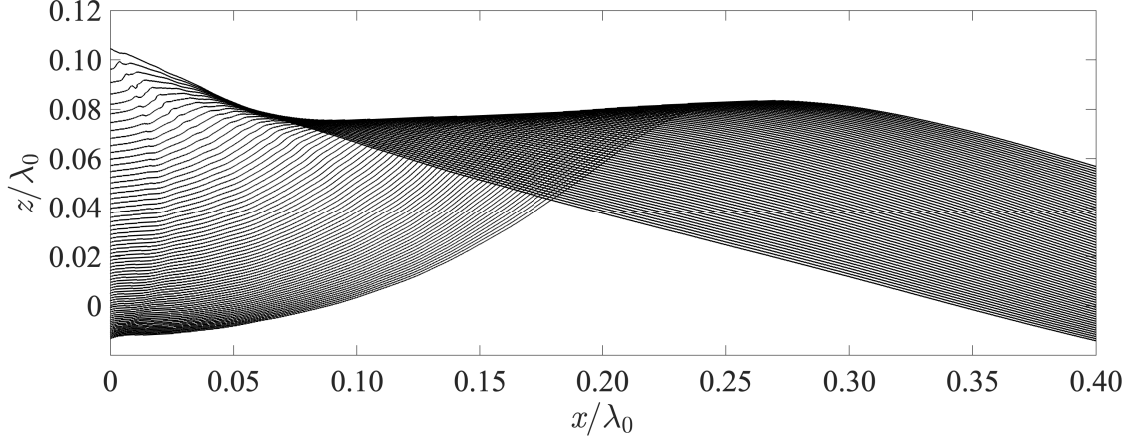


Figure 3.2: Surface profile histories plotted in the laboratory reference frame for $x_c = 5.431\lambda_0$. The time between profiles is 2.67 ms ($=0.00307/f_0$) with a total covered time of 200 ms. The final profile plotted coincides closely with the instant of impact. The mean water level corresponds to $z/\lambda_0=0$.

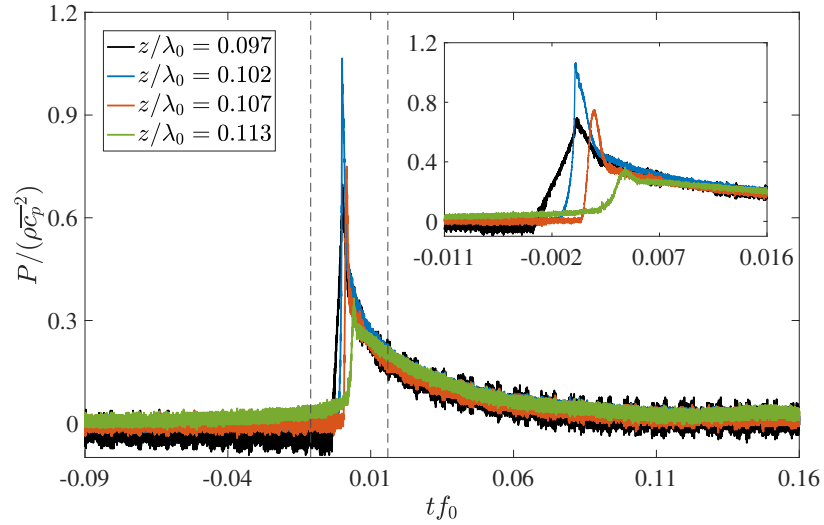


Figure 3.3: Pressure-time histories for the sensors located in the impact zone for the condition $x_c = 5.431\lambda_0$. The vertical dashed black lines indicate the time interval corresponding to the inset, which highlights the pressure peaks recorded at each plotted elevation.

3.1.2 Low-aeration impact

With further downstream positioning of the cube, the impact transitions to a low-aeration regime. In this case, the plunging wave becomes steeper, and a small pocket of air is likely entrapped upon impact, resulting in a more intense interaction compared to the no-aeration case. The surface profiles in Figure 3.4 indicate that the wave crest propagates toward the cube in a manner similar to the no-aeration cases. However, a key distinction is the formation of a small air pocket between the jet emerging from the steep wave and the wall at the moment of impact.

For the low-aeration regime, the condition $x_c = 5.483\lambda_0$ is considered for discussion, where the highest $P_m = 12.64\rho\bar{c}_p^2$, is recorded at $z/\lambda_0 = 0.091$. Figure 3.5 illustrates the wave impact sequence for this condition. A jet emerges from the tongue of the wave and strikes the wall, marking the impact point. Frame-by-frame analysis of the LIF recordings suggests that a small volume of air is likely trapped between the jet and the rising water at the contact point. This inference is further supported by pressure records from sensors in the impact zone ($z/\lambda_0 = 0.086$ to $z/\lambda_0 = 0.102$), shown in Figure 3.6, which—unlike the pressure records under low-aeration conditions—exhibit sub-atmospheric pressures followed by synchronized, damped oscillations after the peak pressure spike. These oscillations are characteristic of alternate compression and expansion of an entrapped air pocket (Peregrine, 2003a). A more detailed discussion of this phenomenon is provided later. After the impact, a vertical jet ascends along the wall, similar to what is observed in the no-aeration impact scenario.

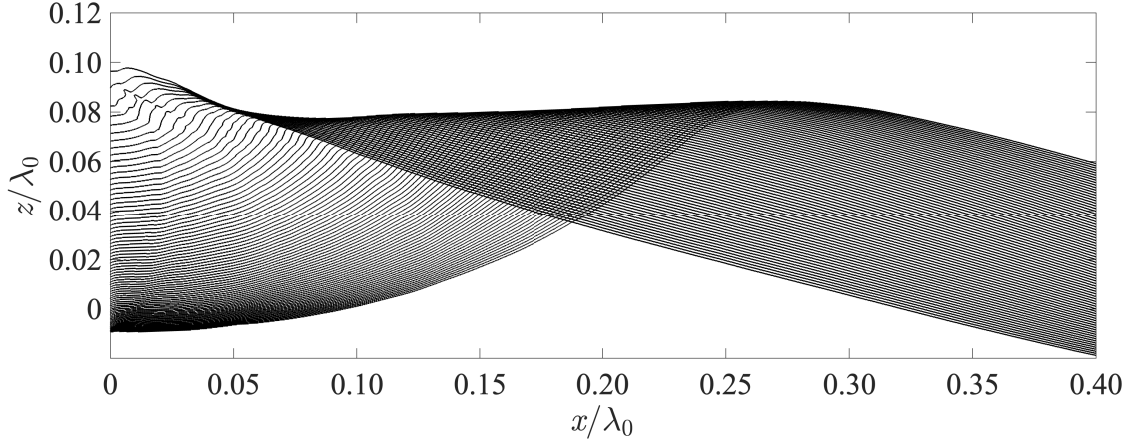
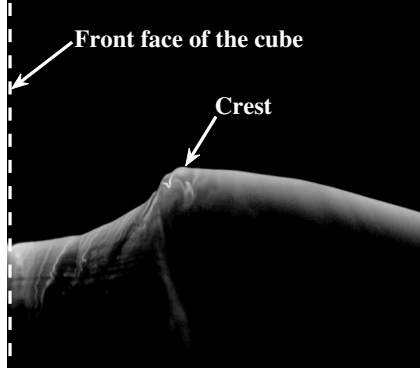


Figure 3.4: Surface profile histories plotted in the laboratory reference frame for $x_c = 5.483\lambda_0$. The time between profiles is 2.67 ms ($=0.00307/f_0$) with a total covered time of 211 ms. The final profile plotted coincides closely with the instant of impact. The mean water level corresponds to $z/\lambda_0=0$.

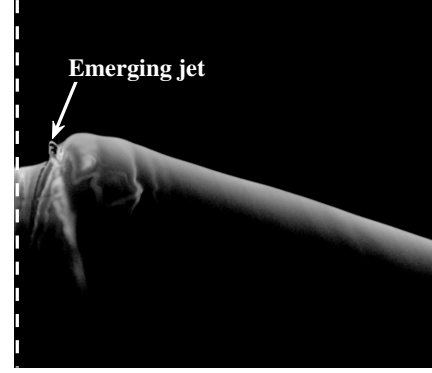
3.1.3 Broken-wave impact

For streamwise positions classified under the broken-wave impact regime, the cube is located sufficiently downstream such that the plunging crest overturns and breaks onto the upstream water surface before reaching the cube front. As a result, a highly aerated and turbulent air–water mixture interacts with the vertical face of the structure. The evolution of the plunging breaker leading to a broken-wave impact at $x_c = 5.680\lambda_0$ is illustrated in Figure 3.7. This position is selected for discussion as it corresponds to the farthest downstream location where the breaking process and the subsequent impact of the fully developed broken wave on the structure are clearly observable. It provides a representative example of a broken-wave impact and supports a clear understanding of the transition from a coherent plunging crest to a chaotic, aerated wave–structure interaction.

The surface profiles obtained from the LIF measurements for this broken-wave



(a)



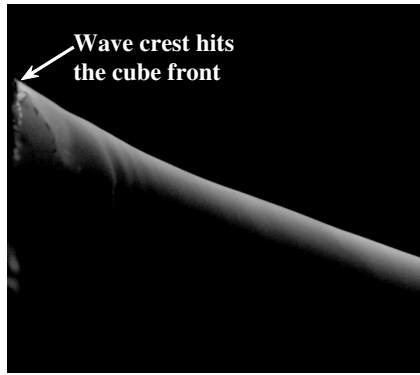
(b)



(c)



(d)



(e)



(f)

Figure 3.5: A series of frames from a high-speed LIF recording captures the evolution of the free surface for $x_c = 5.483\lambda_0$. The front face of the structure is positioned along the left edge of the images. The recordings were obtained at a frame rate of 4500 fps, with a field of view measuring approximately $16\text{ cm} \times 14\text{ cm}$. The frames in panels (a) to (f) correspond to $tf_0 = -0.0247, -0.0056, -0.0019, 0, 0.0048$, and 0.0419 , respectively.

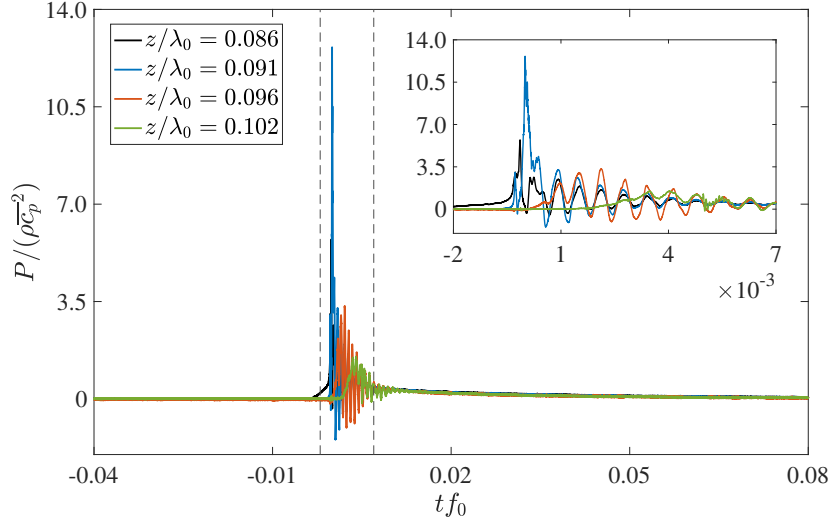


Figure 3.6: Pressure-time histories for the sensors located in the impact zone for the condition $x_c = 5.483\lambda_0$. The vertical dashed black lines indicate the time interval corresponding to the inset, which highlights the pressure peaks recorded at each plotted elevation.

impact condition are shown in Figure 3.8. In contrast to the profiles observed in the no-aeration and low-aeration cases, the downstream placement of the cube in this case allows the wave to reach its maximum surface elevation prior to breaking. Following the onset of breaking, the wave crest becomes increasingly unsteady and disintegrates, forming a secondary jet that crashes onto the upstream water surface. This secondary jet impacts the vertical structure at a significantly lower elevation than that observed in the no-aeration and low-aeration scenarios.

Figure 3.9 presents the pressure-time history for an experimental run at $x_c = 5.680\lambda_0$, where the highest P_m of $5.88\rho\bar{c}_p^2$ is recorded at $z/\lambda_0 = 0.043$. The pressure histories for sensors within the impact zone exhibit oscillations following the peak P_m , resulting from the interaction of the highly aerated flow with the cube. However, unlike the synchronized pressure variations observed in Figure 3.6 for the low-aeration case, the oscillations here are irregular and appear out of phase across

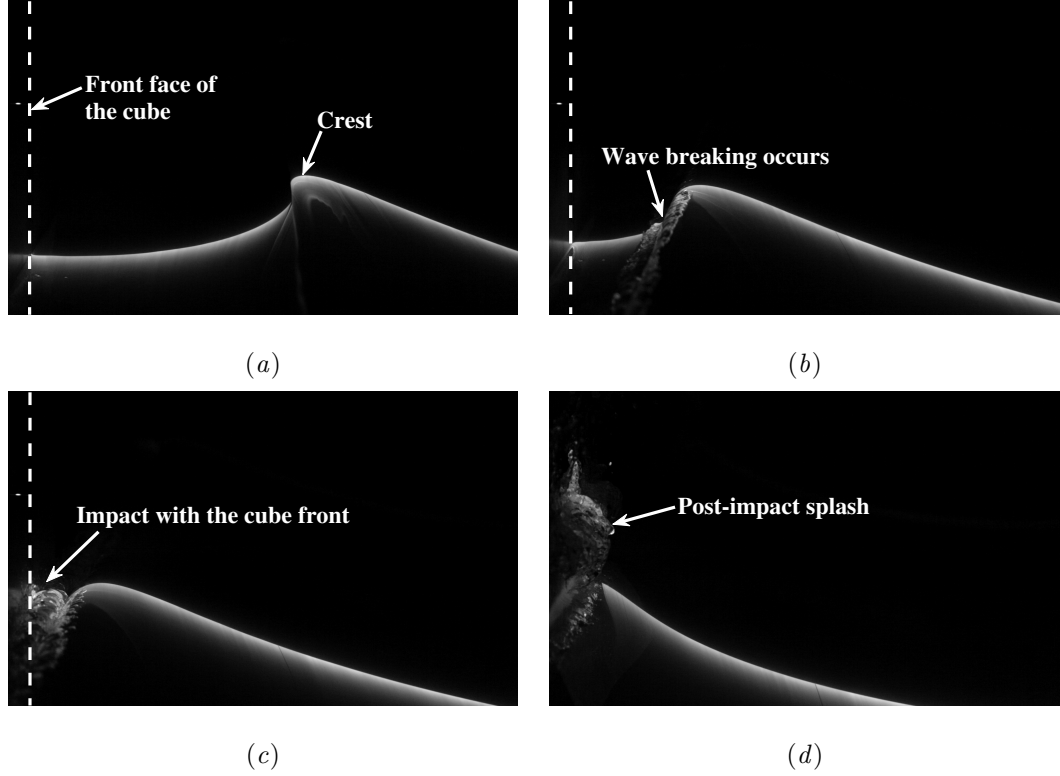


Figure 3.7: A series of frames from a high-speed LIF recording captures the evolution of the free surface for $x_c = 5.68\lambda_0$. The recordings were obtained at a frame rate of 1500 fps, with a field of view measuring approximately $54 \text{ cm} \times 35 \text{ cm}$. The frames in panels (a) to (f) correspond to $tf_0 = -0.161, -0.049, 0$, and 0.083 , respectively.

different sensors. This highlights the increased complexity of impact dynamics under broken-wave conditions, where aeration effects introduce significant variability in pressure distributions.

3.2 Evolution of the approaching wave surface

3.2.1 Surface Profile Envelopes

A useful metric to compare the evolution of the water surface under the three impact scenarios is by analyzing the top envelope of the surface profiles, as shown in Figure 3.10. For any streamwise location, points lying above this envelope remain in

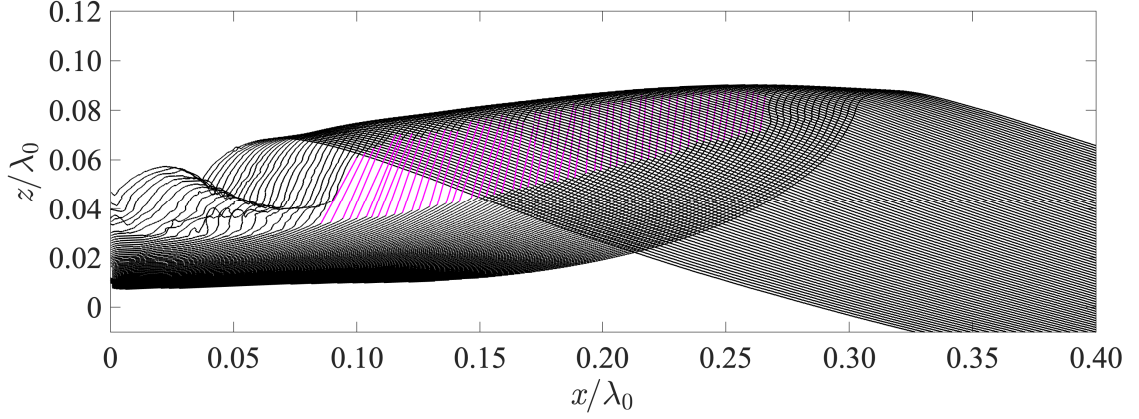


Figure 3.8: Surface profile histories plotted in the laboratory reference frame for $x_c = 5.680\lambda_0$. The time between profiles is 2.67 ms ($=0.00307/f_0$) with a total covered time of 257 ms. The mean water level corresponds to $z/\lambda_0=0$. Magenta-colored segments in some profiles indicate portions of the surface that were obscured by the plunging breaker crest passing overhead, making those regions untrackable in the recorded images. The gaps were interpolated using straight lines to connect the missing sections, thereby preserving spatial continuity and enabling coherent visualization of the surface evolution.

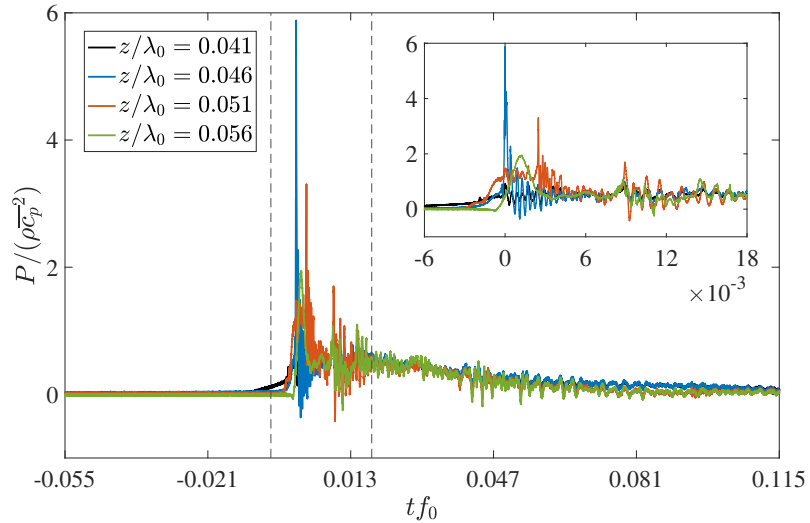


Figure 3.9: Pressure-time histories for the sensors located in the impact zone for the condition $x_c = 5.680\lambda_0$. The vertical dashed black lines indicate the time interval corresponding to the inset, which highlights the pressure peaks recorded at each plotted elevation.

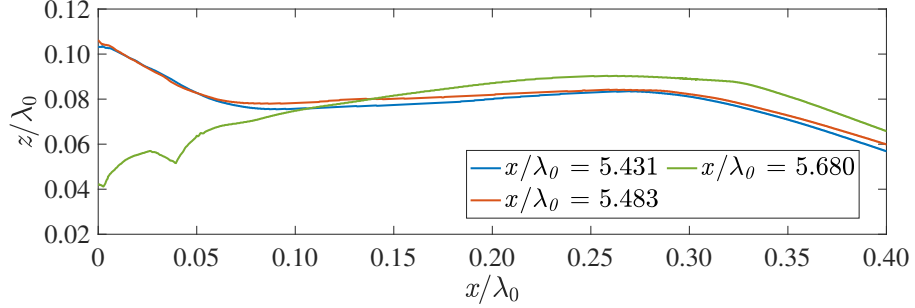


Figure 3.10: Top envelopes of the surface profiles up to the moment of crest impact for the no-aeration and low-aeration cases, and up to the moment of secondary jet impact for the broken-wave condition.

air throughout the recorded duration, while those below the envelope are submerged for at least some amount of time. In the two non-breaking cases, corresponding to $x_c = 5.431\lambda_0$ (no-aeration) and $x_c = 5.483\lambda_0$ (low-aeration), the top envelopes exhibit a characteristic rise–dip–rise pattern. The first crest is observed between $0.25 < x/\lambda_0 < 0.28$, followed by a depression within $0.06 < x/\lambda_0 < 0.08$, and a second peak occurs near the vertical face of the cube. This final peak results from the interaction between the vertically rising water column adjacent to the cube face and the approaching wave crest, a phenomenon discussed in more detail in a later section. Although both envelopes share the same overall structure, slight differences exist in the locations and magnitudes of the features. For the $x_c = 5.431\lambda_0$ case, the first crest reaches a maximum elevation of $0.083\lambda_0$ around $x/\lambda_0 = 0.27$, while for $x_c = 5.483\lambda_0$, a slightly higher elevation of $0.084\lambda_0$ is reached near $x/\lambda_0 = 0.26$.

In contrast, the top envelope under the broken-wave condition at $x_c = 5.680\lambda_0$ displays a substantially different structure, characterized by a single prominent peak within the plotting window. This peak reaches a maximum elevation of $0.09\lambda_0$ at $x/\lambda_0 = 0.258$, which is noticeably higher than the other two impact cases. The

observed differences in envelope profiles are closely tied to the streamwise position of the cube front. As the cube is placed further downstream—transitioning from $x_c = 5.431\lambda_0$ to $x_c = 5.483\lambda_0$ and finally to $x_c = 5.680\lambda_0$ —the plunging breaker is afforded progressively more space to develop and steepen. This trend is evident in the top envelope comparisons. In the low-aeration case at $x_c = 5.483\lambda_0$, the envelope remains slightly elevated compared to the no-aeration case, indicating more pronounced crest development. Although neither of these two conditions shows clear plunging behavior on the upstream water surface, the additional space in the low-aeration case supports slightly greater steepening.

When the cube is positioned farther downstream at $x_c = 5.680\lambda_0$, the plunging breaker has sufficient space to complete its steepening process and undergo breaking. The top envelope for this condition exceeds those of the other two cases until it begins to dip, indicating that the breaker has reached its maximum elevation. The subsequent decline in the envelope profile is consistent with wave breaking. A small rise in the envelope can then be observed around $x/\lambda_0 = 0.027$, corresponding to the formation of a secondary jet that emerges after the plunging crest impacts the upstream surface. This secondary jet propagates forward and strikes the wall at a considerably lower elevation than that observed in either the no-aeration or low-aeration cases.

3.2.2 Rising water column and crest interaction

For the first two impact categories, where the wave reaches the cube front without prior breaking on the upstream surface, an interesting aspect to examine is the rate of water-surface rise along the cube face as the crest approaches. At the streamwise position of $x_c = 5.431\lambda_0$, no air entrainment is observed during the impact, which can be attributed to the absence of developed wave breaking. In this case, a thin water

column rises at the contact point and intercepts the advancing wave crest, resulting in a no-aeration impact. By contrast, for $x_c = 5.483\lambda_0$, slight air entrapment is evident, as earlier highlighted in Figure 3.5*c*. These differences are further examined by comparing the velocity of the rising water surface at the contact point along the cube face.

Based on the surface profiles presented in Figure 3.2 and Figure 3.4, the vertical position (z) and velocity (V_c) of the rising water surface along the front face of the cube are plotted in Figure 3.11 for the no-aeration and low-aeration cases. These parameters are tracked from the initial rise of the water surface at the contact point until the instant of impact (t_i). The velocity V_c is computed using finite differences of the crest elevation over time, based on experimental data. A notable observation is the difference in the initial elevation (z) from which the water surface begins to climb along the cube face: $z = -0.011\lambda_0$ for $x_c = 5.431\lambda_0$, and $z = -0.008\lambda_0$ for $x_c = 5.483\lambda_0$.

Another interesting observation is the elevation of the contact point just before it is intercepted by the oncoming wave crest. For $x_c = 5.431\lambda_0$, the elevation at the face of the cube reaches $z = 0.105\lambda_0$, whereas for $x_c = 5.483\lambda_0$, it reaches $z = 0.092\lambda_0$. Thus, although the rising water surface begins at a slightly higher z for the case $x_c = 5.483\lambda_0$, its elevation just before t_i is lower than in the former case, resulting in a steeper curve for $x_c = 5.431\lambda_0$ in Figure 3.11*a*. The closer proximity of the wall to the wave-breaking initiation in the case of $x_c = 5.431\lambda_0$ results in a higher z before impact.

The V_c curves presented in Figure 3.11*b* exhibit consistently higher values for $x_c = 5.431\lambda_0$ before impact, although both cases show a similar overall trend. The consistently higher V_c observed for $x_c = 5.431\lambda_0$ helps explain the onset of air entrapment at $x_c = 5.483\lambda_0$. Because the contact point rises more slowly in the

latter case, it reaches a slightly lower elevation when the crest impacts the cube face, allowing a thin air pocket to become trapped between the two.

Another possible explanation for air-pocket formation in the case of $x_c = 5.483\lambda_0$ relates to the wave-breaking process itself. A wave breaks when it becomes sufficiently steep and the water particles near the crest travel faster than those near the trough. As the wave continues to steepen, the particle velocity at the crest further increases. Given that the cube face in condition $x_c = 5.483\lambda_0$ is placed slightly further downstream compared to $x_c = 5.431\lambda_0$, it is likely that in this additional distance, the wave has steepened further, leading to an increase in the velocity of the crest particles. As a result, the faster-moving wave tip strikes the wall before the water surface along the contact point has fully risen to the level of crest impingement, thus trapping an air pocket between the wave and the wall

For the third condition, when the cube is located at $x_c = 5.68\lambda_0$, more developed wave breaking is observed before the impact with the cube front, as seen in Figure 3.7. This condition results in significant mixing and spray upstream of the wave crest just prior to its impingement on the cube's front surface. Furthermore, the wave crest is observed to plunge toward a lower elevation, and the corresponding zone of impact is also lower compared to the no-aeration and high-aeration cases.

The above discussion underscores that in scenarios where a vertical cube front serves as an obstruction in a breaking wave impact process, the kinematics are intricately influenced by the streamwise position of the obstruction relative to the wave breaking location. Noteworthy are two interdependent phenomena: the development of the wave as it plunges towards the cube, and the rising water surface at the contact point of the vertical face. When wave breaking is less developed before the impact, as in $x_c = 5.431\lambda_0$, the rising velocity of surface elevation along the vertical boundary is greater, resulting in no air entrapment and a higher elevation of the contact point

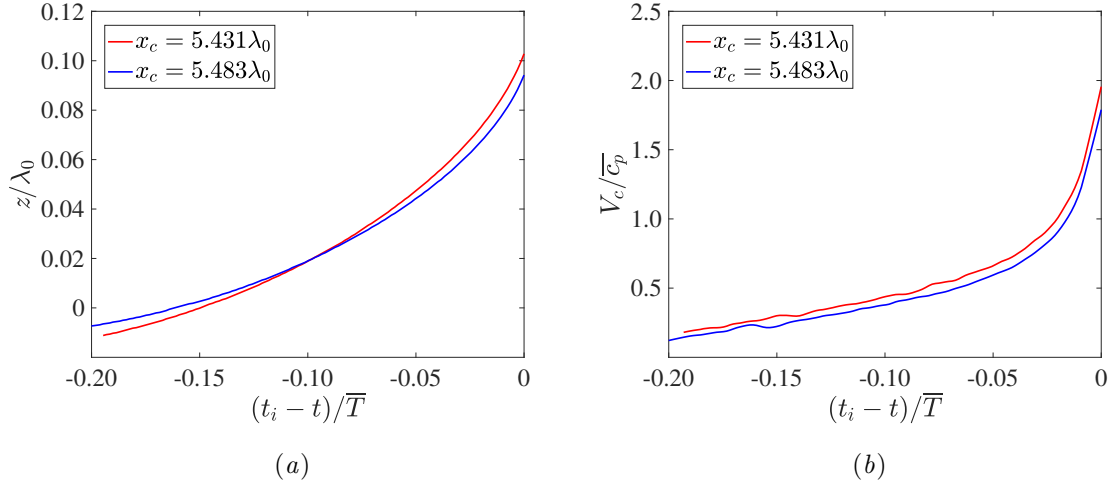


Figure 3.11: (a) Comparison of the location of the rising contact point along the face of the cube before the time of impact (t_i) for the cases $x_w = 5.431\lambda_0$ (no-aeration) and $x_w = 5.483\lambda_0$ (low-aeration). (b) Comparison of the velocity of the rising contact point along the wall before the instant of impact (t_i) for the cases $x_w = 5.431\lambda_0$ (no-aeration) and $x_w = 5.483\lambda_0$ (low-aeration).

at the moment of impact. In contrast, a developed plunging wave, as observed in $x_c = 5.68\lambda_0$, leads to substantial air entrainment and mixing. The intermediate condition, $x_c = 5.483\lambda_0$, presents a scenario where the elevation of the rising surface at the contact point is lower than the elevation of the impinging crest at the instant of impact, leading to potential air entrapment by the impinging wave front upon impact.

3.3 Spatial Variability of the Local Period (T_i) Near Impact

Figure 3.12 presents the variation in the local wave period (T_i) as the wave packet progresses toward the cube face. These local wave periods are calculated from time-resolved surface elevations extracted from the tracked free surface profiles. In the no-aeration and low-aeration impact conditions shown in Figures 3.12a and 3.12b, a gentle increase in T_i is observed as the wave moves closer to the cube. This trend is consistent with the dispersive nature of focused wave groups, which consist of a

broadband spectrum of frequency components. In such wave groups, shorter-period (higher-frequency) components travel faster and arrive earlier, while longer-period (lower-frequency) components follow. As a result, locations closer to the cube become increasingly influenced by the trailing long-period components, which leads to a gradual increase in the measured local period. This smooth variation is a characteristic feature of dispersive wave focusing, where energy accumulates near a focal region and the wave structure evolves continuously in both space and time.

For the broken-wave impact case shown in Figure 3.12c, the measurements of T_i display significantly higher spatial variability. This variability is primarily due to the increased steepness of the wave crest prior to and during breaking. The streamwise positions of the cube for the no-aeration and low-aeration cases are $x_c/\lambda_0 = 5.431$ and 5.483, respectively, whereas the broken-wave impact occurs at a more downstream location of $x_c/\lambda_0 = 5.680$. In the no-aeration and low-aeration cases, the wave does not attain enough steepness to break before reaching the cube, resulting in relatively smooth surface profiles and consistent T_i values throughout the streamwise range. However, in the broken-wave case, the wave has already experienced substantial steepening by the time it enters the analyzed segment spanning $0.02 \leq x/\lambda_0 \leq 0.20$. This steepening introduces nonlinear effects such as crest asymmetry, sharper vertical accelerations, and enhanced crest curvature, which collectively distort the zero-crossing intervals used to compute the local period. Consequently, even before the wave undergoes visible breaking, the extracted T_i values exhibit substantial irregularity across the streamwise domain.

A notable trend in Figure 3.12c is the change in the fitted line representing T_i : the slope remains nearly flat before breaking, but decreases noticeably in the post-breaking region. This behavior may be explained through the lens of unsteady nonlinear wave group dynamics. Banner et al. (2014) demonstrated that in disper-

sively focused wave packets, the wave speed of a breaking crest—evaluated using linear wave theory from the horizontal velocity of the local maximum surface elevation, as is done in the present analysis—decreases as the crest approaches breaking. This deceleration was attributed to unsteady leaning, crest sharpening, and localized energy concentration. This crest slowdown occurs even before overturning and reflects the nonlinear evolution of the wave profile. During this pre-breaking stage, although the wave has steepened significantly, the crest kinematics remain quasi-stationary, leading to relatively stable values of T_i and a near-horizontal trend in the fitted line. After breaking, however, the wave structure becomes strongly disrupted by plunging jets, air entrainment, and turbulent mixing. These processes fragment the surface profile and lead to the emergence of higher-frequency, shorter-scale components. As a result, the local crest-trough spacing compresses, and the measured local period decreases. The downward slope in the fitted T_i line following the breaking event therefore reflects the transition from a coherent dispersive wave packet to a highly disturbed, aerated flow field characterized by irregular wavelets and localized turbulence. This transition is consistent with widely observed post-breaking behavior (for example in Na et al. (2020)), where crest deceleration and the subsequent reorganization of the wave field lead to reductions in phase speed and an increase in spectral energy at higher frequencies.

3.4 Finite-time scaling of the rising water at the contact point

For the streamwise locations classified as no-aeration impacts, namely $x_c/\lambda_0 = 5.423$ and 5.431, an upward motion of the water surface at the contact point is observed as the crest approaches the cube front. In fact, the crest never makes direct contact with the cube face but rather converges onto this rising water surface at the contact point.

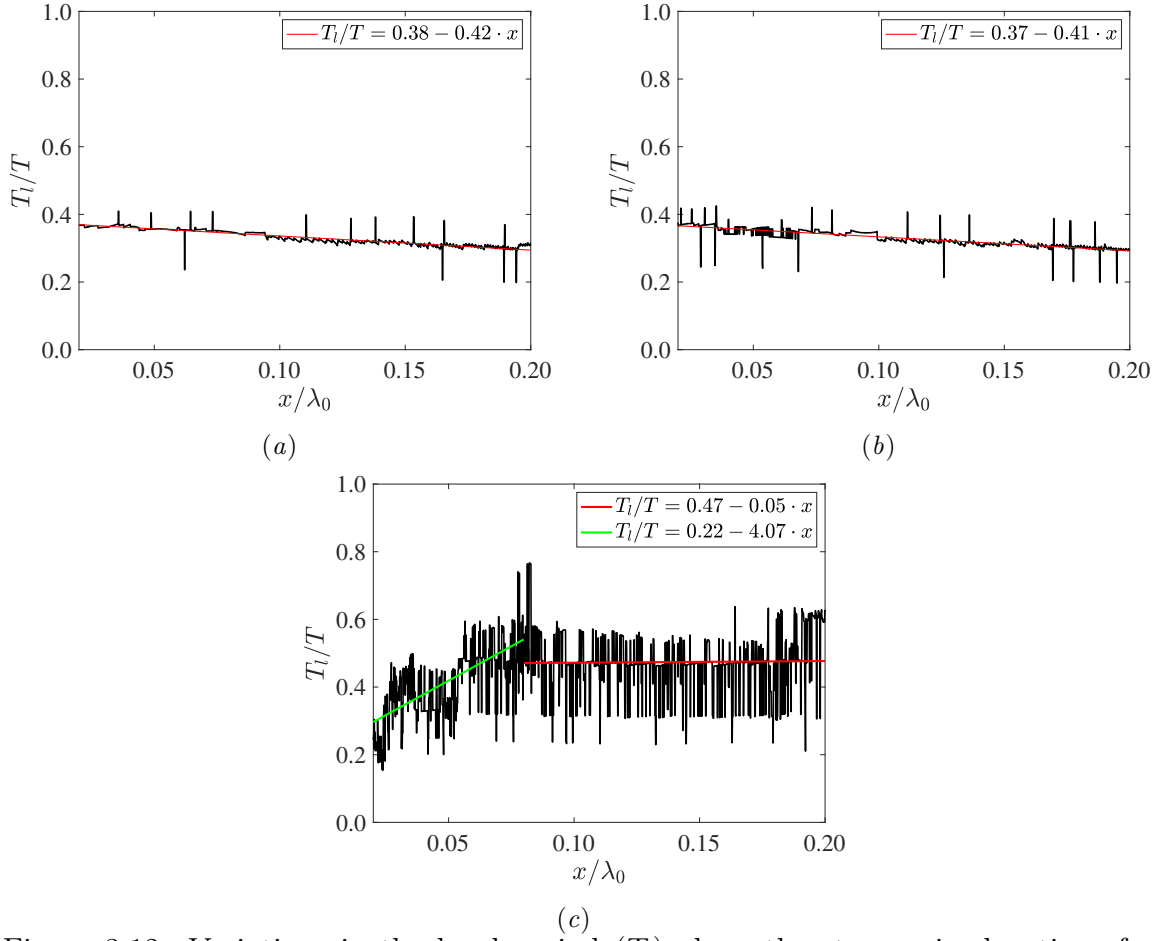


Figure 3.12: Variations in the local period (T_l) along the streamwise locations for the three impact scenarios: (a) no-aeration impact ($x_c = 5.431\lambda_0$), (b) low-aeration impact ($x_c = 5.483\lambda_0$), and (c) broken-wave impact ($x_c = 5.680\lambda_0$). Linear trend lines are fitted to highlight the slope of T_l for each impact scenario. For the broken-wave impact case ($x_c = 5.680\lambda_0$), where the wave breaks before reaching the cube, two distinct linear trend lines are shown: red for the pre-breaking streamwise locations and green for the post-breaking locations, highlighting the contrasting trends in T_l .

This behavior contrasts with that observed under low-aeration conditions, where the crest reaches the cube face before the rising water surface attains the impact point. Under no-aeration conditions, the water adjacent to the cube front exhibits a distinct jet-like rise as t_i approaches, resulting from the redirection of horizontal momentum into vertical motion along the wall. As discussed earlier in Section 3.2.2, the vertical velocity of the contact point, denoted by V_c , increases sharply immediately before impact.

To quantify this rapid acceleration, the temporal evolution of V_c was analyzed using several fitting approaches, and it was found that a power-law relationship effectively represents the growth of V_c as the impact time is approached. Since the interaction between the crest and the water surface at the contact point could alter the local kinematics, the fitted power-law curves were truncated shortly before the moment of impact ($t < t_i$). The resulting relationships for the two streamwise locations are expressed as:

$$\frac{V_c}{\bar{c}_p} \propto 0.379(t_i - t)^{-0.296}, \quad x_c/\lambda_0 = 5.423, \quad (3.4.1)$$

$$\frac{V_c}{\bar{c}_p} \propto 0.290(t_i - t)^{-0.329}, \quad x_c/\lambda_0 = 5.431. \quad (3.4.2)$$

These power-law fits, shown in Figure 3.13a, capture the rapid increase in $V_c/\bar{c}_p$ as $t \rightarrow t_i$. The fitted exponents of -0.296 and -0.329 correspond to the upstream ($x_c/\lambda_0 = 5.423$) and downstream ($x_c/\lambda_0 = 5.431$) positions, respectively, and indicate a nearly singular growth of the vertical velocity as the impact instant is approached. The slightly more negative exponent for the downstream case signifies a steeper asymptotic scaling; however, because the upstream fit has a larger prefactor (0.379 compared with 0.29), the contact-point velocity remains higher for $x_c/\lambda_0 = 5.423$ over most of the pre-impact duration. This behavior suggests that as the crest

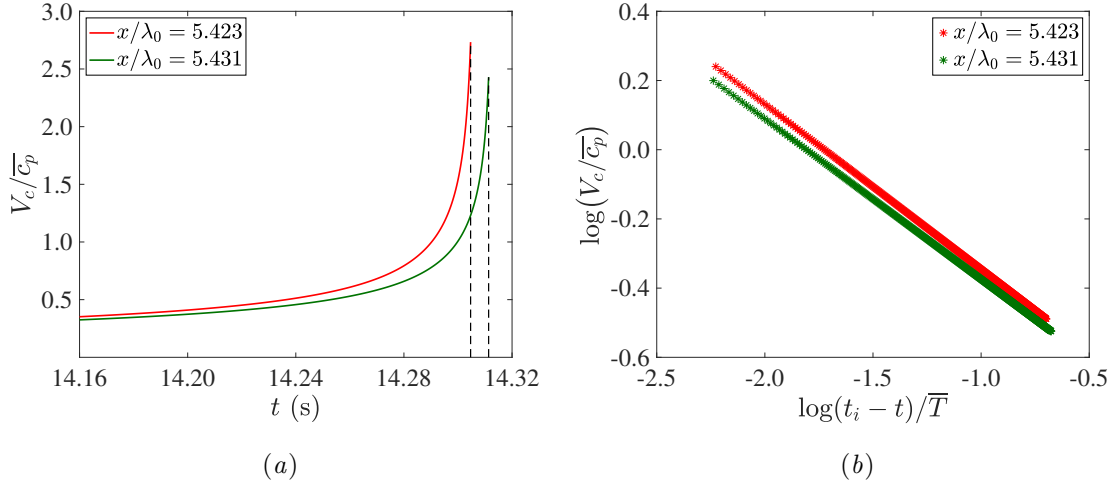


Figure 3.13: Velocity (V_c) of the water surface at the contact point at $x=0$ as a function of time for condition $x_c = 5.431\lambda_0$: (a) Plotted on a linear scale (b) Plotted along logarithmic scale ($t < t_0$)

advances, the vertical momentum exchange near the wall is more intense and localized at the upstream position, resulting in a more vigorous upward motion of the confined water column before impact.

The same relationships are presented on logarithmic axes in Figure 3.13b, where $\log(V_c / \bar{c}_p)$ is plotted against $\log[(t_i - t) / \bar{T}]$. The data align closely along straight lines, confirming that the temporal evolution indeed follows a power-law dependence. The near-linearity of these log-log plots highlights the self-similar nature of the rising motion, implying that the dynamics of the contact point near impact exhibit scale-invariant behavior. Such behavior is characteristic of free-surface singularities, where the vertical velocity increases unboundedly as the time to impact decreases—analogous to finite-time singularity behavior (Zeff et al., 2000).

3.5 Vertical Velocity Field Evolution Near the Cube Front

A notable observation from a detailed examination of the surface elevation profiles is that the rear portions of the free surface, trailing behind the main crest, often appear

as nearly straight segments spaced at regular intervals (see Figures 3.2, 3.4, and 3.8). This led to the hypothesis that these rear portions can be made to coincide in a reference frame moving with an appropriate constant vertical velocity. To explore this behavior, a vertical translation was applied such that the position of the structure front remained fixed along a vertical line. For the broken wave impact condition, where the wave crest overturns prior to contact, the alignment of rear profiles was carried out only up to the point just before breaking occurred, since post-breaking surface features deviate significantly from the quasi-linear behavior observed in the rear segments. After testing different velocities through visual comparison, it was found that a downward reference frame speed of 0.394 m/s, corresponding to $0.27 \bar{c}_p$, provided good alignment of the rear profiles for all three impact conditions. An illustration of this alignment is presented in Figure 3.14, which shows the back-surface-collapsing wave profiles for one of the no-aeration cases when the cube is located at $x/\lambda_0 = 5.431$. In this moving frame, the rear portions of the surface collapse into a narrow cluster, while the main variation during the impact is observed close to the cube front, i.e. between the contact point and the crest.

To examine the variations in vertical surface velocity V_v near cube front under different impact conditions, the velocity field was calculated within the streamwise range $x/\lambda_0 = 0$ to 0.10 for all three cases. The corresponding contour plots, shown in Figure 3.15, capture the flow dynamics during the final moments preceding impact. For both the no-aeration and low-aeration conditions, the contours exhibit a consistent trend: along any constant-time horizontal line, tracing from $x/\lambda_0 = 0$ toward the crest reveals a progressive increase in V_v , followed by a sharp drop past the crest toward the rear face of the advancing plunging breaker. In both cases, the V_v values near the crest continue to increase consistently as the instant of impact at $tf_0 = 0$ approaches. Although this general structure is shared between the two conditions,

the magnitudes of V_v near the crest differ substantially. The low-aeration case exhibits significantly higher vertical velocities, reflecting a more steepened wave front and aligning with earlier observations of elevated peak pressures P_m in this case.

In contrast to the no-aeration and low-aeration cases, the broken-wave impact condition exhibits a markedly different vertical velocity contour. During the time window immediately preceding impact, the plunging breaker has already collapsed onto the upstream water surface, forming a broken wave. As a result, the contour reveals two distinct diagonal bands of elevated vertical velocity V_v extending from $x/\lambda_0 = 0$ toward increasing x/λ_0 . The first diagonal band is associated with the early formation of a secondary jet generated by the initial impact of the plunging crest. Although wave breaking induces energy dissipation, this jet retains considerable vertical momentum. The corresponding surface profiles in Figure 3.15*d* show a sudden surface elevation rise where this nascent jet emerges, resulting in V_v values as high as $4.38 \bar{c}_p$. This abrupt increase in velocity arises due to the rapid vertical displacement of the water surface at a location that had previously exhibited minimal disturbances. The strength of this jet is also consistent with the higher peak pressure values P_m recorded for the broken-wave condition, highlighting the secondary jet's role in concentrating vertical momentum during impact.

As the secondary jet develops further and moves toward the cube front, the vertical elevation of the surface begins to increase more gradually. Consequently, the corresponding V_v values exhibit a moderate decline, marking the end of the initial diagonal band. Immediately following this high-velocity zone, a distinct dip in V_v is observed. This dip corresponds to the downward-sloping rear face of the secondary jet as it descends toward the cube, clearly captured in the trailing edge of the surface profiles in Figure 3.15*d*. The dip does not imply sudden energy loss but instead reflects the localized reduction in elevation rate across the back face of the jet. A similar

sequence of dynamics occurs as the second diagonal band emerges in the contour. Following the dip in vertical velocity caused by the rear face of the secondary jet, the main crest arrives. This arrival induces another abrupt increase in surface elevation, which is again accompanied by a sharp rise in V_v , reaching up to $3.1 \bar{c}_p$. As with the secondary jet, this initial spike is followed by a gradual relaxation of vertical velocity as the main crest continues to propagate forward and the elevation change becomes smoother with time.

These two diagonal bands of elevated vertical velocity therefore reflect two distinct mechanisms: the first is driven by the rapid emergence of the secondary jet over a surface that had previously exhibited minimal disturbances, and the second follows the descent of the rear face of secondary jet and the sudden arrival of the main crest. In both cases, the initial sharp increase in V_v is due to abrupt local surface elevation changes, while the following relaxation in velocity reflects smoother elevation transitions as the respective flow structures develop. Importantly, both high- V_v events highlight the strongly localized nature of the vertical accelerations in broken-wave impact scenarios.

3.6 Dynamics of the Rising Water Column and Crest Interaction

A detailed examination of the LIF experiments reveals a distinct two-part structure in the free surface upstream of the advancing crest during no-aeration and low-aeration impact conditions, where the plunging breaker remains intact before reaching the structure. The free surface in this regime can be segmented into two regions: a vertically rising water column adjacent to the frontal face of the cube, and an upstream region that forms part of the approaching crest, propagating horizontally toward the structure. This segmentation is consistently observed across all streamwise positions

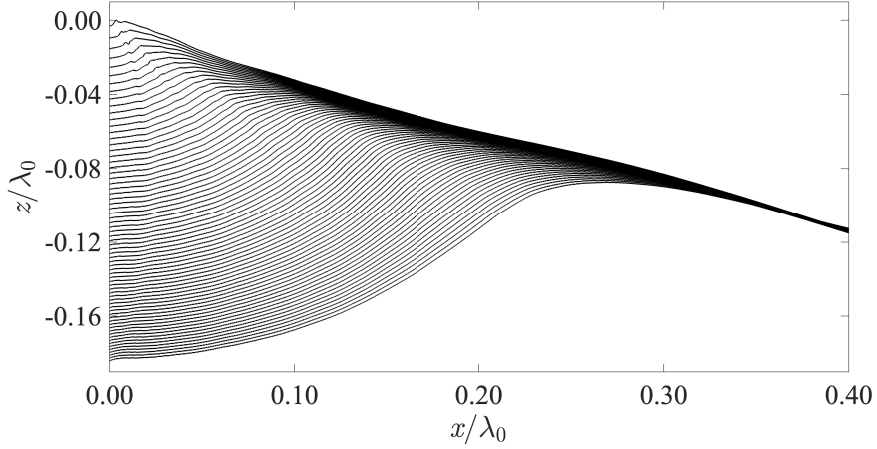


Figure 3.14: The plotted surface profiles for the case $x/\lambda_0 = 5.431$ are shown in a vertically translating reference frame moving downward at a velocity of $V_r = 0.27 \bar{c}_p$. Successive profiles are spaced by 2.67 milliseconds, which corresponds to $0.0031/f_0$, and the dataset ranges from $tf_0 = -0.212$ (approximately 183 milliseconds prior to impact) to the impact instant ($tf_0 = 0$). The vertical axis intersects the front boundary of the structure at $x/\lambda_0 = 0$. The coordinate origin is established at the crossing point of this front boundary with a straight line fitted through the quasi-linear segment of the wave profiles on the upstream side.

associated with non-breaking impacts, and is visualised in Figure 3.16 for the representative cases $x_c = 5.423\lambda_0$ and $x_c = 5.483\lambda_0$. These two locations span the limits of the non-breaking regime, beyond which the wave breaks prior to reaching the structure, resulting in highly aerated and irregular impacts where the two-region behaviour is no longer maintained.

In Figure 3.16, the red-highlighted region corresponds to the vertically rising water column, whereas the green-highlighted region indicates the upstream portion of the crest. These visualisations are arranged in vertically stacked pairs for each case: the upper image shows the raw LIF snapshot, and the lower image displays the same frame with the two regions explicitly traced. This arrangement provides a clear basis for identifying the temporal evolution and interaction of the two distinct free surface segments.

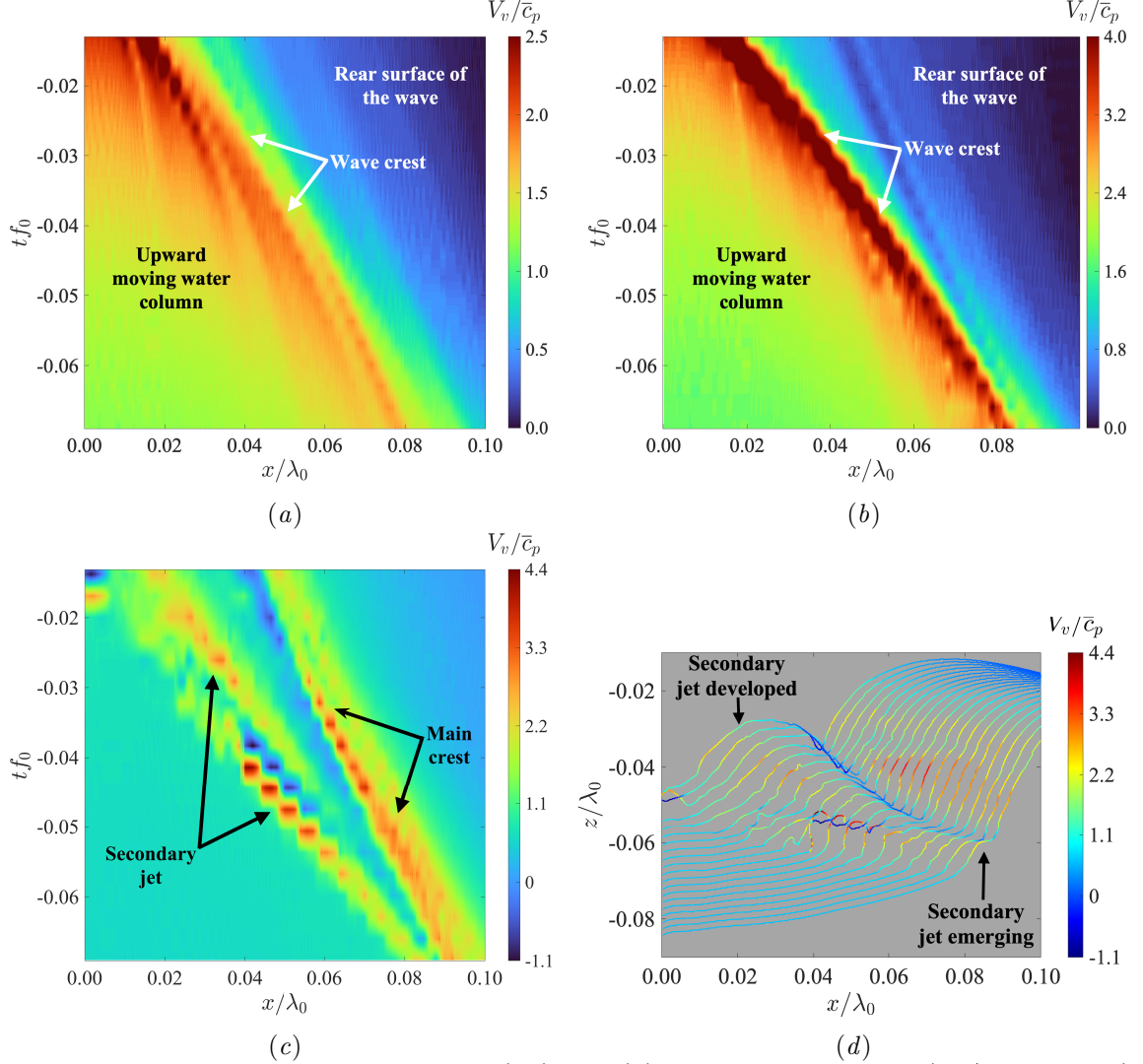


Figure 3.15: Vertical surface velocity (V_v) for: (a) no-aeration impact ($x_c/\lambda_0 = 5.431$), (b) low-aeration impact ($x_c/\lambda_0 = 5.483$), and (c) broken-wave impact ($x_c/\lambda_0 = 5.680$). All three contours are shown in a reference frame moving vertically downward with a velocity $V_v = 0.27\bar{c}_p$. (d) Instantaneous surface profiles for the broken-wave impact case ($x_c/\lambda_0 = 5.680$), plotted in the same moving reference frame and aligned such that the origin corresponds to the intersection of a fitted upstream linear segment with the front boundary ($x/\lambda_0 = 0$). The color along each profile indicates the local vertical velocity magnitude, emphasizing regions of strong upward and downward motion.

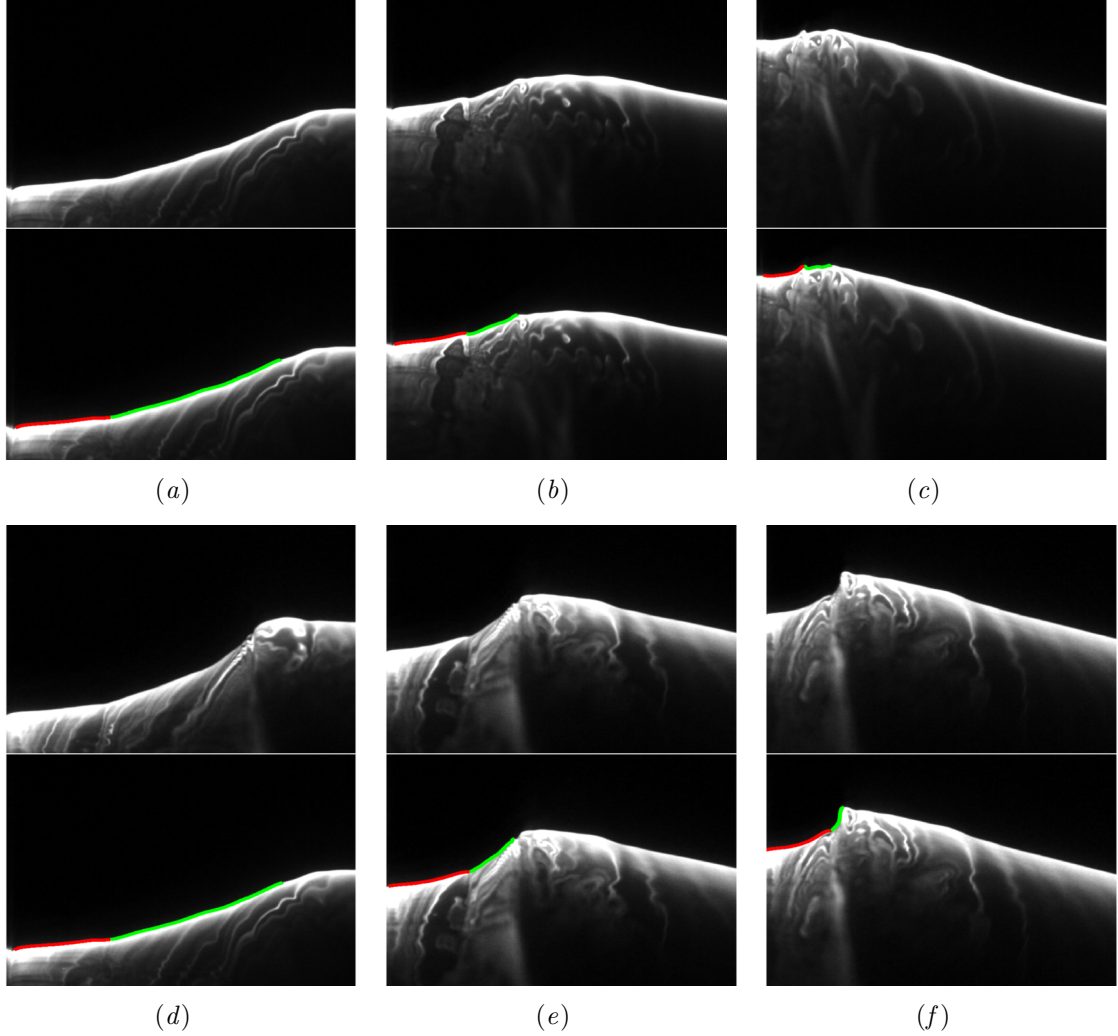


Figure 3.16: Sequential evolution of the wave surface profile near the frontal face of the cube during the advancing phase of a plunging wave crest. Subfigures (a)–(c) correspond to the condition $x_c = 5.423\lambda_0$, and (d)–(f) to $x_c = 5.483\lambda_0$. For each condition, the leftmost frame captures an earlier stage when the crest is farther from the cube, with subsequent frames showing its progression toward impact. Each subfigure consists of two vertically stacked snapshots: the top frame shows the raw LIF image, while the bottom highlights the traced surface profile divided into two distinct segments. The red region, located adjacent to the frontal face of the cube, exhibits a relatively lower curvature, representing the upward motion of a localized water column influenced by pressure redistribution along the vertical wall. In contrast, the green segment farther upstream corresponds to the approaching crest with unaltered forward momentum.

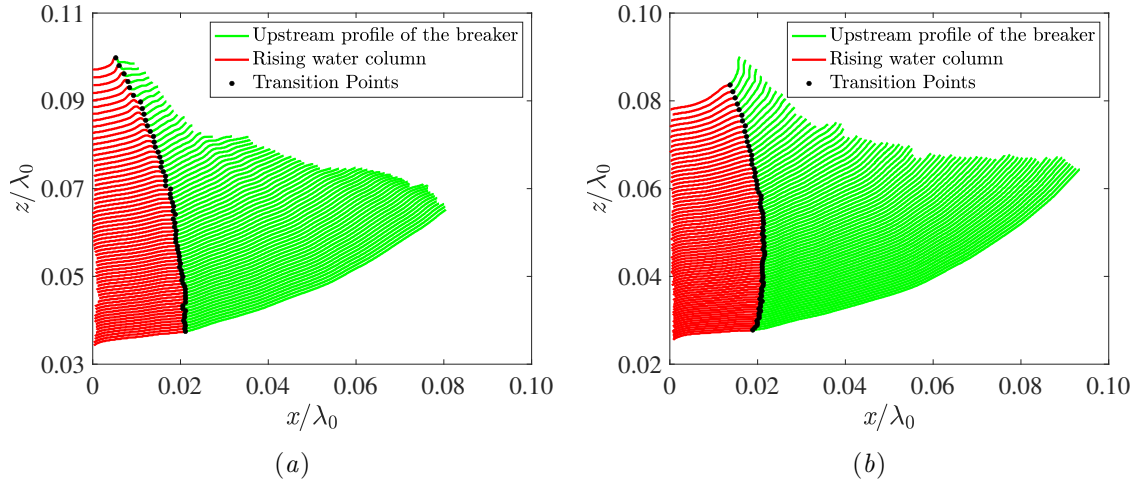


Figure 3.17: Evolving surface profiles of the wave crest as it propagates towards the cube for conditions (a) $x_c = 5.423\lambda_0$ and (b) $x_c = 5.483\lambda_0$. The surface profiles are segmented into two distinct regions: the rising water column extent (red) adjacent to the frontal face of the cube, characterized by a relatively smooth and less steep curvature, and the upstream profile of the advancing wave crest (green), which exhibits a sharper curvature. The transition points between these two regions, marked by black dots, indicate the boundary where the less steep curvature transitions into the steeper upstream profile.

The formation of the vertical water column results from the redirection of horizontal momentum by the impermeable vertical face. As the crest approaches the cube, horizontal momentum is redistributed vertically, generating a smooth upward fluid motion near the wall. The pressure field initially exhibits spatial distribution over a broader region but becomes increasingly localized near the base as the crest advances. This evolving pressure field gives rise to the observed vertical column and establishes a dynamic transition point that separates it from the crest segment. The time-dependent evolution of this transition point is shown in the Figure 3.17 and further quantified through polynomial fits to its trajectory in Figure 3.18a.

Figure 3.18a presents two important observations from these profiles. First, the vertical position of the transition point at the moment of segment convergence systematically decreases with downstream placement of the cube. For example, the

convergence occurs at $z/\lambda_0 \approx 0.097$ for $x_c = 5.423\lambda_0$, while it takes place at $z/\lambda_0 \approx 0.084$ for $x_c = 5.483\lambda_0$. Second, the curvature of the incoming crest increases with downstream cube location. At earlier positions such as $x_c = 5.423\lambda_0$, the crest and water column segments merge gently, maintaining similar profiles. In contrast, at $x_c = 5.483\lambda_0$, the crest becomes noticeably steeper relative to the rising water column, leading to a more pronounced convergence.

The horizontal thickness of the water column also evolves during its rise. Initially narrow due to limited pressure extent, the column thickens as the crest approaches, driven by enhanced vertical momentum transfer. Just prior to convergence, a narrowing is again observed, reflecting the localization of the pressure forcing near the base. While all cases exhibit similar qualitative behaviour, the maximum thickness is marginally greater for downstream positions. The largest thickness is observed for $x_c = 5.483\lambda_0$, where the column extends to $x/\lambda_0 = 0.043$ at $z/\lambda_0 = 0.0214$.

Further insight into these dynamics is gained by examining the third-order polynomial fits to the vertical velocities (V_t) of the transition point, shown in Figure 3.18*b*. Although all five cases exhibit a similar rising trend as the convergence moment approaches, the fitted curves reveal that V_t tends to be marginally higher for upstream positions. For instance, just before convergence, V_t for $x_c = 5.423\lambda_0$ approaches approximately $1.30 \overline{c_p}$, while for $x_c = 5.483\lambda_0$, it is closer to $1.05 \overline{c_p}$. While these differences are relatively small, they indicate a subtle but consistent shift in vertical acceleration characteristics with streamwise location. The decreasing trend in V_t aligns with the visual increase in column thickness observed in the LIF profiles, suggesting a gradual transition from a sharper, jet-like vertical response to a relatively slower, plume-like motion.

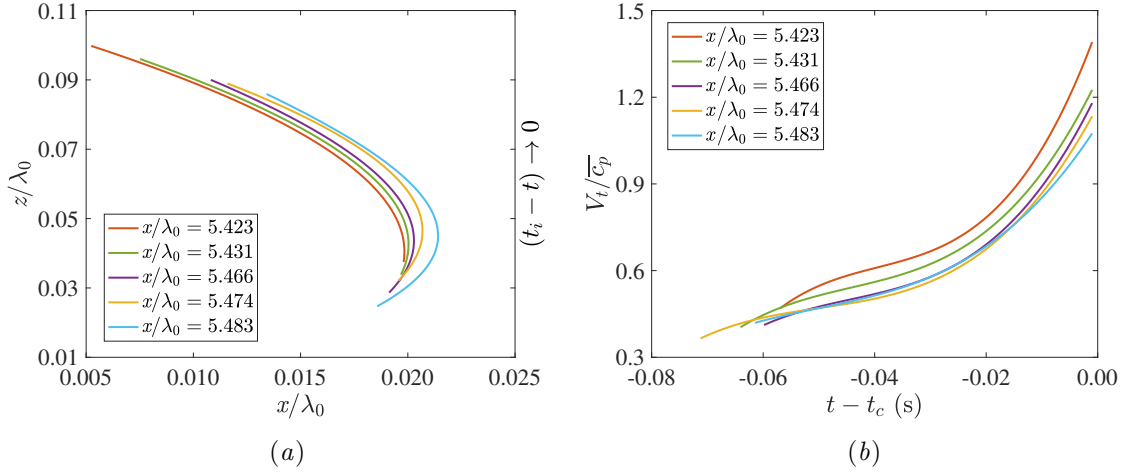


Figure 3.18: Third-order polynomial fits to: (a) the position of the transition point between the upward moving water column and the upstream surface of the horizontally advancing crest as both move toward convergence; and (b) the vertical velocity (V_t) of the transition point up to the moment of convergence (t_c) between the upward moving water column and the upstream surface of the crest. Results are shown for streamwise locations of the cube front ranging from $x_c = 5.431\lambda_0$ to $x_c = 5.483\lambda_0$. In (a), the evolution of the transition point position is plotted as a function of time, with a secondary axis indicating the diminishing time gap ($t_c - t \rightarrow 0$) approaching convergence. The trajectories illustrate how both the elevation and vertical velocity of the transition point evolve as the advancing crest nears the front face of the cube. *Note: the horizontal axis in (a) is not to scale relative to the vertical axis to enhance visibility of the transition-point evolution.*

To identify the underlying forcing responsible for this variation, Figure 3.19a presents the non-dimensional average pressure $P_{\text{avg}}/(\rho\bar{c}_p^2)$ recorded by the lower sensors during the time interval used for computing transition point evolution. Although this chapter focuses on the kinematic evolution of the free surface, a brief examination of the accompanying pressure field is necessary here because the vertical motion of the water column is closely tied to the underlying pressure forcing. At elevations near and just above the mean water level ($z/\lambda_0 < 0.026$), the upstream cases (e.g., $x_c/\lambda_0 = 5.423$ and 5.431) consistently exhibit the highest average pressures, exceeding $0.41 \rho\bar{c}_p^2$. These pressures, concentrated near the base, indicate a highly localized and intense vertical impulse responsible for the faster and narrower upward motion

observed in these cases.

Conversely, downstream locations such as $x_c/\lambda_0 = 5.483$ display relatively lower P_{avg} values near the base. However, beginning around $z/\lambda_0 = 0.026$ and extending upward, the average pressures begin to converge across all cases, with some downstream cases slightly surpassing the upstream values at higher sensors. This vertical redistribution of pressure suggests a broader and more sustained forcing pattern along the front face for the downstream positions. As a result, while the base-level pressure impulse is reduced, more fluid is gradually entrained upward over a larger region, supporting the formation of a thicker water column at reduced velocity.

This interpretation is further supported by Figure 3.19*b*, which shows the non-dimensional duration $T_{\text{sd}}/\bar{T}$, defined as the time interval during which the instantaneous pressure at each sensor remains above 10% of its respective P_m value. In the lower elevations, T_{sd} is consistently longer for downstream positions (e.g., $x_c/\lambda_0 = 5.483$), reaching durations above $0.16\bar{T}$, compared to approximately $0.13\bar{T}$ for $x_c/\lambda_0 = 5.423$. These results indicate that although the instantaneous pressures are lower downstream, the fluid is subjected to vertical forcing over a longer time interval. This extended duration contributes to a more gradual yet broader upward response, reinforcing the plume-like morphology observed in the LIF profiles. In contrast, the upstream cases exhibit shorter T_{sd} but higher P_{avg} , consistent with a more impulsive, jet-like vertical motion.

Together, Figures 3.19*a* and 3.19*b* provide a physically coherent framework for understanding the observed evolution in water column thickness and velocity. At upstream locations, the forcing is sharp and localized, producing a thinner but faster-rising water column. As the cube is positioned farther downstream, the nature of the pressure loading becomes more distributed in both space and time, resulting in a thicker but slower vertical column. These subtle shifts in the vertical momen-

tum transfer mechanism are captured both qualitatively in the LIF visualizations and quantitatively through pressure and velocity measurements, offering a unified interpretation of non-breaking wave–structure interaction dynamics.

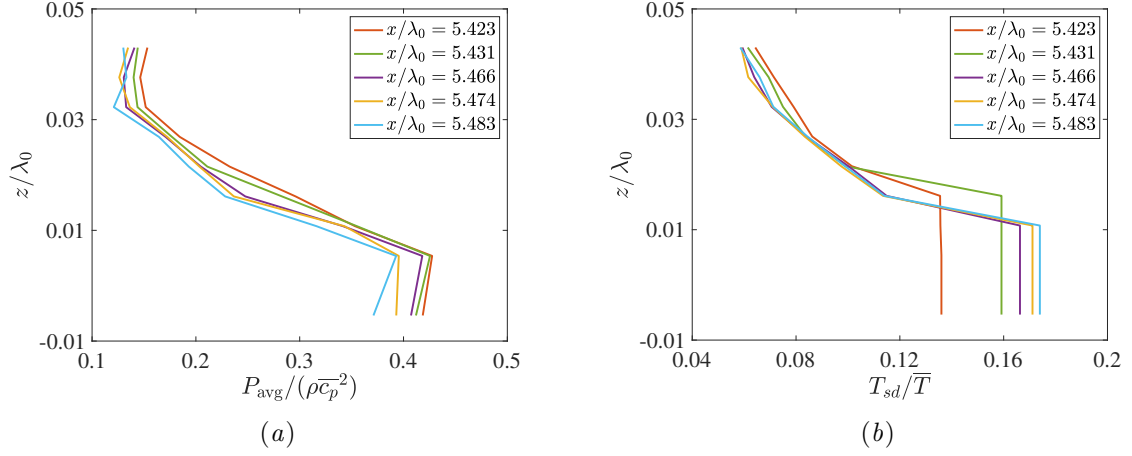


Figure 3.19: Vertical variation of pressure-based forcing quantities over the elevation range of the rising water column during non-breaking impacts. (a) Nondimensional average pressure $P_{\text{avg}}/(\rho \bar{c}_p^2)$ and (b) Duration of significant pressure $T_{\text{sd}}/\bar{T}$, defined as the time during which the local pressure remains above 10% of its peak value. Both quantities are computed over the same time interval used for evaluating the transition point position and velocity, corresponding to the upward motion of the water column before convergence with the incoming crest.

Chapter 4

Dynamics of Plunging Breaker Impact

This chapter focuses on the dynamic response of the structure during wave impact, characterized through time-resolved pressure measurements on the front face of the cube. The discussion quantifies how peak pressures, impulses, and post-impact oscillations vary across impact regimes and links these behaviors to the underlying kinematic evolution.

4.1 Impact pressures

4.1.1 Pressure variations with change in cube location

A notable trend in the impact pressure characteristics across the three categories is the decreasing elevation of the wave impact zone as the cube location shifts downstream. This pattern is more evident in Figure 4.1, which presents the elevations where the highest P_m values were recorded for the 8 different cube locations. These locations span a range of conditions, from an unbroken no-aeration phenomenon to a fully broken-wavefront. While the pressure time series characteristics, with time scales exceeding $0.01\bar{T}$, are consistently reproducible, the peak pressure features—characterized by short rise times of less than $0.01\bar{T}$ —exhibit considerable variability when experiments are repeated under identical conditions. Despite the variability in P_m , a systematic change in the elevation of P_m with respect to the cube location is clearly visible in Figure 4.1.

The dashed boundaries in the Figure delineate three distinct zones corresponding to the three impact categories: Zone I, extending from $5.423 < x_c/\lambda_0 \leq 5.431$,

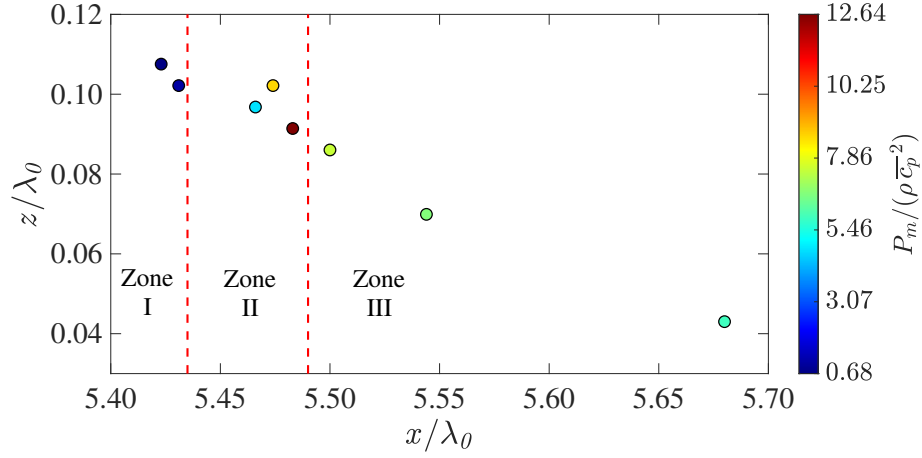


Figure 4.1: Variation of P_m elevation with cube location. Zone I represents cube locations where no-aeration occurs, Zone II corresponds to low-aeration impacts, and Zone III denotes broken-wave impacts.

represents the phenomenon with no air entrapment; Zone II, covering $5.431 < x_c/\lambda_0 \leq 5.483$, includes locations where low-aeration impacts occur; and Zone III, from $5.483 < x_c/\lambda_0 \leq 5.680$, corresponds to cases where the wave crest plunges into the upstream water surface before impacting the cube, resulting in broken-wave impacts. The specific impact pressure characteristics within each of these zones are discussed in detail in later subsections.

4.1.2 Regime-dependent characteristics of impact pressures

Figure 4.2 shows the characteristics of the impact pressure, illustrating the variation of P_m with elevation for the three impact categories: no-aeration, low-aeration, and broken-wave. The values of P_m , rise time (t_r), and the relative timing of P_m across elevations with respect to the elevation where the highest P_m occurs (denoted as $t - t_i$) are extracted from the representative experimental runs for each category introduced in Section 3.1, namely $x_c = 5.431\lambda_0$ (no-aeration), $5.483\lambda_0$ (low-aeration), and $5.680\lambda_0$ (broken-wave).

For the no-aeration case, the highest P_m is $1.07\rho\bar{c}_p^2$, recorded at $z = 0.102\lambda_0$. In comparison, the low-aeration and broken-wave impacts exhibit highest P_m values of $12.64\rho\bar{c}_p^2$ and $5.88\rho\bar{c}_p^2$, occurring at $z = 0.091\lambda_0$ and $z = 0.043\lambda_0$, respectively. These elevations, marked by dashed horizontal lines in Figure 4.2 for every P_m curve, correspond closely to the crest impact elevations observed in the LIF recordings, where higher P_m values are expected. However, the impact crest does not always coincide exactly with any sensor location, as the pressure sensors are mounted at discrete elevations. This mismatch introduces a slight discrepancy between the observed crest impact location and the sensor recording the highest P_m value. As a result, there stands a possibility that the actual highest P_m may be greater than what is recorded.

A comparison across the three impact conditions reveals significant variation in P_m magnitudes. In the no-aeration condition ($x_c = 5.431\lambda_0$), where no distinct impact is observed and a vertical water jet forms near the intersection of the wave crest and the structure surface, the recorded P_m is considerably lower than that in the low-aeration condition ($x_c = 5.483\lambda_0$), where a small volume of air is entrapped. In the broken-wave condition, where the structure is positioned further downstream at $x_c = 5.680\lambda_0$, the peak P_m is lower than that in the low-aeration case but still higher than in the no-aeration case. These trends are also reflected in the average of the highest P_m values calculated over multiple runs at the corresponding streamwise positions: $0.88\rho\bar{c}_p^2$ for no-aeration, $8.75\rho\bar{c}_p^2$ for low-aeration, and $6.68\rho\bar{c}_p^2$ for broken-wave conditions. These results suggest that increased air entrainment in the broken-wave scenario reduces the magnitude of the impulsive pressure imparted to the structure during impact. In contrast, when only minimal air is entrained, as in the low-aeration case, the impulsive loading on the structure is significantly enhanced.

Analysis of the vertical pressure distribution reveals three distinct zones: the wave impact zone (shaded regions in Figure 4.2c), the zone below wave impact, and

the zone above wave impact. In the wave impact zone, defined as the elevations around the peak P_m , sensors record notably low t_r values (as low as $0.001\overline{T}$) and substantial peak pressures. This zone spans approximately $z = 0.016\lambda_0$ downward from the elevation where the highest P_m is recorded. After the initial impact, the time series frequently exhibits oscillations, particularly in the low-aeration and broken-wave categories where the structure is positioned further downstream.

Moving downward from the impact zone, a steep increase in t_r values is observed with increasing distance from the region of wave impact (Figure 4.2*b*). Concurrently, a consistent decrease in P_m values occurs as the elevation decreases below the impact zone. However, as shown in Figure 4.2*c*, which presents the relative occurrence time of P_m at different elevations with respect to the elevation where the highest P_m is recorded, P_m occurs almost simultaneously across the zone of impact suggesting coherent hydrodynamic behavior. In contrast, at elevations above the highest P_m location, a noticeable delay is observed in the occurrence of P_m compared to its occurrence at the peak impact elevation and certain locations within the impact zone. This delay is consistent with the transition of the incident wave into an upward sloshing motion, as previously illustrated in Figures 3.1 and 3.5.

In the region above the wave impact, sensors register longer t_r , accompanied by a steeper decline in P_m compared to the region below the impact. For all the three impact categories, the peak pressures measured above the impact zone are comparable in magnitude to those recorded below. Despite differences in the magnitudes of highest P_m , all three conditions exhibit a consistent trend: the peak pressure decreases progressively with increasing distance from the wave impact zone.

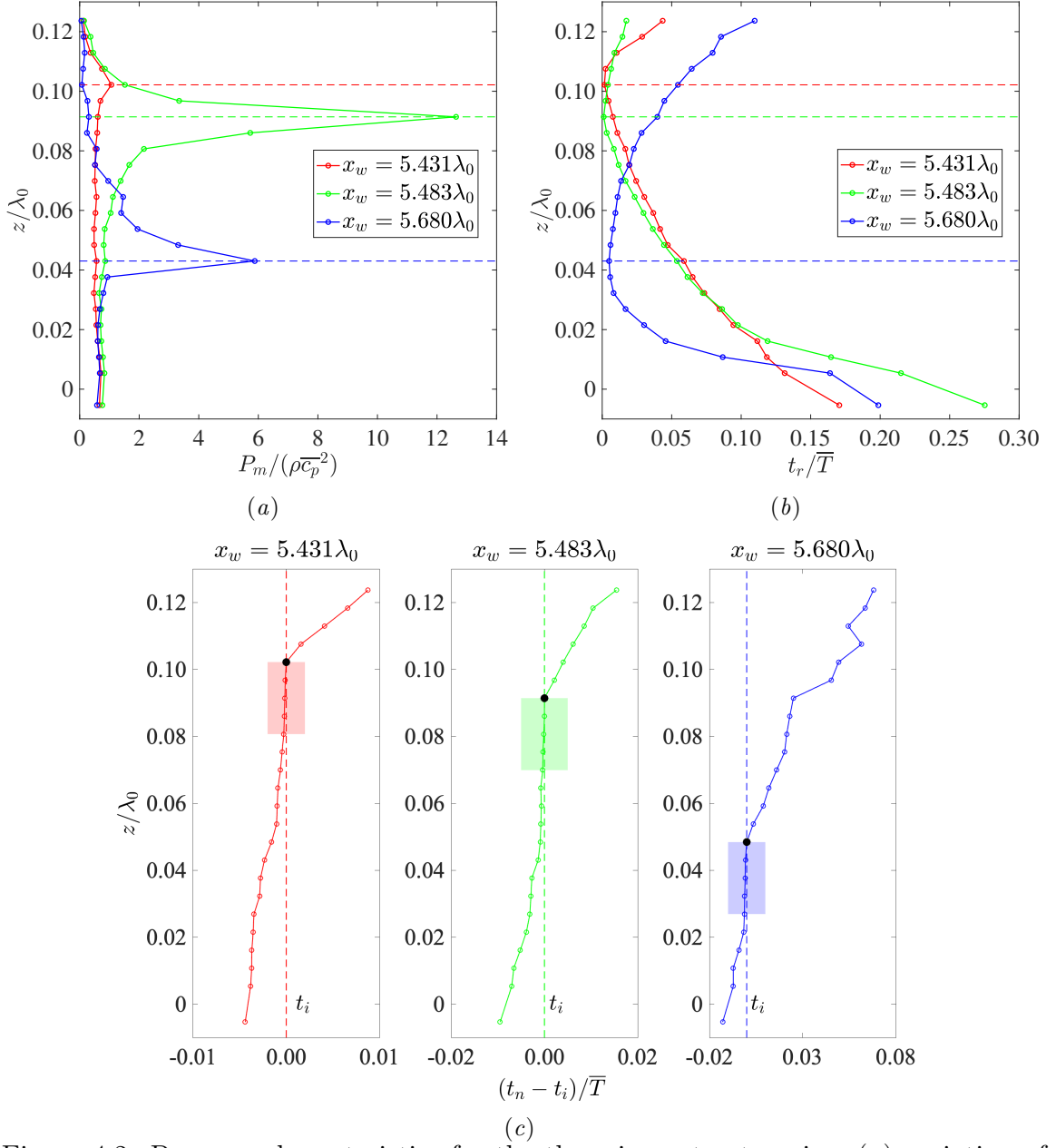


Figure 4.2: Pressure characteristics for the three impact categories: (a) variation of P_m with elevation, (b) variation of t_r with elevation, and (c) relative occurrence time of P_m at different elevations with respect to the elevation of impact. The horizontal dashed lines in (a) and (b) indicate the elevations where the highest P_m values were recorded for each impact category, corresponding to the crest impact location. The vertical dashed lines in (c) represent the instant of impact, t_i . The black solid markers in (c) denote the wave impact elevations where the highest P_m values were measured. The shaded regions in (c) highlight the range of elevations where P_m occurs nearly simultaneously for each impact scenario. The pressure characteristics in each subplot correspond to simultaneous measurements from a single experimental run. The mean water level is at $z/\lambda_0 = 0$.

4.2 Pressure dynamics within distinct impact zones

To gain a clearer understanding of how the pressure characteristics evolve under varying degrees of wave breaking, the impact process is examined separately within each of the three identified zones: no-aeration (Zone I), low-aeration (Zone II), and broken-wave (Zone III). Each zone exhibits a unique combination of pressure features, including variations in P_m , t_r , and post-impact oscillatory behavior.

4.2.1 Zone I: No-aeration impact

Figure 4.3 presents the characteristics of P_m for two streamwise locations of the cube where no-aeration impacts occur. Based on the pressure distributions shown in Figure 4.3a, the highest P_m values of $0.68\rho\bar{c}_p^2$ and $1.07\rho\bar{c}_p^2$ are recorded at $x_c = 5.423\lambda_0$ and $x_c = 5.431\lambda_0$, respectively. The lowest rise times (t_r) at these locations range from $0.0016\bar{T}$ to $0.0028\bar{T}$. Although only two cases are reported for the no-aeration condition, a discernible trend is evident from multiple experimental runs: positioning the cube front farther downstream from the wave maker results in an increase in P_m . This trend is supported by the peak P_m values recorded across repeated runs for each case, which range from $0.66\rho\bar{c}_p^2$ to $0.68\rho\bar{c}_p^2$ at $x_c = 5.423\lambda_0$ and from $0.69\rho\bar{c}_p^2$ to $1.07\rho\bar{c}_p^2$ at $x_c = 5.431\lambda_0$.

Additionally, a consistent trend is observed for t_r , as shown in Figure 4.3b: lower values of t_r generally correspond to higher values of P_m . As will be discussed in a later section, this inverse relationship appears to correlate more strongly with the magnitude of P_m than with sensor elevation, impact classification, or the streamwise position of the structure.

When compared with the peak P_m values recorded for low-aeration and broken-wave impact categories, the P_m values for no-aeration impacts are moderate, and the

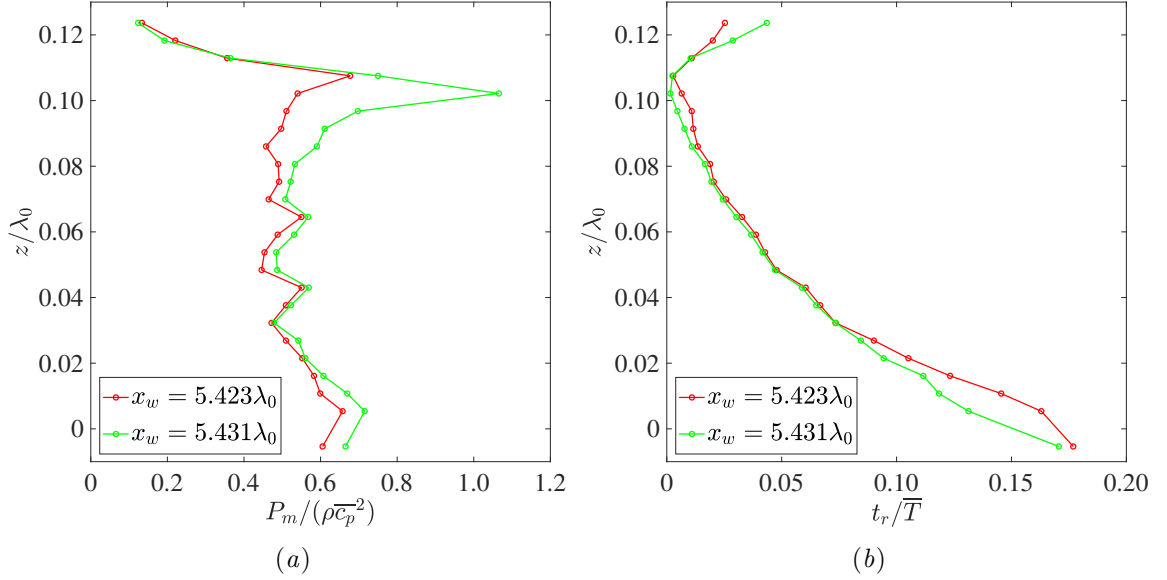


Figure 4.3: Pressure characteristics at the three streamwise locations of the cube associated with no-aeration type impacts: (a) P_m at varying elevations, (b) t_r at varying elevations. The pressure characteristics shown in each subplot represent simultaneous measurements from a single experimental run. The mean water level corresponds to $z/\lambda_0 = 0$.

associated t_r values are longer. This indicates a slower rise and fall of the pressure peak relative to the other two categories, consistent with the absence of a direct wavefront impingement on the cube front. The observed variations in P_m across sensor elevations can be attributed to the region of free-surface convergence. As reported by Cooker and Peregrine (1992), high P_m values measured at elevations coinciding with surface convergence play a key role in driving the no-aeration impact mechanism. Subsequent studies further suggest that a more confined region of surface convergence leads to the formation of a smaller but more intense jet, thereby producing higher P_m values. Within Zone I, this impact category is thus characterized by a single dominant pressure peak at each elevation, without the subsequent oscillations that are present in aerated impact conditions.

4.2.2 Zone II: Low-aeration impact

The P_m and t_r trends for the cube streamwise positions within Zone II of the low-aeration impact category are presented in Figure 4.4. Low-aeration impacts are characterized by substantially higher values of P_m , ranging from $4.80\rho\bar{c}_p^2$ to $12.64\rho\bar{c}_p^2$, concentrated in the localized region of wave impact. The corresponding t_r values lie between $0.001\bar{T}$ and $0.0017\bar{T}$, with the shortest t_r observed across all cases occurring in this category at $x_c = 5.483\lambda_0$. As shown earlier in Figure 4.1, the elevation of wave impact descends as the structure is shifted downstream. The peak pressure recorded at $x_w/\lambda_0 = 5.483$ ($12.64\rho\bar{c}_p^2$) is notably high, exceeding by more than an order of magnitude the P_m values observed in the no-aeration cases without direct impact.

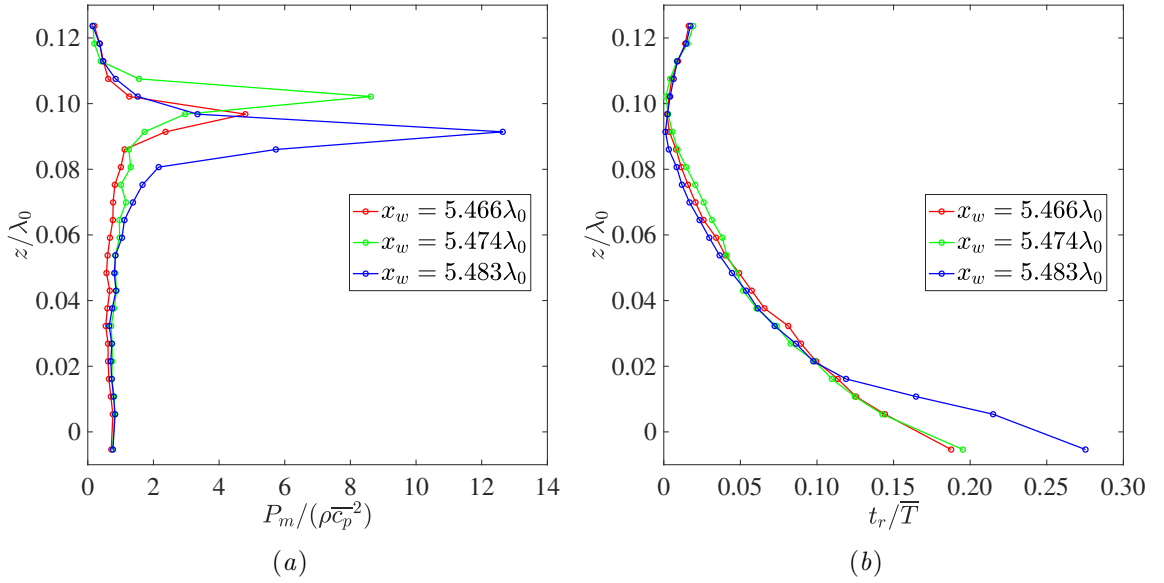


Figure 4.4: Pressure characteristics at the three streamwise locations of the cube associated with low-aeration type impacts: (a) P_m at varying elevations, (b) t_r at varying elevations. The pressure characteristics shown in each subplot represent simultaneous measurements from a single experimental run. The mean water level corresponds to $z/\lambda_0 = 0$.

Pressure records for the three low-aeration conditions show that the pressure does not immediately reach its peak value upon impact of the wave crest on the cube's front face. Instead, precursor spikes appear before the main peak in the measurements from the sensors that registered the highest P_m values at each streamwise position within this impact regime. For the cases $x_w/\lambda_0 = 5.466$ and 5.483 , these precursor spikes are followed by oscillations before the primary pressure peak appears. A detailed comparison of the pressure measurements with the synchronized video frames from the LIF movies reveals that these precursor spikes occur precisely when the jet emerging from the wave crest first interacts with the front face of the cube, as highlighted in the LIF snapshot in Figure 3.5*d*. The subsequent pressure oscillations indicate that the entrapped air bubble undergoes compression and expansion even before the main wave crest impacts the structure. Following these precursor pressure spikes and oscillations, the impact of the main wave crest results in the primary pressure peak, corresponding to the highest P_m in each case. Figure 4.5 illustrates this sequence, with insets providing a magnified view of the pressure evolution in the region of interest. The ability to capture such highly transient phenomena, including the intricate dynamics of wave impact within extremely short time intervals, can be attributed to the high sampling rate of the pressure sensors used in the present study.

Pressure measurements obtained during wave impact in Zone II reveal distinct oscillations in the pressure signal immediately after impact, as shown in Figure 4.6. These oscillations arise from the cyclic compression and expansion of entrapped air bubbles. The dominant oscillation frequencies were measured to be 7031 Hz, 3515 Hz, and 1757 Hz for the streamwise locations $x_w/\lambda_0 = 5.466$, 5.474 , and 5.483 , respectively. A clear trend emerges: as the cube face is positioned farther downstream, the frequency of post-impact oscillations decreases. According to Topliss et al. (1993), who investigated pressure oscillations during wave-impact events, os-

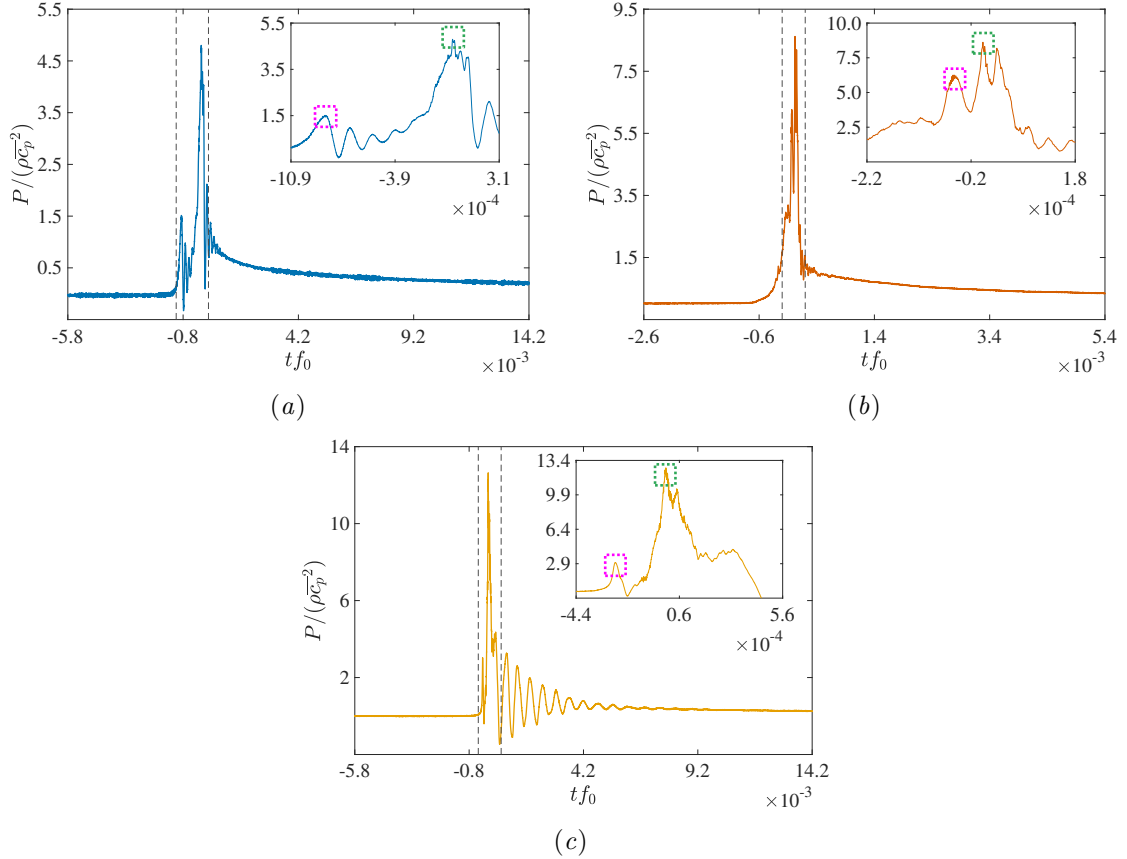


Figure 4.5: Pressure time histories recorded by the sensors that measured the highest P_m for the three low-aeration cases: (a) at $x_w = 5.466\lambda_0$, (b) at $x_w = 5.474\lambda_0$, and (c) at $x_w = 5.483\lambda_0$. The insets in each subfigure provide a magnified view of the pressure variations within the time interval marked by the vertical dashed lines, offering a detailed perspective on the pressure evolution during the impact. The magenta box highlights the initial pressure spike corresponding to the jet's interaction with the structure, while the green box marks the subsequent pressure peak associated with the impact of the following wave crest.

cillation frequency and air entrainment are inversely related, with higher frequencies indicating smaller entrapped air volumes. This relationship establishes a strong link between the cube's streamwise position and the dynamic pressure response following impact.

A further noteworthy feature of these oscillations is their phasing behavior across sensor elevations. For certain downstream positions, particularly at $x_w/\lambda_0 = 5.483$, the oscillations occur nearly in phase at and immediately below the elevation corresponding to the highest P_m . This phase coherence is also discernible, though less pronounced, at $x_w/\lambda_0 = 5.474$, while it is absent at the more upstream location of $x_w/\lambda_0 = 5.466$. At the latter position, oscillations are confined mainly to the sensor struck by the crest and the one immediately below it. In contrast, as the cube is shifted downstream, an increasing number of sensors exhibit in-phase oscillations.

The emergence of such phase coherence at lower frequencies for the more downstream cases can be explained by the relatively larger air pocket formed during impact. When the cube is positioned farther downstream, the approaching crest is slightly less steep at contact, promoting the entrapment of a more voluminous air cavity between the wave front and the structure face. A larger cavity oscillates at a lower natural frequency because the effective stiffness of the enclosed air decreases with increasing volume. It also enhances the spatial coherence of pressure fluctuations, as the continuous air region behaves as a coupled acoustic domain that transmits pressure waves more uniformly across neighboring sensors. The in-phase oscillations observed at $x_w/\lambda_0 = 5.483$ therefore likely result from the resonant response of this larger cavity, combined with the downward propagation of acoustic waves initiated at the impact point.

The emergence of strong, low-frequency, in-phase oscillations for $x_w/\lambda_0 = 5.483$ indicates that the entrapped air acts as a single compressible cavity rather than a

collection of discrete bubbles. In this regime, the oscillations represent a collective “breathing mode” of the air–water mixture, with pressure fluctuations transmitted coherently throughout the pocket despite the reduced sound speed associated with higher aeration. This differs from the more localized, high-frequency bubble oscillations observed at $x_w/\lambda_0 = 5.466$, where the smaller air content results in less coupling between sensors. A similar transition from localized to cavity-scale acoustic behavior was noted by Bredmose et al. (2009), who observed that wave impacts can excite both high-frequency local resonances and lower-frequency collective modes within aerated regions.

The present observations therefore suggest that the vertical region of acoustic influence corresponds to the zone of pronounced low-aeration effects, extending from the point of impact to several centimeters below it. The amplitude of these oscillations, however, decays progressively with depth, reflecting the attenuation of acoustic energy as it travels through the bubbly mixture. The strong phase alignment observed for $x_w/\lambda_0 = 5.483$, where the oscillation frequency is lowest, further implies that structural vibrations contribute negligibly to the measured pressure variations. Instead, the dominant mechanism is the fluid-dynamic coupling between the entrapped air cavity and the surrounding liquid. Collectively, these results underscore the pivotal role of air entrapment in controlling both the frequency and coherence of post-impact pressure oscillations, and illustrate the transition from localized, high-frequency bubble oscillations to spatially coherent, low-frequency acoustic modes as the trapped air volume increases.

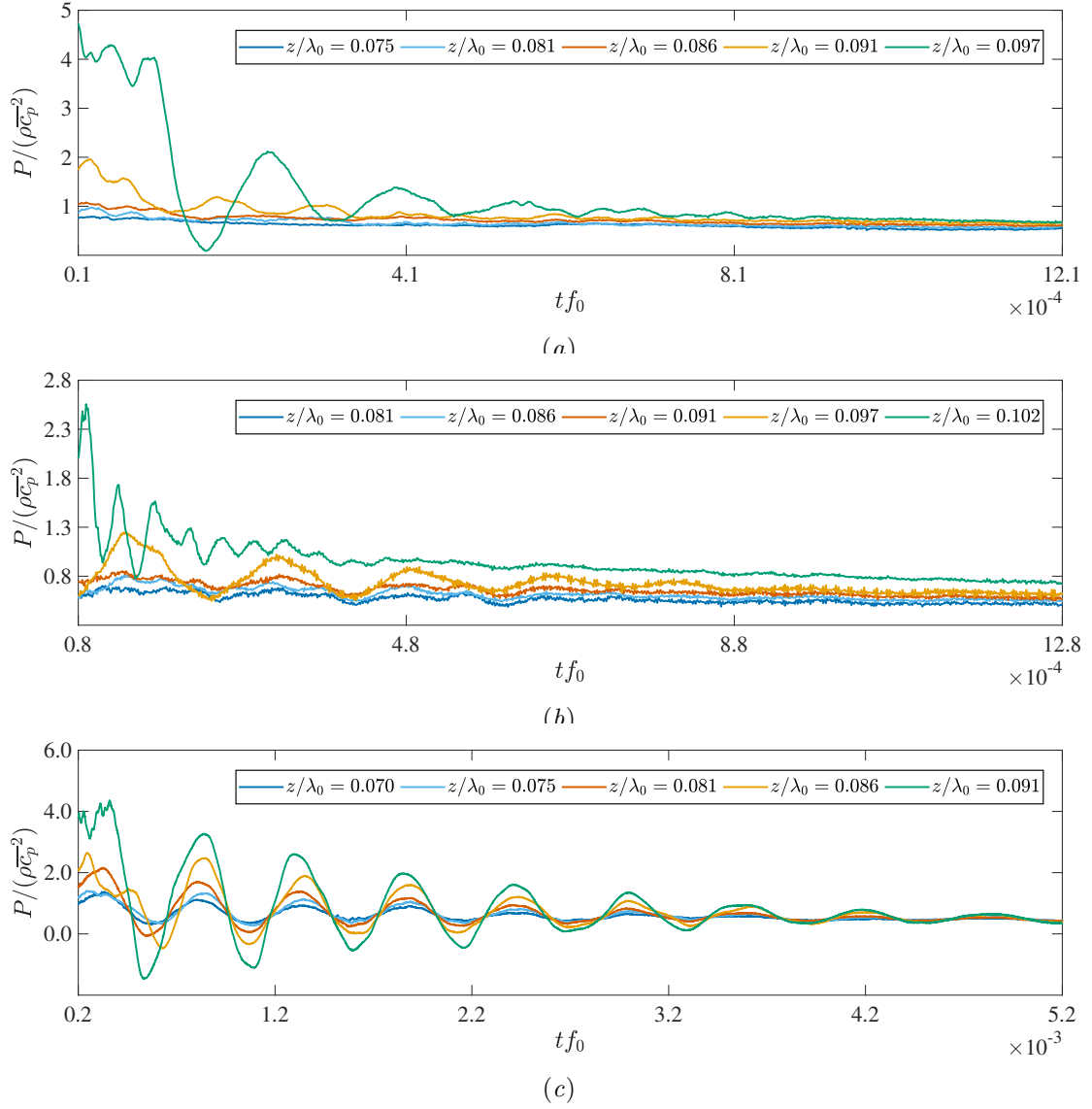


Figure 4.6: Pressure oscillations at and below the impact elevation for the three low-aeration conditions: (a) $x_c = 5.466\lambda_0$, (b) $x_c = 5.474\lambda_0$, and (c) $x_c = 5.483\lambda_0$. The corresponding elevations of the highest pressure, P_m , are $z/\lambda_0 = 0.096$, 0.102 , and 0.091 , respectively.

For the streamwise positions within the low-aeration zone, the elevations above the impact zone exhibit P_m values ranging from $3.34\rho c_p^2$ proximate to the impact zone, to $0.14\rho c_p^2$ at $z/\lambda_0=0.124$. Additionally, the t_r values are observed to range from

0.002 $\bar{T}$ (just above the impact zone) to 0.019 $\bar{T}$ (further above the impact zone). A clear trend emerges where P_m decreases with increasing elevation above the impact zone. Associated with this decrease in P_m is an increase in t_r and a corresponding delay in the occurrence of P_m compared to the timing of peak P_m observed at the elevation where it is maximal in each of the three cases. Consequently, the starting times—the moment when pressure values begin to rise before reaching peak P_m —are also delayed as the measurement location moves upward from the impact zone.

For elevations below the zone of wave impact, P_m values range from $5.73\rho\bar{c}_p^2$ to $0.72\rho\bar{c}_p^2$ with t_r ranging from 0.0032 $\bar{T}$ to 0.13 $\bar{T}$. As observed with elevations above the impact zone, a decrease in P_m is noted as the measurement location moves further below the impact zone. Interestingly, as the location of the cube shifts upstream, P_m decreases at the same corresponding elevation. However, at elevations below the impact zone, P_m can be observed to occur simultaneously. This pattern contrasts with observations from the zone above the impact point, where pressure peaks were delayed as the measurement location moved upward from the impact zone. The delayed occurrence of P_m at elevations above the impact zone can be a result of the upward movement of the water surface along the structure post-impact, rising toward higher elevations post-impact. On the other hand, the simultaneous occurrence of P_m at elevations below the impact zone can be attributed to the intense pressure impact experienced within the impact zone during the event.

Negative sub-atmospheric pressure values were observed at the elevations where the highest P_m values was recorded for the case $x_c = 5.483\lambda_0$ ($12.64\rho\bar{c}_p^2$). This phenomenon was noted at the elevation where the highest P_m occurred, i.e., at $z/\lambda_0=0.0914$, as well as at elevations above and below this point, as shown in Figure 4.7. The rapid fall in pressure values results from the expansion of the air pocket compressed between the impinging jet and the rising water surface along the contact

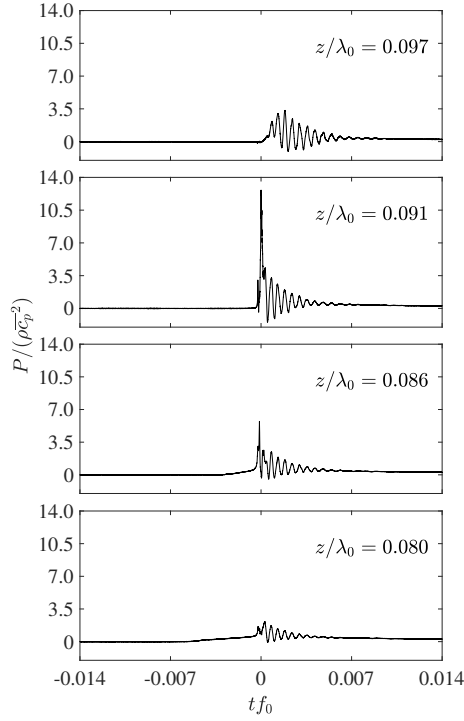


Figure 4.7: Pressure measurements of the case $x_c/\lambda_0 = 5.483$ for the elevations where the highest negative pressure was recorded

point. Following the negative pressure, damping of the pressure oscillations occurs with a similar trend at the presented elevations, indicating the subsequent expansion and compression of the entrapped air bubble. The highest negative pressure, $-1.47\rho\bar{c}_p^2$, corresponding to about 12% of the highest positive P_m , occurs at the end of the first expansion phase at the same elevation as the highest positive P_m . According to Yui et al. (1988), the following suppression of the oscillations occurs because the entrapped air pocket collapses into smaller air bubbles.

4.2.3 Zone III: Impact of broken-wave

The variation of P_m with elevation for the three cases in Zone III is shown in Figure 4.8. As the cube is placed farther downstream within Zone III, the plunging process becomes more developed (see Figure 3.7), the wave breaks upstream of the

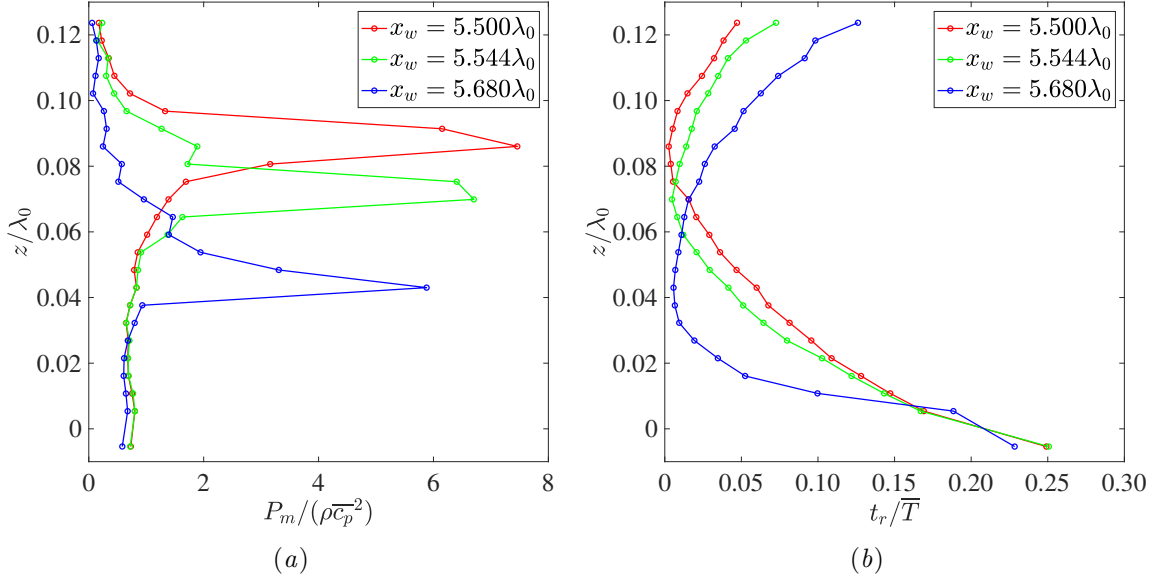


Figure 4.8: Pressure characteristics at the three streamwise locations of the cube associated with broken-wave impacts: (a) P_m at varying elevations, (b) t_r at varying elevations. The pressure characteristics shown in each subplot represent simultaneous measurements from a single experimental run. The mean water level corresponds to $z/\lambda_0 = 0$.

structure, the overturning crest impacts the upstream free surface, and a secondary jet forms and propagates ahead of the main crest. Consequently, the secondary jet impinges on the wall prior to the arrival of the main crest, establishing a two-stage loading sequence.

Following overturning, the wave crest traps an air pocket that rapidly fragments into smaller bubbles as breaking progresses. The resulting bubble dynamics—cyclic compression and expansion—generate post-impact pressure oscillations that begin immediately after the secondary-jet impact and persist beyond the main-crest collision. Similar to Zone II, sub-atmospheric pressures frequently occur after the initial peak, indicating substantial air entrainment both before and during impact. However, the vertical phasing of these oscillations differs from that in Zone II: while sensors located

near the elevation of maximum P_m may exhibit partial phase alignment (Figure 4.9), the oscillations in Zone III are generally less synchronized than those observed in the low-aeration case at $x_c = 5.483\lambda_0$. A plausible explanation is that, in Zone III, the entrapped air becomes dispersed into a spatially non-uniform bubbly mixture with a high void fraction. This medium exhibits a substantially reduced and spatially variable effective sound speed, leading to strong scattering and attenuation of pressure waves, which prevents coherent acoustic propagation with depth. Additionally, the coexistence of multiple excitation sources—the secondary jet and the subsequent crest impact—together with spatial variations in bubble size distribution and local aeration levels, induces phase dispersion and de-correlation among sensors. In contrast, the low-aeration regime supports a more cavity-like, coherent response, producing in-phase oscillations across adjacent elevations.

Another distinct feature of Zone III is the presence of two oscillation modes in the pressure signal (Figure 4.10). Immediately after the secondary-jet impact, high-frequency oscillations are observed, while following the main-crest impact, the post-impact oscillations occur at a lower frequency. An example is presented in Figure 4.10c, which shows the pressure record for the sensor located at the impact elevation for the case $x_c = 5.68\lambda_0$. Here, the dominant frequency following the secondary-jet impact is approximately 1345 Hz, whereas that after the main crest is about 330 Hz, indicating that the jet-induced oscillations are more than four times as frequent. Such two-mode behavior is evident across all streamwise positions corresponding to the broken-wave impact regime. The higher initial frequency is consistent with the rapid, localized deformation and ventilation associated with the nascent jet, whereas the subsequent lower-frequency response reflects a larger, more distributed aerated region excited by the main crest.

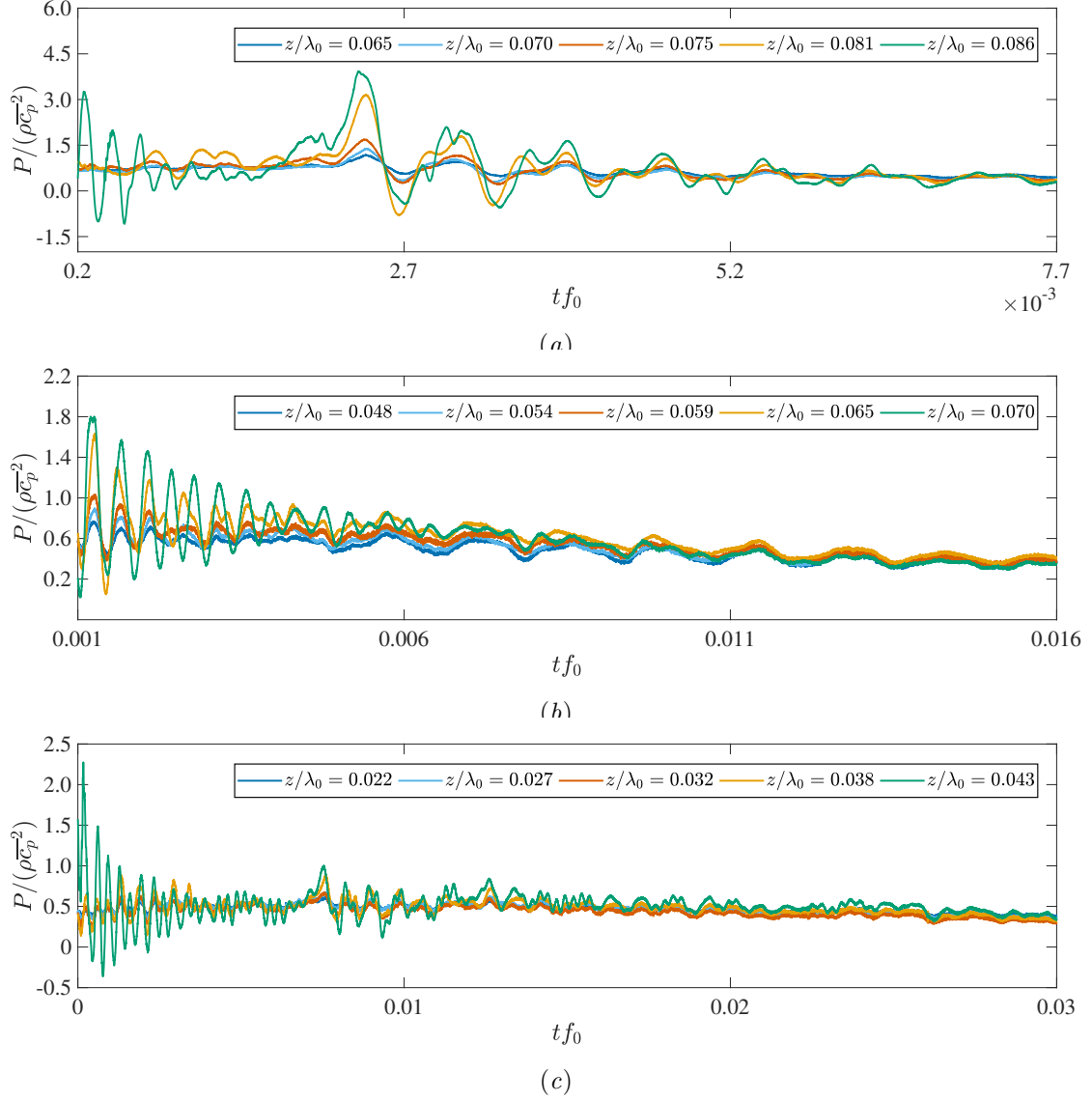


Figure 4.9: Pressure oscillations at and below the impact elevation for the three broken-wave conditions: (a) at $x_c = 5.500\lambda_0$, (b) at $x_c = 5.544\lambda_0$, and (c) at $x_c = 5.680\lambda_0$. In (a), the highest P_m is recorded at $z/\lambda_0 = 0.081$, whereas in (b) and (c), it is recorded at $z/\lambda_0 = 0.054$ and $z/\lambda_0 = 0.048$, respectively.

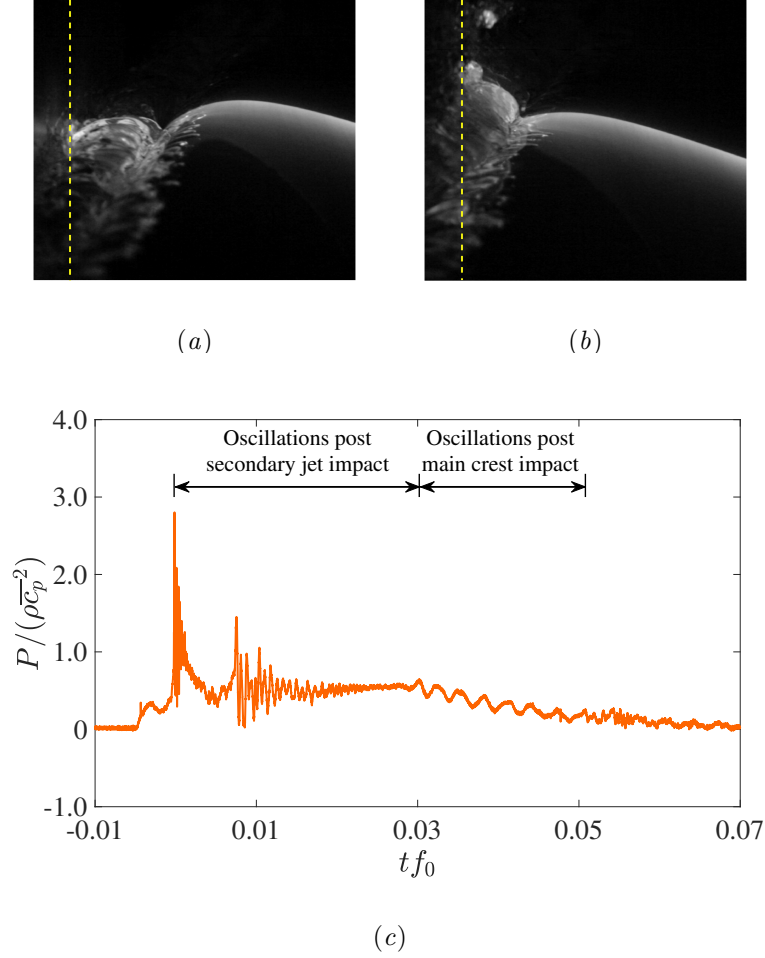


Figure 4.10: For the broken-wave condition ($x_c/\lambda_0 = 5.68$): (a) Snapshot from the LIF movie capturing the moment when the secondary jet emerging from the broken-wave impacts the front face of the cube. (b) Snapshot from the LIF movie showing the instant when the main wave crest strikes the front face of the cube. (c) Pressure measurements illustrating the two distinct oscillation modes observed post-impact. The dashed vertical lines in (a) and (b) indicate the position of the cube's front face.

Peak pressures in Zone III reach values exceeding $9\rho\bar{c}_p^2$, comparable to the upper range observed in Zone II. The corresponding rise times, ranging from $0.0014\bar{T}$ to $0.0052\bar{T}$, also overlap partially with those in Zone II. The broader range of t_r can be attributed to the turbulent air–water mixture present during impact, where the presence of entrapped air prolongs the loading process. Because the trapped air must undergo compression followed by re-expansion, the pressure peak persists over

a longer duration. As shown earlier in Figure 4.1, the elevation of maximum loading shifts downward in Zone III relative to Zone II as the cube is positioned farther from the wavemaker, consistent with the plunging crest striking the surface at progressively lower z/λ_0 before impingement on the wall. Compared with Zone I, the elevated P_m values observed in Zone III highlight the amplifying influence of air entrainment on impulsive loading, even though the resulting pressure oscillations are less vertically coherent than those in the low-aeration regime.

4.3 Force measurements for the three different impact conditions

By integrating the recorded pressure values over the elevation range of the installed sensors, the resultant force per unit width (F) acting on the frontal face of the cube is obtained for the entire measurement duration. Figure 4.11 compares the time histories of pressure and the corresponding F for the cube locations that exhibited the highest P_m in each of the impact categories discussed previously in Section 4.1.2.

It is emphasized that this comparison does not imply a direct proportionality between the absolute magnitudes of pressure and force, since F represents an integrated response over the wetted surface where pressures may vary spatially and are only partially captured by the instrumented region. Instead, the comparison focuses on the temporal relationship between the two quantities—how the evolution of local pressures at the impact zone influences the overall force response. Although unmonitored regions between adjacent pressure sensors may contain localized pressure peaks, the present measurements provide a reliable representation of the overall force evolution and its primary trends during impact.

Analysis of the curves in each plot reveals that the time-dependent behavior of $F(t)$ closely follows that of the pressure at the primary impact elevation. Specifically,

for the no-aeration impact, a distinct force peak coincides with the occurrence of P_m , though the corresponding rise and fall durations of F are more gradual. For the low-aeration and broken-wave conditions, the temporal variations in force and pressure remain nearly synchronous, occurring within comparable time intervals. Both impact regimes also display matching oscillation frequencies in the force and pressure signals during impact, indicating that the oscillatory pressures are acting coherently across a significant portion of the cube's frontal face. This temporal alignment underscores the role of high-frequency pressure fluctuations in modulating the instantaneous force and highlights their potential influence on the structural response during impact.

An instructive example is provided by the broken-wave condition at $x_c = 5.680\lambda_0$ (Figure 4.11*c*), where large disparities in local peak pressures are not proportionally reflected in the net force. The secondary-jet impact produced $P_m \approx 5.80\rho\bar{c}_p^2$, while the subsequent main-crest impact reached only $P_m \approx 0.42\rho\bar{c}_p^2$. In contrast, the corresponding nondimensional forces were $F/(\rho\bar{c}_p^2\lambda_0 w) = 0.086$ and 0.049 at $tf_0 = 0$ and $tf_0 = 0.04$, respectively. Thus, a pressure ratio of roughly 14:1 yields a much smaller force ratio of about 1.8:1, showing that the secondary-jet loading is highly localized in both elevation and duration—producing very high P_m without a proportional increase in the integrated force. This finding demonstrates that spatially confined pressure peaks do not necessarily translate into proportionally larger total forces, and that the spatial distribution of pressure over the structure plays a decisive role in determining the net loading.

Among all cube positions examined, the highest nondimensional force, $F_{\max} = 0.155\rho\bar{c}_p^2\lambda$, occurred for the low-aeration impact at $x_c = 5.483\lambda_0$, coinciding with the position of the highest measured P_m . This peak force exceeded those obtained for the other two impact regimes— $F_{\max} = 0.07\rho\bar{c}_p^2\lambda$ for the no-aeration case and $0.086\rho\bar{c}_p^2\lambda$ for the broken-wave case. Although the low-aeration peak force was roughly twice

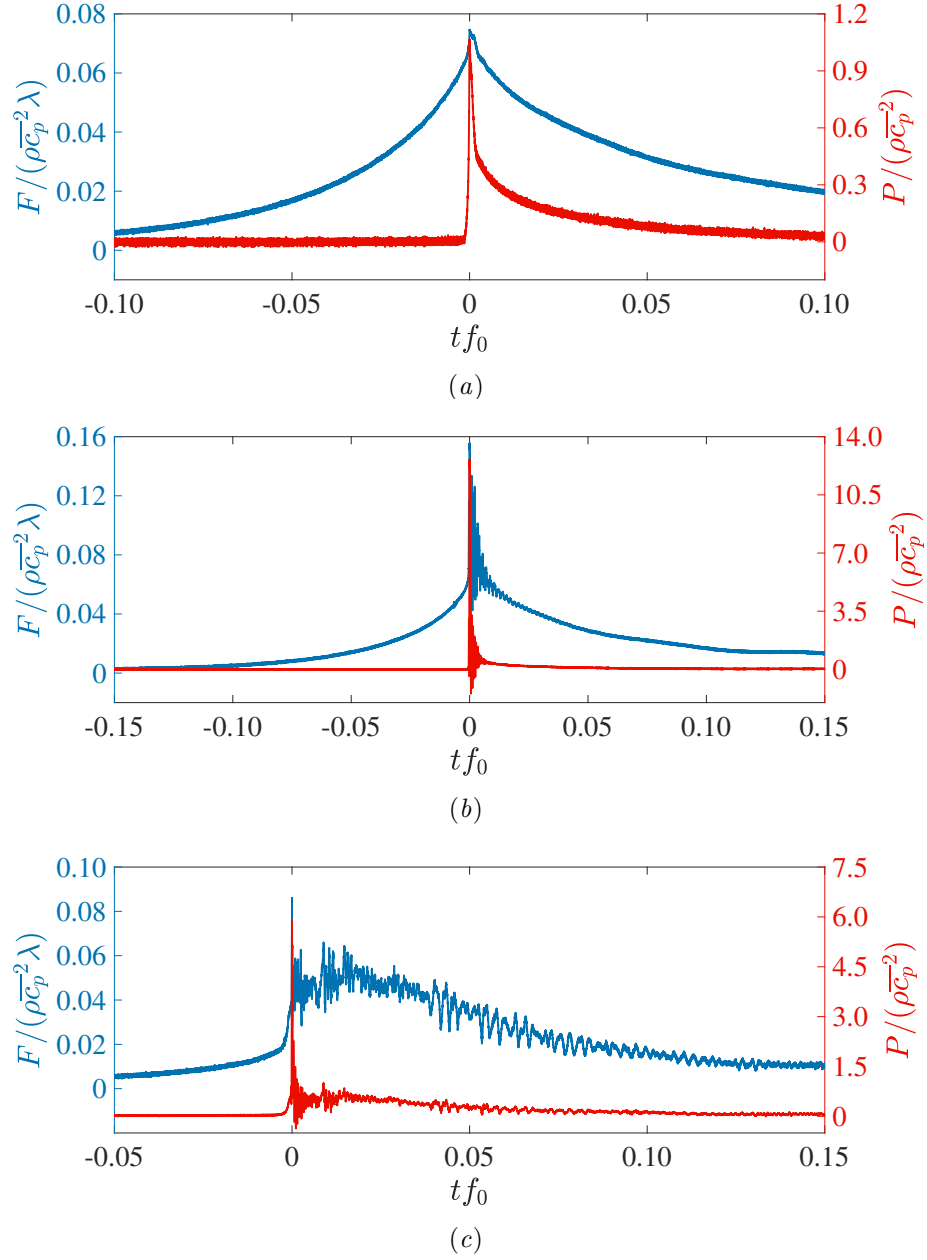


Figure 4.11: Time history of the horizontal force experienced by the frontal face of the structure during the impact event for the three different impact categories given by $x_c/\lambda_0 = 5.431$, 5.483 , and 5.68 , shown in (a) no-aeration impact, (b) low-aeration impact, and (c) broken-wave impact, respectively. The force values (in red) were obtained by integrating the pressure measurements over a unit width, as recorded by the pressure sensors across the range of elevations where the sensors are located. The pressure-time history for the sensor which recorded the maximum pressure for each of the respective conditions is also presented in blue.

Table 4.1: Maximum nondimensional force ($F_{\max}$) and force impulse (F_I) for different impact conditions.

Impact condition	$F_{\max}$	F_I
No-aeration ($x_c = 5.431\lambda_0$)	$0.077\rho\bar{c}_p^2\lambda$	0.0093 s
Low-aeration ($x_c = 5.483\lambda_0$)	$0.155\rho\bar{c}_p^2\lambda$	0.0081 s
Broken-wave ($x_c = 5.680\lambda_0$)	$0.086\rho\bar{c}_p^2\lambda$	0.0056 s

that of the other two regimes, the disparity is far smaller than that observed in P_m , where the ratios were approximately 14.5:1 (low-aeration to no-aeration) and 4.4:1 (low-aeration to broken-wave). Notably, at $x_c = 5.483\lambda_0$, the force increased sharply by nearly $0.06\rho\bar{c}_p^2\lambda$ within $0.0007\bar{T}$, emphasizing the distinctly impulsive nature of this loading compared with the much slower rise time of $0.106\bar{T}$ observed in the no-aeration condition.

The total force transmitted to the structure during impact, quantified as the force impulse (F_I), is obtained by integrating the force over time. This parameter is crucial because even moderate forces acting over longer durations can induce significant structural deflections. Table 4.1 lists the F_I values for the three impact categories. The results reveal that the no-aeration and low-aeration cases yield nearly identical F_I values, despite large differences in their corresponding $F_{\max}$ and P_m . This similarity reflects the close resemblance in their pre-impact wave evolution: in both cases, the incident crest steepens and rises along the cube face, but only in the low-aeration case does a small jet form and entrap an air pocket immediately before impact. In contrast, the no-aeration case proceeds without air capture. Thus, while the presence of trapped air leads to a much sharper peak, the overall momentum transfer—represented by F_I —remains comparable.

For the broken-wave condition, F_I is substantially lower, reflecting the influence of turbulence and air–water mixing in reducing force transmission to the cube face.

The peak impact elevation also shifts downward from $z/\lambda_0 = 0.107$ and 0.091 in the no- and low-aeration cases to $z/\lambda_0 = 0.048$ in the broken-wave regime, indicating that the wave interacts with a smaller vertical section of the cube. This reduction in the effective impact area, combined with the dissipative nature of the aerated mixture, accounts for the lower F_I values. The consistent observation across multiple runs—that low-aeration impacts produce the highest $F_{\max}$ while no-aeration impacts yield the highest F_I —confirms the robustness of this trend across the tested regimes.

4.4 Impact pressure variations over repeated experiments

Considering the entire region above the still-water level where the wave interacts with the front face of the structure, the actual impact zone is relatively limited. The wave impact dynamics are strongly influenced by the degree of breaker development immediately prior to contact. Consequently, even small variations in the wave evolution at that instant can produce substantial differences in the recorded impact pressures. This sensitivity was clearly evident in the present experiments, where repeated runs conducted at the same streamwise position of the cube exhibited noticeable variability in the measured pressures.

To illustrate this variability, a representative case corresponding to the streamwise position $x_c/\lambda_0 = 5.483$ is examined in detail. This location, classified as a low-aeration impact in which an air pocket is likely entrapped during impact, lies near the critical transition between the non-breaking (no-aeration) and fully breaking (broken-wave) regimes, and produced the highest measured peak pressure, P_m , among all tested conditions. Because of its transitional nature, this case is particularly sensitive to subtle differences in the wave-breaking process, making it well suited for assessing the repeatability and variability of impact pressures under nominally identical

conditions.

For this configuration, pressure time histories from repeated runs are presented in Figure 4.12. The figure compares pressure records at five elevations (z/λ_0) near the impact zone, where Runs 1, 2, and 3 are compared with Run 4, which produced the highest P_m . Each subfigure highlights variations in impact behavior, including the presence of post-impact oscillations and the elevation of the dominant pressure peak. Figure 4.12*a* compares Runs 1 and 4. In Run 1, a relatively modest peak pressure of $1.21\rho\bar{c}_p^2$ occurs at $z/\lambda_0 = 0.102$, with no evidence of post-impact oscillations, suggesting an impact event free of air entrapment. These characteristics—lower P_m and a smooth decay without oscillations—closely resemble the no-aeration impacts discussed in Section 4.2.1.

By contrast, Runs 2, 3, and 4 exhibit higher P_m values accompanied by distinct post-impact oscillations, indicating that air entrapment likely occurred just before impact. In Runs 2 and 3, where oscillations are particularly pronounced, the impacts occur at slightly higher elevations than in Run 4, as the corresponding highest P_m values appear at larger z/λ_0 positions (Figures 4.12*b* and 4.12*c*). These higher-elevation impacts also occur marginally later in time relative to the primary peak observed in Run 4. Although a general correlation is observed between impact elevation and timing, the magnitude of P_m does not vary monotonically with elevation. The recorded P_m values for Runs 1 through 4 are $1.21\rho\bar{c}_p^2$, $7.24\rho\bar{c}_p^2$, $6.36\rho\bar{c}_p^2$, and $12.64\rho\bar{c}_p^2$, respectively—demonstrating more than an order-of-magnitude difference in peak pressures under nominally identical initial conditions. The differences in post-impact oscillations are also significant, with dominant frequencies of 3515 Hz, 2197 Hz, and 1757 Hz for Runs 2, 3, and 4, respectively.

Bagnold (1939) observed that the mechanics of wave impact are largely governed by the compression of the intervening air layer and its effective thickness. He

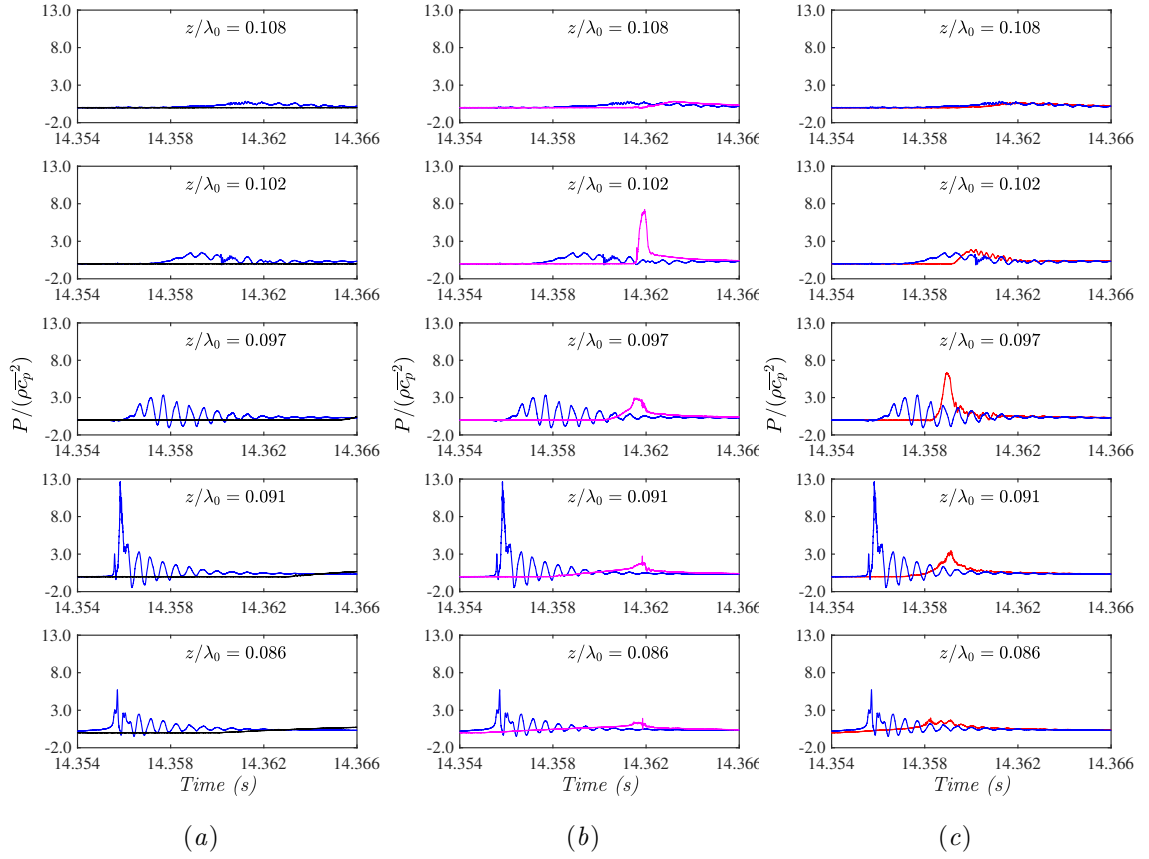


Figure 4.12: Comparison of pressure measurements obtained over repeated runs of experiment at $x_c/\lambda_0=5.483$: (a) Run 1 (black) and Run 4 (blue), (b) Run 2 (magenta) and Run 4 (blue), and (c) Run 3 (red) and Run 4 (blue). The pressures at each wall location are simultaneous measurements obtained from a single run of the experiment.

further noted that the air layer thickness is difficult to define precisely because of irregularities in the approaching wave front. Such non-uniformity can produce spatial variations in pressure across the impact area. However, as discussed above, the variability extends not only to the magnitude of P_m but also to the elevation of impact, revealing the high sensitivity of the process to even minute differences in breaker development immediately before contact.

Despite the substantial variability in instantaneous pressures, the pressure impulse (P_i), obtained by integrating the pressure record over time, exhibits relatively better consistency. The impulse, evaluated over a duration of $0.02\bar{T}$ centered on the impact peak at each elevation, varies by approximately 20% near the impact zone and by less than 13% elsewhere among the different runs. Although these variations are not negligible, they are considerably smaller than those observed in the peak pressures, which can differ by a factor of two or more between runs. This consistency indicates that, while instantaneous pressures and oscillatory features are highly sensitive to local aeration and free-surface evolution, the overall momentum transfer remains comparatively stable. Consequently, P_i provides a reliable descriptor of impact severity even under conditions of pronounced run-to-run variability in pressure magnitude and air entrapment.

4.5 Characteristics of pressure time series based on mean

This subsection discusses the characteristics derived from the mean values. The mean quantities presented here are obtained by averaging the respective measurements at each sensor elevation across all repeated runs. Figures 4.13, 4.14, and 4.15 show the mean distributions of P_m , t_r , and P_i , respectively, for the three impact categories. In each figure, the grey band surrounding the mean trend line represents the uncertainty

range, defined as one standard deviation above and below the mean at each elevation. The results are shown for representative streamwise positions corresponding to each regime: $x_c/\lambda_0 = 5.431$ (no-aeration), 5.483 (low-aeration), and 5.680 (broken-wave).

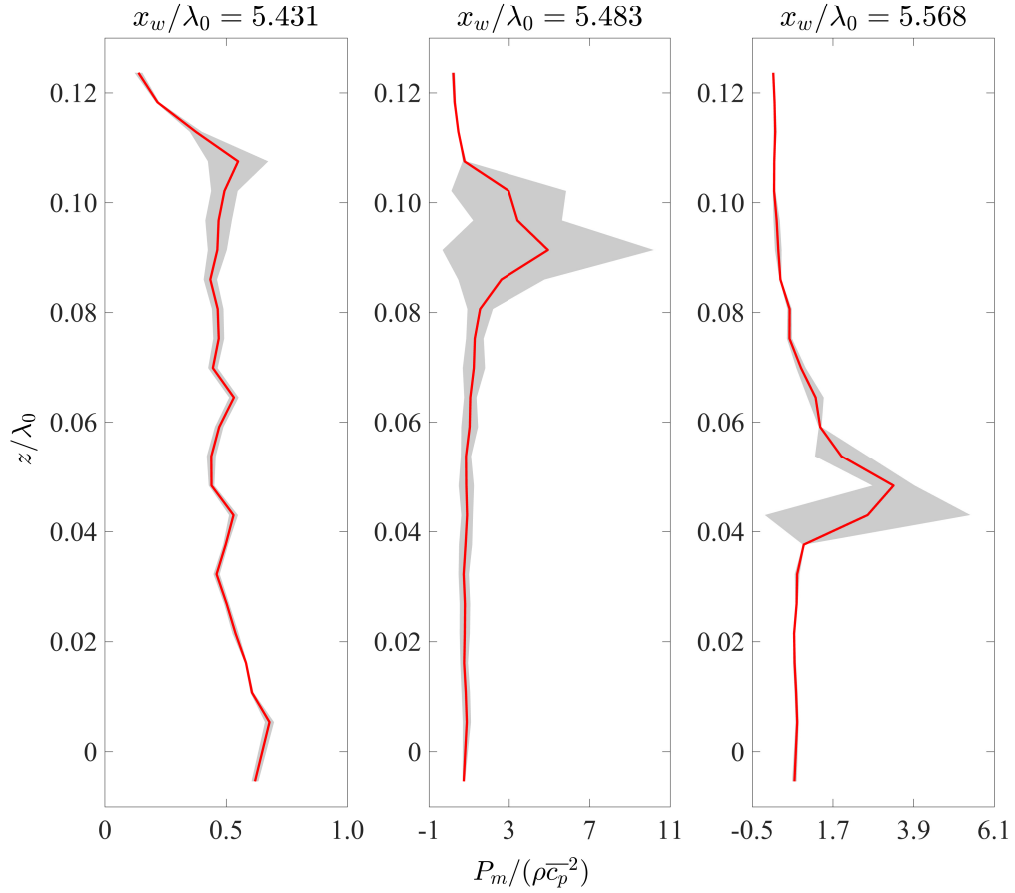


Figure 4.13: Mean and standard deviation of P_m for the three different impact conditions. The red trend line shows the average of P_m values recorded at various elevations along the front face of the cube across multiple experimental runs. The surrounding grey band illustrates the uncertainty range, corresponding to one standard deviation above and below the mean at each elevation. Note that the x-axis range differs across the three subplots to best highlight the variability within each condition; direct comparisons of absolute magnitude across subplots should therefore be made with caution.

Figure 4.13 illustrates the variation of mean P_m with elevation. The highest mean values of $0.78\rho\bar{c}_p^2$, $4.93\rho\bar{c}_p^2$, and $3.35\rho\bar{c}_p^2$ correspond to the three impact regimes.

The shaded envelopes indicate the uncertainty range, defined as one standard deviation above and below the mean at each elevation. Pressure variability is clearly most pronounced within the impact zone and diminishes both below and above it. Relative to the peak region, the P_m values near or below the still-water level are significantly smaller—typically below $0.75\rho\bar{c}_p^2$ —and a similar attenuation is observed at elevations well above the impact zone.

The corresponding mean rise times (t_r), plotted in Figure 4.14, show that the smallest t_r values coincide with the elevations where the largest P_m values occur. The mean t_r ranges from about $0.0014\bar{T}$ within the impact zone to roughly $0.276\bar{T}$ below the still-water level. In contrast to the trend for P_m , greater variability in t_r is observed near and below the mean water level. Above the impact zone, variations in t_r are less pronounced but remain noticeable for the broken-wave condition ($x_c/\lambda_0 = 5.680$), where the region is typically dominated by spray and fragmented droplets following impact. Together, the patterns of mean P_m and t_r clearly highlight the impulsive nature of plunging-wave impacts, characterized by large peak pressures and short rise durations.

With respect to the pressure impulse (P_{imp}), the high-impact phase persists for only about $0.0015\bar{T}$, much shorter than the characteristic wave period of 0.87 s. Consequently, time integration of the pressure over the positive loading duration shows that the short-lived peaks contribute relatively little to the overall impulse. Figure 4.15 presents the elevation-dependent distribution of P_{imp} for all three impact regimes. The results indicate that the largest impulse values occur near the still-water level, even though the corresponding pressure maxima at those elevations are considerably smaller than those in the main impact zone. The inverse relationship between P_m and P_{imp} suggests that sustained, moderate pressures can contribute more significantly to the overall momentum transfer than brief, high-intensity peaks.

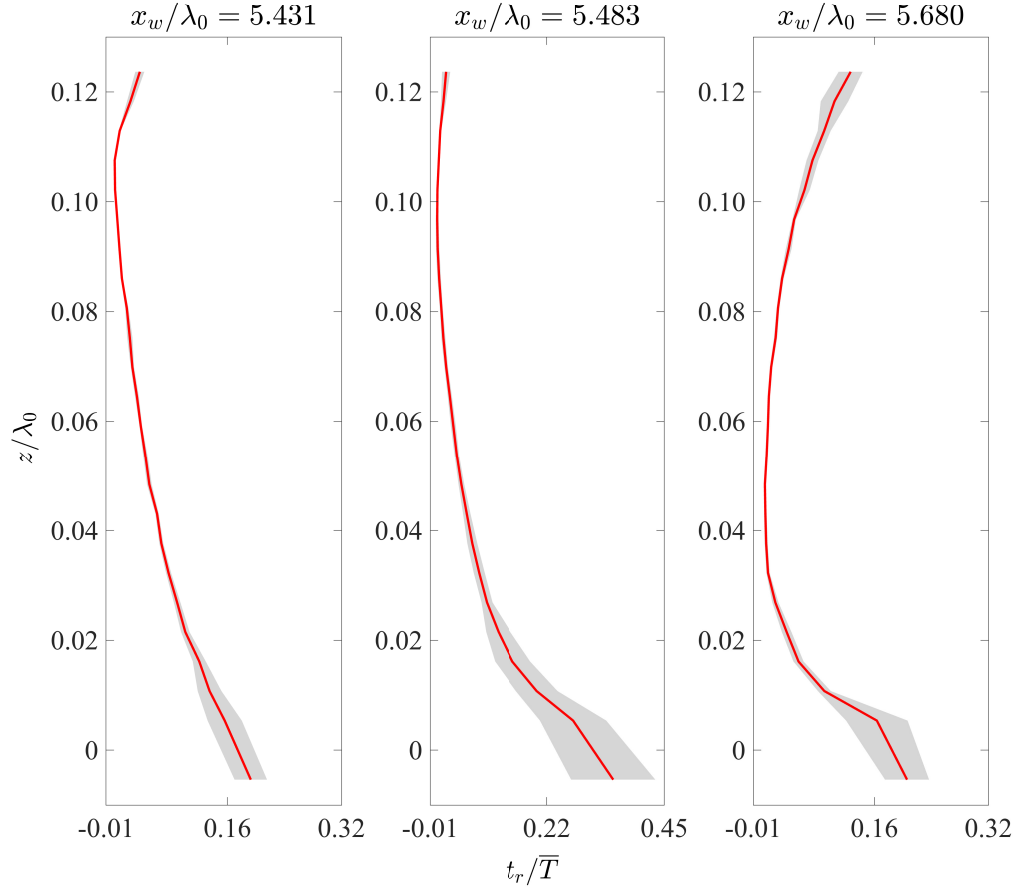


Figure 4.14: Mean and standard deviation of t_r for the three different impact conditions. The red trend line shows the average of t_r values recorded at various elevations along the front face of the cube across multiple experimental runs. The surrounding grey band illustrates the uncertainty range, corresponding to one standard deviation above and below the mean at each elevation. Note that the x-axis range differs across the three subplots to best highlight the variability within each condition; direct comparisons of absolute magnitude across subplots should therefore be made with caution.

A comparison of Figures 4.14 and 4.15 further reveals that P_{imp} tends to increase with t_r across elevations, implying that regions subjected to longer loading durations are more likely to induce appreciable structural deflections.

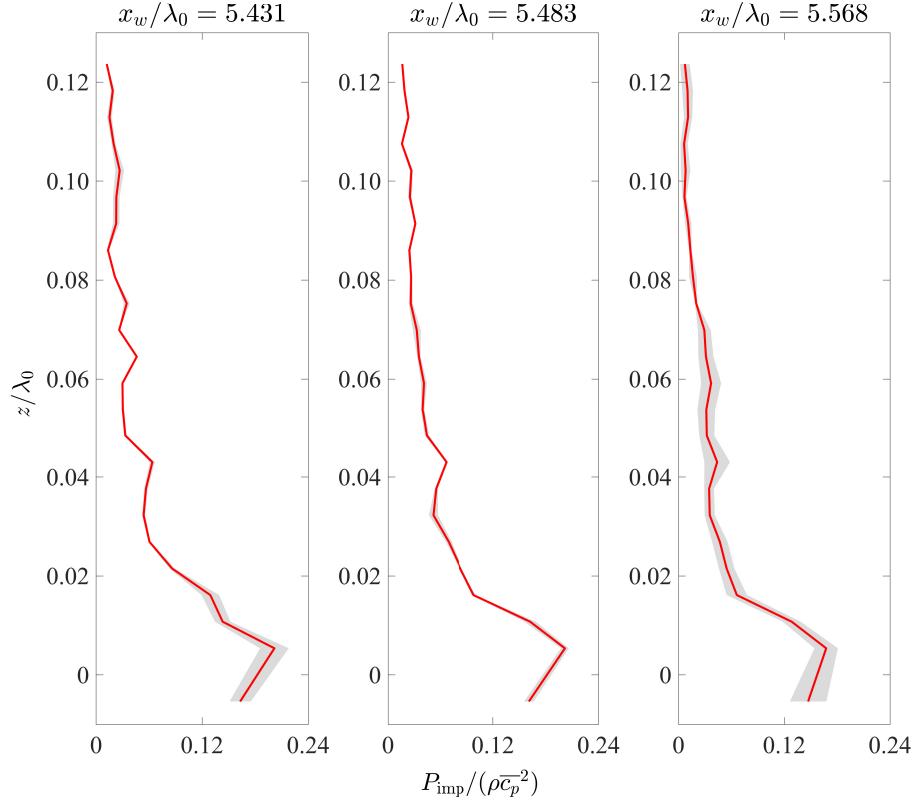


Figure 4.15: Mean and standard deviation of P_{imp} for the three different impact conditions. The red trend line shows the average of P_{imp} values recorded at various elevations along the front face of the cube across multiple experimental runs. The surrounding grey band illustrates the uncertainty range, corresponding to one standard deviation above and below the mean at each elevation.

Regarding variability, P_{imp} shows the highest scatter for the broken-wave condition ($x_c/\lambda_0 = 5.680$), moderate variability for the no-aeration condition ($x_c/\lambda_0 = 5.431$), and minimal scatter for the low-aeration condition ($x_c/\lambda_0 = 5.483$). This pattern differs from that of P_m and t_r , where the low-aeration case exhibited the greatest variability. The relatively high variability in P_{imp} for the broken-wave im-

fact is consistent with the inherently chaotic and turbulent nature of the bubbly mixture striking the cube face, where compressibility and aeration effects fluctuate strongly between runs. Despite these differences, all three datasets exhibit a common feature: the largest deviations occur primarily at elevations corresponding to the highest magnitudes of P_m , t_r , or P_{imp} .

4.6 Relationship between P_m and t_r

The results discussed so far indicate that both the magnitude and temporal characteristics of impact pressures are closely linked. The parameters P_m and t_r together define the impulsive nature of the impact— P_m representing the load intensity and t_r the characteristic time over which the pressure rises. Shorter t_r values are typically associated with more abrupt impacts and higher local accelerations of the free surface, leading to larger P_m , whereas longer t_r values correspond to more gradual pressure developments influenced by air entrainment and cushioning effects.

Figure 4.16 presents the measured P_m values plotted against their corresponding t_r from all experimental runs. The markers indicate the type of wave impact associated with each data point. Although the data exhibit noticeable scatter, a clear distinction can be observed among the different impact regimes in the insets presented in Figure 4.16. No-aeration impacts show moderate P_m values within a narrow range of t_r , while low-aeration impacts exhibit larger P_m values and shorter t_r , consistent with strongly impulsive behavior. Broken-wave impacts, in contrast, generally correspond to higher t_r values and a wider spread in P_m , reflecting the influence of aeration and jet-induced fluctuations.

A power-law function was fitted to the dataset to represent the overall trend

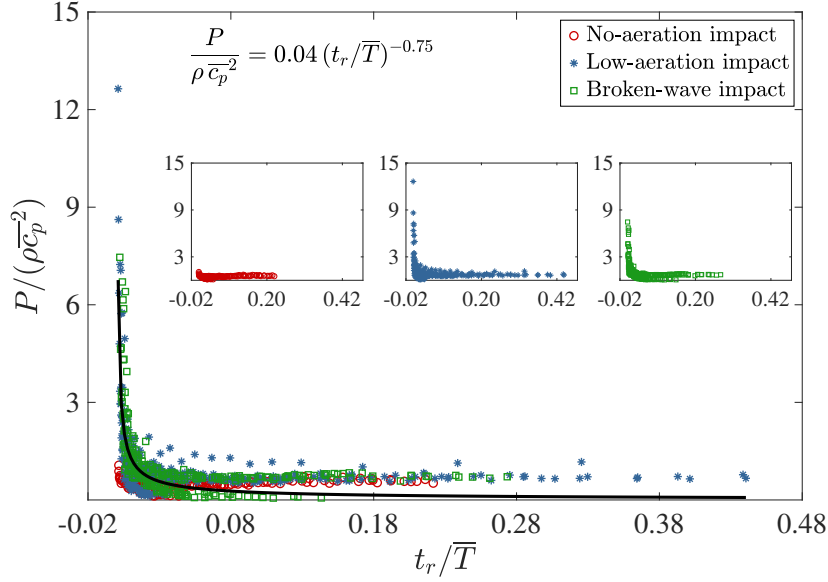


Figure 4.16: Relationship between P_m and t_r based on the measurements from various experimental runs.

between P_m and t_r , expressed as

$$\frac{P_m}{\rho \bar{c}_p^2} = 0.04 t_r^{-0.75}. \quad (4.6.1)$$

This fitted curve captures the general tendency of P_m to increase with decreasing t_r , illustrating the inverse relationship between impact magnitude and rise time. Similar power-law behavior has been reported in other liquid-impact and wave-structure interaction studies, though with some variability in the exponent—for instance, values ranging from -0.15 to -0.85 depending on the impact conditions (Kisacik et al., 2012; Song et al., 2015; Chuang et al., 2017). The present exponent of -0.75 lies well within this empirically observed range and reinforces the conceptual understanding that a rapid pressure rise (small t_r) is characteristic of impulsive loading.

Indeed, in the measured dataset, the most extreme low-aeration impacts lie close to or slightly above this trend—consistent with the idea that minimal but effec-

tive air entrapment provides just enough cushioning to modulate the rise time without completely quenching the impulsive effect. To frame this finding within broader theoretical context, one may relate it to pressure-impulse theory (e.g. Cooker and Peregrine, 1995), which connects peak pressures to integrated impulse over impact durations under assumptions of incompressible, inviscid flow. While pure impulse theory does not explicitly prescribe a power law between P_m and t_r , the presented empirical relation can be viewed as a consistency check—showing that in a diverse dataset, the pressure peaks and rise durations still co-vary in a manner compatible with the dominance of short-duration inertial pressure effects.

Moreover, the observed scatter and regime separation in the insets underline important caveats: the fitted power law does not supersede the need for regime-specific caution. For instance, in broken-wave impacts where air fragmentation, jet dynamics, and bubble oscillations come into play, t_r may not fully capture all the temporal complexity (e.g. secondary oscillations or delayed re-compression). Thus, while the power law provides a useful “backbone” trend, detailed mechanistic interpretation still requires careful consideration of local aeration, geometry, and kinematics. Deviations from this trend are themselves informative, pointing toward physical processes (e.g. air cushioning, fragmentation, turbulence) that depart from simple impulsive assumptions.

4.7 Theoretical Interpretation of Peak Pressure Distributions

Cooker and Peregrine (1991) proposed an idealized pressure impulse model to predict the pressure distribution during the impact of a breaking wave against a vertical wall. The model assumes a flat free surface at the instant of impact and considers the flow to be incompressible and inviscid. The pressure impulse, denoted by I , is defined

as the integral of pressure over the impact duration Δt . Air entrainment effects are not included in the formulation. Figure 4.17 illustrates the boundary conditions and governing equations associated with this model. The expression for the pressure impulse is given by:

$$I(y) = -\frac{2\rho U}{h} \sum_{n=0}^{\infty} \left[\frac{(1 - \cos \mu \lambda_n h) \sin(\lambda_n y)}{\lambda_n^2} \right], \quad (4.7.1)$$

$$\lambda_n = \left(n + \frac{1}{2} \right) \frac{\pi}{h}, \quad (4.7.2)$$

where U is the velocity of the impinging wave front, h is the vertical height from the wave crest to the reference depth, μ is the non-dimensional extent of the impact zone ($\mu = \Delta z/h$), and y denotes the vertical coordinate measured from the channel bed.

Assuming a triangular temporal profile of the impact pressure at each elevation, the peak pressure $P_m(y)$ is computed as:

$$P_m(y) = \frac{2I}{\Delta t}. \quad (4.7.3)$$

In the present study, a detailed frame-by-frame examination of the LIF recordings revealed that the impacting wave front for the condition $x_c/\lambda_0 = 5.483$ was nearly vertical immediately prior to impact, closely aligning with the assumptions of the theoretical model. This observation is supported by a snapshot from the LIF movie shown in Figure 4.18. While the model assumes a vertical wall extending through the entire water column, the experimental configuration differs in that the bottom face of the cube extends only to a depth of $z_b/\lambda_0 = -0.1113$ from the still water level. Accordingly, the parameter h is defined as the vertical distance between the crest elevation and this reference depth. Based on the LIF observations and ex-

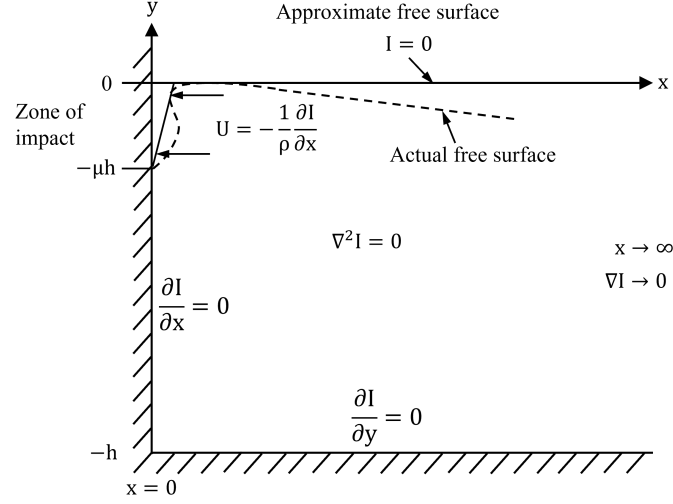


Figure 4.17: Boundary conditions and resulting governing equations for the theoretical pressure impulse model proposed by Cooker and Peregrine (1991).

tracted surface-elevation profiles, the following values were obtained for the critical case $x_c/\lambda_0 = 5.483$: $U = 1.73$ m/s ($1.2\bar{c}_p$), $\Delta t = 0.0018$ s ($0.0021\bar{T}$), $\mu = 0.037$ ($\Delta z = 0.032\lambda_0$), and $h = 0.248$ m ($0.21\lambda_0$). These parameters were subsequently used in equation 4.7.1 to compute the theoretical peak pressure distribution.

Figure 4.19 presents the comparison between the theoretical profile of P_m and the experimentally measured values across multiple runs, including their mean. Pressure data corresponding to elevations above the incident crest location ($z = 0.102\lambda_0$) have been excluded, as they pertain to post-impact dynamics rather than the primary impact event. This ensures the analysis focuses solely on the pressure response attributable to the initial wave impact. It can be observed that the computed results closely align with the experimental data on mean P_m . The theoretical model employs a time interval, Δt , which represents the period during which the momentum of the impinging wave front transitions from a horizontal to a vertical direction upon striking the structure. At the impact elevations, the presence of trapped air influences the pressure, leading to a nonlinear rise over time. This rate of pressure increase,

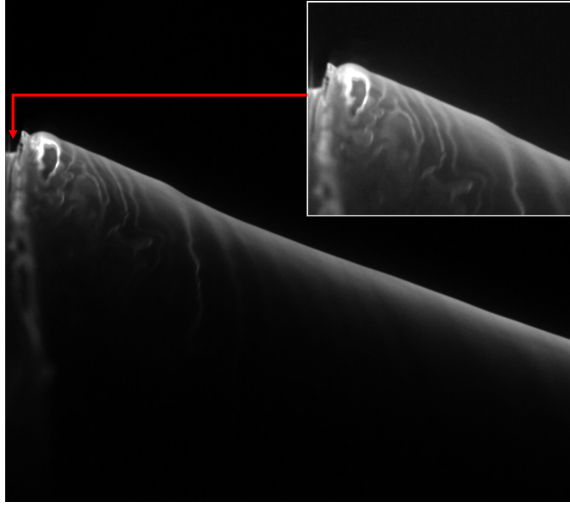


Figure 4.18: Snapshot from the LIF movie for the case $x_c/\lambda_0 = 5.483$, captured just prior to impact. The inset presents a magnified view of the nearly flat wave front immediately before contact with the structure.

similar to the results earlier discussed in 4.2.2, is not uniform due to the dynamics of trapped air, which the model oversimplifies by assuming a triangular pressure variation. Consequently, the model yields an "average" peak pressure within this interval. Therefore, the general agreement between the theoretical results and the experimental mean values, particularly at the impact zones, is not surprising. Thus, although originally formulated for a wall extending to the bed, this idealized pressure-impulse framework can also be adapted for structures of finite draft to provide a reasonable prediction of peak impact pressures. When applying the model to such configurations, it is essential to redefine the parameter h based on the actual submergence depth of the structure below the still-water level, ensuring that the resulting pressure distribution remains physically consistent with the finite-draft geometry.

The model, however, does not sufficiently capture the pressures at elevations below the impact zones. At these lower elevations, pressures arise from ongoing wave dynamics—alterations and movements in the water that exist before and continue past the impact event. These pressures stem from the general movement and behavior of

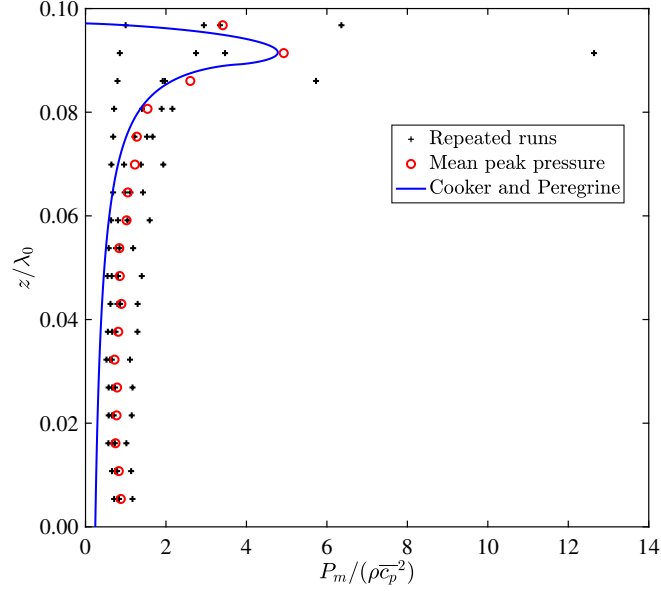


Figure 4.19: Peak pressure distribution comparison between the experimentally measured values and theoretical estimations of Cooker and Peregrine

the wave rather than solely its collision with the structure. As the theoretical model focuses on direct impacts using the impulse method, it overlooks these additional pressures. Consequently, while the model aligns accurately with average pressures at points of direct impact, it fails to consider the broader hydrodynamic effects at lower elevations. This aspect highlights a limitation in the model's ability to fully capture the complete range of hydrodynamic pressures experienced around a structure during wave impacts.

The theoretical predictions of the peak pressure distributions, derived from equations 4.7.1 and 4.7.3 heavily rely on the input values of U , h , μ , and Δt . These parameters are experimentally determined for the comparisons presented in this study. The values of U , h , and μ are obtained from surface profile data captured in LIF movies, while Δt is derived from pressure records, specifically calculated as twice the mean t_r for the P_m at the elevation where the highest pressures occur. Estimating these values prior to the experiments is challenging due to the complex dynamics of

wave breaking and interaction with the structure, particularly when air entrapment is involved, which further complicates the mechanics.

The estimations of U and Δt , which linearly scale the peak pressures as a function of the ratio $U/\Delta t$, can introduce subjective biases into the analysis of the peak pressure distribution. Despite these values being derived from realistic experimental observations, their direct influence on peak pressure calculations means that minor variations can disproportionately affect the outcome. To counteract this subjectivity and enhance the objectivity of the analysis, a normalization process can be implemented for the theoretical peak pressures. By normalizing these pressures against the maximum P_m value found in the data set, the impact of any variability in U and Δt on the calculated results can be reduced. This normalization scales the peak pressures to a uniform reference point, minimizing the effects of the initial parameter estimations. Similarly, this normalization technique can also be applied to the experimental data by adjusting the values based on the highest value of mean P_m recorded. This approach not only harmonizes the experimental data with the theoretical model but also ensures a more consistent comparison across various test runs. By focusing on the overall distribution of impact pressures rather than the absolute values, a better understanding of the underlying physical phenomena can be achieved without the bias introduced by the initial estimations of U and Δt .

To demonstrate the sensitivity of the theoretical predictions to variations in μ , the pressure distribution profiles are presented in Figure 4.20, with constant values for U , Δt and h . A larger μ , indicating a wider zone of impact, leads to a noticeable increase in both the P_m and the overall pressure distribution profiles. Such a trend is consistent with the observation that greater momentum is transferred to the wall within the same time period.

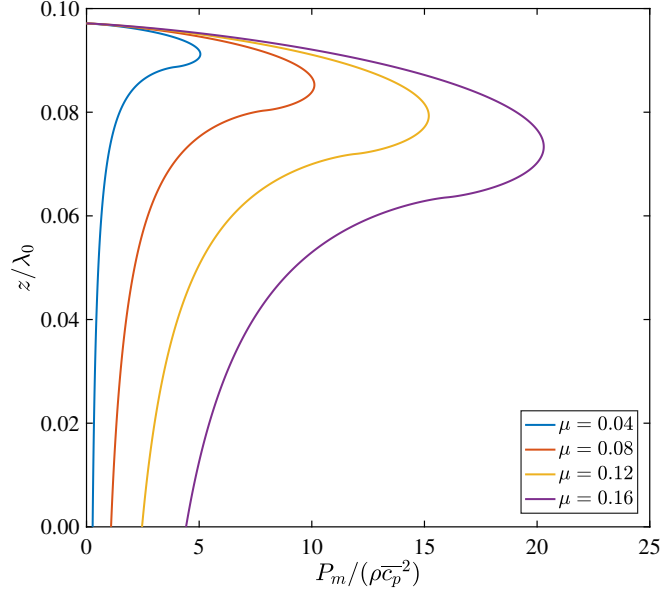


Figure 4.20: Variations in the theoretical estimate of the peak pressure distribution based on μ

4.8 Estimation of characteristic bubble size from oscillation frequency

The high-frequency pressure oscillations measured in the pressure readings indicate the presence of localized air cavities or bubble clusters entrapped near the impact zone. These air inclusions play a crucial role in modulating the transient pressure response by locally altering the compressibility of the fluid and introducing additional stiffness into the air–water mixture. However, despite the availability of LIF visualizations and other imaging efforts, the flow region containing the entrapped air bubbles could not be resolved, and thus direct observation of their size or distribution during the impact process remains unavailable. Estimating the characteristic bubble radius based on the measured oscillation frequency therefore provides an indirect means to quantify the scale of entrapped air and its potential influence on the pressure dynamics.

To overcome the limitation of direct optical measurements, the oscillation fre-

quencies extracted from the pressure time series are interpreted using bubble oscillation theory. Following the formulation of Plesset and Prosperetti (1977), the natural frequency of a spherical bubble in water is given by

$$f_n = \frac{1}{2\pi R_0} \sqrt{\frac{3kP_{eq}}{\rho} - \frac{2\sigma}{\rho R_0}}, \quad (4.8.1)$$

where R_0 denotes the equilibrium bubble radius, ρ is the density of water, $k = 1.4$ is the heat capacity ratio of air, and $\sigma = 73 \text{ mN m}^{-1}$ represents the surface tension of water. The pressure inside the bubble at equilibrium, P_{eq} , is related to the ambient pressure, P_∞ , by

$$P_{eq} = P_\infty + \frac{2\sigma}{R_0}. \quad (4.8.2)$$

Using equation (4.8.1), the oscillation frequency of 1757 Hz observed for the streamwise location $x_c/\lambda_0 = 5.483$, which corresponds to the highest measured pressure of $P_m = 12.64 \rho \bar{c}_p^{-2}$, can be associated with the natural frequency of a spherical air bubble with an approximate diameter of 3.8 mm under an ambient pressure of 1 atm. Such a small entrapped air pocket is therefore believed to have contributed to the highly impulsive pressure peak recorded at $z/\lambda_0 = 0.091$ for this condition. For the other streamwise locations within the low-aeration regime, namely $x_c/\lambda_0 = 5.466$ and $x_c/\lambda_0 = 5.474$, the corresponding dominant oscillation frequencies of 7031 Hz and 3515 Hz yield estimated bubble diameters of 0.94 mm and 1.9 mm, respectively.

Interestingly, these bubble diameters of 0.94 mm, 1.9 mm, and 3.8 mm correspond to the highest P_m values of $4.80 \rho \bar{c}_p^{-2}$, $8.62 \rho \bar{c}_p^{-2}$, and $12.64 \rho \bar{c}_p^{-2}$ for the three respective streamwise locations. The increase in estimated air-pocket diameter with distance from the wavemaker is physically intuitive: as the breaker steepens further downstream, it traps a slightly larger volume of air upon impact with the cube face.

However, this observation appears contrary to earlier studies, notably Bagnold (1939), who proposed—based on a one-dimensional piston model of water compressing an air cushion—that the highest impact pressures occur when the entrapped air layer is minimal. A key condition in Bagnold’s theory, however, is that the compression of the trapped air must be simultaneous over the entire pocket area. In the present experiments, stochastic surface irregularities of the wave crest and small-scale asymmetries prior to impact likely prevented this condition from being fully satisfied. The resulting nonuniform compression may therefore explain why larger estimated air-pocket diameters in our study correspond to higher P_m values, in contrast to the classical prediction.

Moreover, the possibility that these findings result from an isolated or anomalous experimental run is improbable. Even when averaging the highest P_m values across multiple realizations, the trend persists: the averaged peak pressures for $x_c/\lambda_0 = 5.466$, 5.474 , and 5.483 are approximately $3.00 \rho \bar{c}_p^2$, $6.22 \rho \bar{c}_p^2$, and $8.75 \rho \bar{c}_p^2$, respectively. As the plunging breaker becomes progressively steeper with increasing downstream position, the formation of larger entrapped air pockets—and consequently, higher impulsive pressures—appears consistent with the observed dynamics.

Although these estimated bubble sizes are approximate and represent an idealized single-bubble response, they provide valuable order-of-magnitude insight into the scale at which air compressibility effects become significant. Bubbles of this size oscillate within the kilohertz range, aligning well with the frequencies observed in the pressure records. Their presence can locally amplify pressure gradients, induce rapid compression–expansion cycles, and enhance the impulsive characteristics of the impact. These findings emphasize that even small quantities of entrained air can strongly influence short-duration pressure dynamics, reinforcing the crucial role of air compressibility in shaping the overall impact loads.

4.9 Persistence and Frequency of Post-Impact Pressure Oscillations

An interesting trend is observed in the post-impact oscillations for both the low-aeration and broken-wave impact regimes, as briefly discussed in Sections 4.2.2 and 4.2.3. In the low-aeration impacts, the pressure oscillations that follow the main impact peak persist for approximately $0.00081 tf_0$, $0.0009 tf_0$, and $0.0052 tf_0$ at the stream-wise positions $x_w/\lambda_0 = 5.466$, 5.474 , and 5.483 , respectively. The persistence of these oscillations clearly increases as the cube is positioned farther downstream. A plausible explanation for this trend lies in the progressive increase in the size and coherence of the entrapped air pocket between the advancing crest and the cube face. Physically, the trapped air–water region may be viewed as a coupled *mass–spring–damper system*, in which the inertia of the surrounding water and air forms the effective mass M_{eff} , the compressibility of the trapped air provides the stiffness K_{eff} , and various dissipative mechanisms (acoustic radiation, viscosity, turbulence) contribute to the effective damping coefficient C_{eff} . The dynamics of such a system are classically represented by

$$M_{\text{eff}}\ddot{x} + C_{\text{eff}}\dot{x} + K_{\text{eff}}x = 0, \quad (4.9.1)$$

as described in standard vibration theory (De Silva, 2006).

The natural frequency of this oscillator is given by

$$\omega_n = \sqrt{\frac{K_{\text{eff}}}{M_{\text{eff}}}}, \quad f_n = \frac{\omega_n}{2\pi}, \quad (4.9.2)$$

and the effective stiffness of the air pocket depends primarily on its compressibility, approximated as

$$K_{\text{eff}} \approx \beta_{\text{air}} \frac{A^2}{V}, \quad (4.9.3)$$

where A is the wall contact area, V the trapped air volume, and β_{air} the bulk modulus of air ($\beta_{\text{air}} \approx p_0 \gamma$, with $\gamma \approx 1.4$). As the cube moves downstream, the entrapped air volume V increases, leading to a lower stiffness K_{eff} and, consequently, a lower natural frequency. This interpretation is consistent with the measured dominant frequencies of 7031 Hz, 3515 Hz, and 1757 Hz for the same three streamwise positions.

The persistence of oscillations, on the other hand, is governed by the damping ratio

$$\zeta = \frac{C_{\text{eff}}}{2\sqrt{K_{\text{eff}}M_{\text{eff}}}}, \quad (4.9.4)$$

which determines how rapidly the oscillations decay according to

$$p(t) = p_0 e^{-\zeta \omega_n t} \sin(\omega_d t), \quad \omega_d = \omega_n \sqrt{1 - \zeta^2}. \quad (4.9.5)$$

As damping decreases, oscillations persist longer. The effective damping coefficient C_{eff} is influenced by three dominant processes: (i) *radiation damping*, where acoustic energy leaks from the cavity into the surrounding water; (ii) *thermo-viscous damping*, due to heat transfer and viscous losses in small bubbles; and (iii) *turbulent damping*, caused by incoherent pressure fluctuations in fragmented bubble clouds (Ainslie and Leighton, 2011). As the air pocket becomes larger and more contiguous, its acoustic impedance relative to water decreases, reducing radiation losses. Similarly, thermo-viscous damping weakens at lower frequencies, and turbulent losses diminish as oscillations become more coherent (Sojahrood et al., 2019). Together, these effects reduce C_{eff} downstream, producing progressively longer-lived oscillations.

A similar trend in the persistence of oscillations is observed in the *broken-wave impact regime*, where the cube is positioned even farther downstream and the plunging breaker has already collapsed on the upstream water surface before striking the structure. For these cases, the persistence of oscillations increases to approximately $0.012 tf_0$, $0.0195 tf_0$, and $0.074 tf_0$ at $x_w/\lambda_0 = 5.50$, 5.544 , and 5.68 , respectively. Although the oscillations in this regime are no longer perfectly in phase across sensors, they still exhibit partial coherence, indicating that a comparable mechanism may be at play. Here, instead of a single, discrete air pocket, a thick aerated layer forms near the wall—a mixture of water and dispersed bubbles that may act as a *compressible acoustic medium*. Such bubbly layers are known to strongly reduce sound speed and attenuate high-frequency energy (Commander and Prosperetti, 1989; Johnsen and Colonius, 2009). The reduction in sound speed can trap and recycle acoustic energy between the wall and the free surface, causing slower radiative damping and prolonged oscillation decay.

It is therefore possible that, as the system transitions from a compact entrapped air cavity (low-aeration regime) to an extended bubbly layer (broken-wave regime), the fundamental oscillation frequency continues to decrease while the persistence time increases. In the broken-wave cases, the thick, aerated layer may behave as a weakly damped “acoustic cushion,” sustaining low-frequency oscillations over a longer period despite the flow’s turbulence.

4.10 Influence of entrained air on pressure-wave transmission during impacts

In low-aeration impact conditions, a pocket of air is typically entrapped during the wave interaction, which may subsequently disperse into small bubbles or be released

into the surrounding atmosphere. These impacts are characterized by high-intensity peak pressures (P_m) that occur nearly simultaneously over a broad range of elevations. This behavior is attributed to the steepened plunging breaker, which reaches near-critical steepness just before impact and imparts significant horizontal momentum to the water particles interacting with the front face of the cube. Among the 24 pressure sensors used in this study, one sensor located at $z/\lambda_0 = -0.0053$ remains submerged below the still water level throughout the experiment, both before and after the wave impact. Given its position, it is unlikely that this sensor experiences the same level of horizontal water particle velocity as those located at higher elevations, which are more directly impacted by the steepened wave crest. Despite this, analysis of the pressure time series at this submerged location reveals that the P_m at this sensor occurs after the peak pressure is recorded at the elevation where the maximum P_m is observed in each case.

Specifically, the peak pressure at $z/\lambda_0 = -0.0053$ occurs approximately 0.5 ms, 1.1 ms, and 2 ms after the maximum P_m for the low-aeration cases at $x_c/\lambda_0 = 5.466$, 5.474, and 5.483, respectively. Since this sensor is not directly impacted by the high-velocity water near the surface, these delayed peaks are interpreted as resulting from pressure waves propagating downward from the region of primary impact. Using the known elevation difference between the two sensors, the estimated propagation velocities of these pressure waves are approximately 261.65 m/s, 109.36 m/s, and 70 m/s for the three respective cases. Although the speed of sound in pure water is approximately 1500 m/s, even a small concentration of entrained air can substantially reduce the speed of acoustic wave propagation. Entrapped air bubbles act as local resonators, absorbing and slowing the pressure wave. Theoretical studies, including those by Crighton et al. (2012), indicate that a void fraction as small as 1% can reduce the effective sound speed to less than one-tenth of its value in pure water.

Using a theoretical expression proposed by Wallis (1969) for the velocity of an acoustic wave (v_s) in a homogeneous mixture containing small bubbles with a density much lower than that of water, it is possible to estimate the air fraction (α) entrained during low-aeration impacts. The expression is given by

$$v_s = \frac{v_g}{\sqrt{\frac{\rho_w}{\rho_a} \alpha (1 - \alpha)}}, \quad (4.10.1)$$

where v_g is the speed of sound in air, and ρ_w and ρ_a are the densities of water and air, respectively. Based on equation (4.10.1), estimates of α corresponding to different values of v_s measured within the bubble-entrained water mass are presented in Table 4.2.

Using the values of v_s determined from the pressure-wave propagation discussed previously, the estimated air contents for the three low-aeration cases at $x_c/\lambda_0 = 5.466$, 5.474, and 5.483 are approximately 0.2%, 1.1%, and 2.9%, respectively. Consistent with these estimates, as discussed earlier, the corresponding dominant post-impact oscillation frequencies are 7031 Hz, 3515 Hz, and 1757 Hz for the same three cases, clearly showing that the oscillation frequency increases as the aeration level decreases. This trend arises because a lower air content yields a stiffer water–air mixture that behaves more like pure water, thereby allowing the entrapped air pocket or bubbles to oscillate at a higher natural frequency (i.e., the system “rings” faster following impact). Conversely, greater aeration increases the mixture’s compressibility, reducing its effective stiffness and leading to slower oscillations at lower frequencies.

It is important to note that the theoretical expression in equation (4.10.1) is based on the assumption of a homogeneous bubbly mixture composed of small, uniformly distributed gas bubbles within a continuous liquid phase. This assumption

Table 4.2: Sound velocity in water v_s and air content α

Air content α (%)	0.2	0.7	1.2	1.7	2.7	3.5
Sound velocity v_s (m/s)	264.4	141.68	108.49	91.38	72.88	64.27

is reasonably valid in low-aeration scenarios, where the air content remains small and the bubbly structure is relatively uniform. However, in broken-wave impact cases, the flow becomes highly aerated and heterogeneous, often comprising large, irregular air pockets and turbulent air–water mixtures. Under such conditions, the local void fraction can approach 0.5, and the assumptions underlying the theoretical model are no longer satisfied. In particular, the denominator in equation (4.10.1), which scales with $\alpha(1 - \alpha)$, becomes vanishingly small as α approaches unity, leading to unphysical predictions of infinite sound speed. For these reasons, the application of this expression is limited to low-aeration cases, and it is not employed for estimating void fractions under broken-wave impact conditions.

Chapter 5

Summary and Conclusion

In the research presented in this dissertation proposal, the interaction between a deep-water plunging breaker and a finite-draft cube was experimentally investigated in a wave tank measuring 14.8 m in length and 1.22 m in width. The water depth was maintained at 0.91 m throughout the experiments, while the bottom face of the cube was submerged to a fixed depth of $z = -0.133$ m relative to the still-water level. The plunging breaker was generated using a dispersive focusing technique, and its interaction with the cube's front face was examined at eight different stream-wise positions ranging from $x_c = 5.423\lambda_0$ to $x_c = 5.680\lambda_0$, where $\lambda_0 = 1.181$ m denotes the wavelength of the focused wave packet. The free-surface evolution and wave-structure interaction were captured using a cinematic laser-induced fluorescence (LIF) technique, while impact pressures were recorded by an array of 24 fast-response piezoelectric sensors mounted on the front face of the cube.

Based on the synchronized visual observations and pressure measurements, the cube-wave interactions were categorized into three distinct impact regimes, with the elevation of wave contact (z/λ_0) consistently decreasing as the cube was positioned farther downstream. In the first regime ($x_c = 5.423\lambda_0$ and $5.431\lambda_0$), no air entrapment occurred as the plunging breaker reached the cube. The LIF images and pressure data indicated a sloshing-type interaction, where the rising water along the cube face met an underdeveloped breaker crest, producing a single pressure peak with a comparatively gradual rise time. The upward motion of the contact point along the cube face followed a clear power-law relationship with time. The highest impact pressures for this regime occurred near the intersection of the rising water surface and

the approaching crest, even though the crest itself never made direct contact with the cube. This regime also exhibited the highest pressure impulse over the entire duration of the wave–structure interaction.

In the second regime ($5.466\lambda_0 < x_c < 5.483\lambda_0$), the plunging jet impacted the cube before the rising contact surface reached the elevation of impact, entrapping an air pocket between the jet and the structure. This resulted in substantially higher impact pressures due to the compression and subsequent expansion of the trapped air, as evidenced by pronounced high-frequency post-impact oscillations in the pressure records. Both the no-aeration and low-aeration regimes exhibited a two-segment surface evolution: one defined by the curvature of the water surface rising along the cube and the other by the upstream curvature of the approaching crest. As the cube was moved farther downstream, the rising segment retained a gentle curvature, while the upstream segment became progressively steeper with increased breaker development. The velocity of the transition point between these segments decreased with increasing streamwise position, whereas its horizontal extent—representing the effective thickness of the rising water column—grew gradually. The highly impulsive pressures observed in the low-aeration regime were found to agree closely with the predictions of the idealized pressure-impulse theory of Cooker and Peregrine (1990).

In the third regime ($5.50\lambda_0 < x_c < 5.68\lambda_0$), the plunging breaker collapsed onto the upstream free surface before reaching the cube, generating a secondary jet that preceded the main crest. Both features merged into a turbulent air–water mixture before impacting the structure. The corresponding pressure signals exhibited strong post-impact oscillations, indicative of intense air entrapment. Although the peak pressures in this regime were lower than those in the low-aeration case, they remained significantly higher than those in the non-aerated impacts. Because the impact occurred closer to the mean water level and over a more confined vertical

region, this regime registered the lowest values of pressure impulse, maximum force, and force impulse.

Distinct differences across the three regimes were evident in terms of local wave period, post-impact pressure dynamics, and the vertical velocity of the plunging surface. Overall, the most severe and impulsive loads were observed for the low-aeration regime at $x_c = 5.483\lambda_0$, where a small entrapped air pocket and strong surface focusing combined to produce the highest measured pressures and the most energetic impact event.

Chapter 6

Future Work

The experimental results and analyses presented in this proposal have established a strong foundation for understanding the interaction between a plunging breaking wave and a finite-draft vertical structure. Current efforts are focused on finalizing and submitting a comprehensive manuscript to the *Journal of Fluid Mechanics*, which presents detailed findings on impact classification, wave kinematics, and pressure dynamics developed thus far. Building on this foundation, the remaining work toward the completion of the dissertation will focus on extended analysis, comparative modeling, and synthesis of the results into a unified framework. The following tasks are planned for the next six months leading up to the final defense:

1. Finalization and Submission of the *JFM* Manuscript

Complete the ongoing manuscript being prepared for submission to the *Journal of Fluid Mechanics*, which consolidates the key findings on the kinematic and dynamic characteristics of plunging-breaker impacts. The paper will present a comprehensive discussion of the experimental results analyzed and described thus far in this dissertation.

2. Exploration of Transitional Impact Regimes

Examine the critical streamwise positions of the cube at $x/\lambda_0 = 5.44$, 5.449, and 5.457, where both aerated and non-aerated impacts are intermittently observed across multiple runs. The goal is to identify the specific kinematic and geometric conditions that determine whether air entrapment occurs at these transitional locations and to understand the mechanisms governing this variability.

3. Comprehensive Regime-Level Analysis

Most analyses so far have focused on a single representative case from each impact regime (no-aeration, low-aeration, and broken-wave) to establish characteristic behaviors. The next phase will expand this approach by analyzing multiple trials and streamwise positions within each regime. This broader investigation will quantify how key kinematic and dynamic parameters—such as contact-point velocity, free-surface evolution, peak pressure, impulse, and post-impact oscillations—vary across cases within the same regime. The objective is to develop a comprehensive regime-level characterization that captures variability and transitional overlaps, thereby strengthening the physical interpretation and facilitating the development of predictive models.

4. Preparation and Submission of an Additional Journal Manuscript

Following the completion of the extended analyses, prepare and submit another journal manuscript summarizing the new insights gained from the expanded dataset. The paper will integrate the findings from the transitional and regime-level studies, providing a cohesive interpretation of the impact dynamics, scaling behavior, and aeration effects on finite-draft structures.

Completing these tasks will lead to a comprehensive understanding of plunging-breaker impact mechanics on finite-draft structures, helping to bridge the gap between traditional full-depth wall impacts and realistic offshore configurations. The planned work is expected to be finished within the next six months, ensuring full readiness for the final dissertation defense.

Table 6.1: Proposed timeline for completion of remaining dissertation tasks.

Tasks	Dec 2025	Jan 2026	Feb 2026	Mar 2026	Apr 2026	May 2026
Finalize and submit <i>JFM</i> manuscript	*	*				
Explore transitional impact regimes	*	*				
Comprehensive regime-level analysis	*	*	*			
Prepare and submit journal manuscript			*	*	*	
Finalize dissertation writing				*	*	
Defend dissertation						*

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